# Claude Multi-Account Fine Tuning


Changelog
- v1.33.1 (2026-10-27)
  - Also in v1.33.1 (fixes that landed on `main` after v1.33.0)
- v1.33.0 (2026-10-24)
  - Added
  - Changed
  - Fixed
  - Merged from origin/main in this release (d992502, 7ab1ab0)
  - Known limitations / imperfections (tracked, not fixed in this release)
- v1.32.0 (addendum) — jev-setup / Kimi-lock review rounds, already in the `v1.32.0` tag
- v1.32.0 — 2026-10-22
  - BREAKING (security) — read before upgrading
  - Fixed
  - Added
  - Changed
- v1.31.0 — 2026-10-22
  - Changed
  - Added
  - Security
- v1.30.6 — 2026-10-21
  - Changed
  - Added
- v1.30.5 — 2026-10-10
  - Fixed
  - Added
  - Documentation
  - Verification
  - Commits in this release
- v1.30.4 — 2026-10-06
  - Added
  - Fixed
  - Known limitations (not fixed in this release)
  - Verification
  - Commits in this release
- v1.30.3 — 2026-10-05 — `quota/limits` patch: Kimi-leg fixes, live-test gating, doc corrections
  - Fixed

Incident (disclosed)
Testing and validation
Added
Known limitations (not fixed)
v1.30.2 — 2026-10-05 — `quota/limits` patch: review and known-issues fixes
Fixed
Known limitations
Earlier release
v1.30.1 — 2026-10-05 — `quota/limits` patch: post-release review fixes
Fixed
Testing & Validation
Known remaining gaps (not fixed in 1.30.1)
v1.30.0 — 2026-10-04 — `quota/limits`: universal usage reporting across every provider alias and native account
Added
Fixed (found during this feature's own review process)
Testing & Validation
v1.29.0 — 2026-10-03 — llmctl integration hardening: honest limits, safe switching, live Superpowers verification
Added — detection honesty (US1)
Added — on-demand switching hardening (US2)
Added — live Superpowers-command verification (US3)
Added — documentation (US4)
Added — upstream `llmctl` fixes (operator-directed scope expansion)
Testing & Validation
v1.28.0 — 2026-09-11 — Token Router (138+ models) + Pi CLI agent + Helix family for Pi
Added — Token Router integration
Added — Pi CLI agent support (new family layer)
Changed
Testing & Validation
v1.27.0 — 2026-09-05 — Kimi Code CLI accounts + `kimi-<id>` provider aliases (+ one-time legacy `kimi-*` → `kc-*` rename)
Added — Kimi Code support
Breaking — one-time legacy rename (`kimi-*` → `kc-*`)
Unreleased commits since v1.26.8 riding along
Notes
v1.26.8 — 2026-08-28 — autocompact thrashing guard + safety margin for provider aliases
Fixed
Changed
v1.26.7 — 2026-07-27 — vendor-prefixed model ids routable again (16 of 24 aliases 503'd before any network call) + the vendored-Go fallback that never fell back + two misdiagnoses retired
Fixed
Fixed (ccr submodule, vasic-digital/claude-code-router)
Changed

Added (folded from v1.26.5 / v1.26.6 — shipped but never changelogged)

Notes

v1.26.1 — 2026-07-25 — ccr buildable on same-minor toolchains + launch locks that actually lock + honest toolchain-skew guidance

Added

Fixed

Fixed (ccr submodule, vasic-digital/claude-code-router)

v1.26.0 — 2026-07-24 — free-tier-first per-model aliases by default + dynamic account dispatch + provider-scoped gateway paths

Added

Fixed

Notes

v1.25.5 — 2026-07-23 — fail-loud install over broken/missing ccr + interactive-shell stderr fix + anti-fossil r-m-w hardening

Fixed

Verified

v1.25.4 — 2026-07-22 — un-fossilise ephemeral cma-proxy addresses (kill the 502 class) + toon Go port

Fixed

Added

v1.25.3 — 2026-07-22 — trim knob survives generator regen + operator docs for trim/gate/container-mode

Fixed

Added

Changed

v1.25.2 — 2026-07-22 — ccr resolved by stable path (PATH-shadowing doppelgänger), provider trim mode, mandatory live release gate

Fixed

Added

v1.25.1 — 2026-07-22 — router aliases refused to launch on a stale bundled ccr (self-healing rebuild)

Fixed

Verification gap that let it ship (honest)

Verified

v1.25.0 — 2026-07-22 — the compatibility proxy goes Go (cma-proxy) + HelixAgent fully working (routed + capacity-sized + tool-engaging) + provider-verification honesty + rc-safety

Proxy framework — python proxies replaced by one Go binary (`cma-proxy`)

Fixed

Reliability

Safety — shell-rc writes (§11.4.167)

Repo & CI hygiene

Submodules

Verified

v1.24.0 — 2026-07-21 — credit-aware model selection; and the v1.23.0 launch regression + the false "ALL GREEN" claim that hid it

Added — credit-aware model selection (mandatory tier policy)

Fixed

Fixed — the verification that was supposed to catch this

Added — regression guards

Verified

v1.23.0 — 2026-07-20 — Bundled Go claude-code-router is now the SOLE router (JS fully replaced) + full retest

Changed

Fixed

Verified (full live retest — captured evidence in `scripts/tests/proof/`)

v1.22.9 — 2026-07-20 — bundled claude-code-router v0.4.9 (`--upstream-timeout`) + verify_ccr_live.sh auth+proxy live-proof legs (49 checks)

v1.22.8 — 2026-07-20 — bundled claude-code-router v0.4.8 (authenticated outbound proxy config block, redacted password)

v1.22.7 — 2026-07-20 — bundled claude-code-router v0.4.7 (inbound auth switch, `--max-attempts`, start/ui flag forwarding)

v1.22.6 — 2026-07-20 — bundled claude-code-router v0.4.6 (docs correction; no behavior change)

v1.22.5 — 2026-07-20 — bundled claude-code-router v0.4.5 (OpenAI-facade long-context routing symmetry)

v1.22.4 — 2026-07-20 — bundled claude-code-router v0.4.4 (streaming token accounting on both relay paths)

v1.22.3 — 2026-07-20 — bundled claude-code-router v0.4.3 (TLS/HTTP3 CLI flags, OpenAI-facade metric parity, synchronous bind) + verify_ccr_live.sh TLS/HTTP3 live-proof leg

v1.22.2 — 2026-07-19 — bundled claude-code-router v0.4.2 (transport/hot-reload/load live suites)

v1.22.1 — 2026-07-19 — bundled claude-code-router v0.4.1 (metrics polish + live e2e); toolkit verify_ccr_live.sh live proof of the bundled Go ccr

v1.22.0 — 2026-07-19 — bundled claude-code-router v0.4.0 (/metrics + semantic cache wired, Router.think, exhaustive tests)

v1.21.0 — 2026-07-19 — bundled Go claude-code-router built+installed as `ccr` (claude-ccr-build); submodule bumped to v0.3.0 (response cache + cross-provider fallback wired)

v1.20.0 — 2026-07-19 — bump bundled claude-code-router to v0.2.0 (multi-protocol gateway: OpenAI inbound facade, Anthropic passthrough, classifiers, Think/LongContext routing, hot-reload, redacted logging)

v1.19.0 — 2026-07-19 — port deepseek+xiaomi to router (ccr) transport + IPv4 fix

v1.18.0 — 2026-07-19 — HelixAgent/HelixLLM local-model exposure + 128k output-token clamp + ccr launch fixes

Added

Fixed
Testing / validation
v1.17.0 — 2026-07-18 — Cross-alias session & background-agent continuity
Fixed
Added
v1.16.0 — 2026-07-18 — Output-token cap for router providers
Fixed
Added
v1.15.0 — 2026-07-18 — Full Kimi variant support (OAuth subscription models)
Added
Fixed
Testing
v1.14.0 — 2026-07-17 — Anti-bluff provider verification (strict filtering)
Fixed
Added
Changed
v1.13.3 — 2026-07-17 — Session-sharing pipefail fix
Fixed
Added
v1.13.0 — 2026-07-10 — Project-scoped cwd-hook (session-resumption fix)
Fixed
Added
v1.12.3 — 2026-07-05 — Session-name sanitization (kebab-case)
Changed
Added
v1.12.2 — 2026-07-05 — Native alias auto-registration + account-detection hardening
Fixed
Added
v1.12.1 — 2026-07-05 — Judge independence + resolve/ robustness hardening
Changed
Added
Fixed
v1.12.0 — 2026-07-05 — Semantic code-visibility (layer 3) + live TUI verification (layer 4)
Added
Fixed
Notes
v1.11.2 — 2026-07-03 — Token-limit guard + re-enabled provider proxies + Poe tool cap
Fixed
Changed
Added
v1.11.1 — 2026-07-03 — Live per-alias Claude Code verification (CLI + TUI) + provider fixes
Added

Fixed
Verified (live on host)

v1.10.8 — 2026-07-01 — noclobber-safe router-provider config write

Fixed
Added (tests)
Verified
Note (not a toolkit bug)

v1.10.7 — 2026-06-29 — Shared-items drift guard + audit closure

Added (tests)
Audited — no code changes (3 parallel investigators; every finding independently checked)
Verified

v1.10.6 — 2026-06-29 — Committed credential-leak regression test

Added (tests)
Verified

v1.10.5 — 2026-06-29 — Provider 'null' field normalization + coverage

Fixed
Added (tests)
Audited — no change needed (independently verified, not taken on trust)
Verified

v1.10.4 — 2026-06-29 — set -e/pipefail abort fixes + hardened test coverage

Fixed
Changed (tests)
Verified

v1.10.3 — 2026-06-29 — Execution-level wrapper test coverage

Added
Verified

v1.10.2 — 2026-06-29 — Self-healing rc source lines + strict rc tests

Fixed
Added
Verified

v1.10.1 — 2026-06-29 — Robust cma_run wrapper assertions

Fixed
Verified

v1.10.0 — 2026-06-29 — Auto-applied per-alias session color + coverage/wiring

Added
Fixed
Verified

v1.9.2 — 2026-06-29 — Hermetic CLAUDE_BIN resolver test

Fixed
Verified

v1.9.1 — 2026-06-29 — CLAUDE_BIN resolves across per-host install locations

Fixed

Verified

v1.9.0 — 2026-06-29 — Per-project auto-sessions that actually work live + zero-coverage tests

Fixed

Added

Verified

v1.8.1 — 2026-06-29 — Merge-engine correctness + portability hardening

Fixed

Added

Verified

v1.8.0 — 2026-06-29 — Alias isolation + token-limit guard + per-project auto-sessions

Fixed

Added

Verified

v1.7.12 — 2026-06-28 — One-line curl installer

Added

Verified

v1.7.11 — 2026-06-28 — Round-4: coverage-gap regression tests, toon recursion guard, arg validation

Fixed

Added — coverage-gap regression tests (the same class that let enable-plugins

Verified

v1.7.10 — 2026-06-28 — Round-3 audit: enable-plugins bug fix, path-traversal guards, proxy robustness

Fixed

Security (defense-in-depth)

Robustness

Verified

Audit (round 3) — verified clean

v1.7.9 — 2026-06-28 — Hardening round 2: injection-safe alias writes, broadened secret redaction, docs accuracy, shellcheck 0

Security

Fixed

Docs

Quality

Verified

v1.7.8 — 2026-06-28 — Secret hygiene (argv + committed-proof leaks), dead-code fix, coverage tests

Security

Fixed

Added

Verified

v1.7.7 — 2026-06-28 — Portable realpath (BSD portability hardening), set -u edge fix, regression tests

Fixed

Added

Verified

Audit findings (v1.7.6 — no code change required)

v1.7.6 — 2026-06-28 — Always-non-interactive execution, alias-file integrity, macOS/bash-3.2 portability, 4-host rollout

Fixed

Added

Multi-host rollout (nezha · mistborn.local · thinker.local · amber.local)

Verified

v1.7.5 — 2026-06-28 — Cross-provider /resume session visibility fix

Fixed

Root Cause Analysis

Verified

Tests

v1.7.4 — 2026-06-26 — Kimi provider fix + AWS IaC MCP disabled by default

Fixed

Changed

Tests

v1.6.6 — 2026-06-21 — TOON integration for token-efficient prompts

Added

Token Savings

Note

Tests

v1.6.5 — 2026-06-21 — Poe proxy fix (alias file + install)

Fixed

Verified

Tests

v1.6.4 — 2026-06-21 — Poe proxy fix for tool compatibility

Fixed

Verified

Tests

v1.6.3 — 2026-06-21 — Poe provider (382 models, 3 aliases)

Added

Verified

v1.6.2 — 2026-06-21 — Chutes provider documentation + model update

Changed

Verified

v1.6.1 — 2026-06-21 — cache_control fix + E2E tests

Fixed

Added

Verified working (all aliases tested with "Do you see our codebase?")

v1.6.0 — 2026-06-21 — Multi-alias provider system

Added

Changed

Full test suite

Usage

v1.5.1 — 2026-06-20 — Linux stat fix + nezha deployment

Fixed

Deployment

Full test suite
v1.5.0 — 2026-06-20 — Cross-alias session visibility
Added
Changed
Full test suite
How it works
Performance
v1.4.0 — 2026-06-20 — OpenCode Zen provider alias
Added
Changed
Full test suite
Honest notes
v1.3.0 — 2026-06-19 — Xiaomi MiMo provider alias
Added
Changed
Full test suite
Honest notes
v1.2.0 — 2026-06-19 — Z.AI Coding Plan provider alias
Added
Changed
Full test suite
v1.1.0 — 2026-06-16 — Distributed infrastructure + provider verification
Added
Changed
Fixed (LLMsVerifier — shipped as PR #2, applied to deployed builds)
Verified live (real "Do you see my code?" against production APIs)
Safety
v1.0.0 — 2026-06-16 — Dynamic provider-alias generator
v1.6.7 — 2026-06-21 — Poe proxy fix for all aliases
Fixed
Verified
Tests
v1.6.8 — 2026-06-21 — Poe proxy gzip fix
Fixed
Verified
Tests
v1.6.9 — 2026-06-21 — Poe proxy $ref fix for Grok-4
Fixed
Verified
Tests
v1.7.0 — 2026-06-22 — Poe proxy complete fix (all aliases verified)
Fixed
Verified (all three aliases through full Claude Code flow)
Root Cause Analysis
Tests
v1.7.1 — 2026-06-22 — Full validation + release
Fixed
Tests

Release
v1.7.2 — 2026-06-22 — Claude alias verification, full release
Added
Tests

# Changelog

All notable changes to the Claude multi-account toolkit.

## v1.33.1 (2026-10-27)

Patch release on top of v1.33.0 (tags `v1.33.0` / `claude_toolkit-1.33.0`, both at `53bf5fb`). It also carries the two test/launch fixes that landed on `main` after v1.33.0 (`f52fa7f`, `31176b8`; see "Also in v1.33.1" below).

- **Carrier propagation fix** (`CLAUDE.md`, `AGENTS.md`, `GEMINI.md`, `QWEN.md`): each carrier now has the standard `## INHERITED FROM constitution/CLAUDE.md` pointer section (§11.4.35), so the constitution hub propagation gates (11.4.279 / 287 / 288 / 289) classify the four toolkit carriers as pointer-inheritance instead of MISSING. Documentation only; no script or runtime behaviour changes. The toolkit keeps no nested constitution copy (own-org dependencies are resolved, see "Shared dependency layout").
- **New test `scripts/tests/test_carrier_inherited_heading.sh`**: asserts each of the four carriers has a line-anchored, non-fenced `## INHERITED FROM` heading, with the predicate self-validated against six mutants (``` fence, ~~~ fence, wrong case, demoted `###`, indented, inline) plus a golden-good control. Written test-first: RED on the pre-patch carriers (4 failed), GREEN after (11 passed).
- **Honest limit**: these four gates only go fully green for the toolkit together with the separate hub exclusion row for the `superspec` example fixture (`.specify/extensions/superspec/examples/static-landing-page/CLAUDE.md`), which lives in the hub's `config/covenant_propagation_exclusions.tsv`, not in this repository. Until that row lands, each of the four gates still reports that one fixture as MISSING.
- **Release gate**: the `claude-release-gate` LIVE alias-smoke leg needs real provider credentials and was NOT RUN for this prep; only the non-live legs were run.

### Also in v1.33.1 (fixes that landed on `main` after v1.33.0)

- **Launch wrappers no longer abort on a failed `stat -c %Y`** (`scripts/tests/test_jev_env_stat_race.sh`, found by the full-suite run: `test_kimi_alias_file.sh` failed 1 run in 20, `https twin without CA pin still launches`, rc 1). `cma_jev_env` (emitted into every alias file), `_cma_mtime`, the provider-sync TTL check and `cma_jq_edit`'s chmod read a file's mtime or mode as `stat -c ... || stat -f ...`. On GNU coreutils `stat -f` is FILESYSTEM status, not BSD's format flag: when `-c` failed because the endpoint cache did not exist yet and the file appeared before the fallback ran (the function's own detached refresh creates it), the fallback printed `File: ... ID: ... Namelen: ...` on stdout, `echo 0` appended `0`, and `$(( now - mtime ))` died ("syntax error in expression", or "File: unbound variable" under `set -u`), aborting the wrapper before it launched anything. The value is now validated as digits only (octal for the mode); anything else is "unknown" = stale, no chmod. The new test makes the race deterministic with a `stat` shim that fails the first `-c` call (RED on the old code: wrapper aborted, `_cma_mtime` returned the filesystem text; GREEN with the guard) and carries its own paired mutant (guard removed from a copy, killed). 42 further runs of the previously flaky suite: 42 passed.

- **`scripts/tests/test_jev_hosted_live_fixes.sh` T7a is hermetic about `OPENROUTER_API_KEY`** (found by the full-suite run on a host that exports it): the case asserts "no request at all (OPENROUTER_API_KEY is unset)" but nothing unset the variable, so the gate read it, found it set and legitimately sent the request (sink log: 1 line, `want=0 got=1`). Reproduced on the unmodified v1.33.0 tree: fails with the variable exported, passes under `env -u OPENROUTER_API_KEY`. The suite now unsets it up front (cases that need it set pass it inline). Paired mutant (a dummy key exported instead of the unset) fails exactly that assertion, so it still bites.

# v1.33.0 (2026-10-24)

Minor release on top of v1.32.0 (tags `v1.32.0` / `claude_toolkit-1.32.0`, both at `adae02d`). It is a minor, not a patch, release by operator decision because it adds a feature set: the portable `jev_realpath` path helper and the hermetic wrapper tests, alongside fixes. The `jev-setup` validation and Kimi-lock items listed in the "v1.32.0 (addendum)" section below were already shipped in the `v1.32.0` tag and are NOT new in this release.

## Added

- `jev_realpath` (`scripts/jev-kimi-lib.sh`): a portable canonical-path helper that resolves symlink chains without `readlink -f` (BSD/macOS `readlink` has no `-f`); covered by the new `test_jev_portable_path.sh` (20 assertions).
- `test_jev_wrappers_env_hermetic.sh`: proves `test_jev_wrappers.sh` gives the same result under hostile inherited `JEV_*` / `LLMCTL_*` variables.

## Changed

- `scripts/jev-setup.sh` and `scripts/jev-kimi-lib.sh` use `jev_realpath` instead of `readlink -f` (two call sites).
- `docs/CONTINUATION.md` and this changelog corrected to the real tag state; the README tests badge and test-file count now match the measured count (126, including the new `test_jev_hosted_live_fixes.sh` and `test_jev_key_origin_class.sh`).

## Fixed

- `jev_realpath` ignores an exported `CDPATH` (a relative `cd` was redirected and printed an extra line) and treats leading-dash names as paths by passing `--` to `readlink`, `dirname`, `basename` and `cd` (cells P3b/P3c).
- `test_jev_wrappers.sh` scrubs inherited `JEV_*` / `LLMCTL_*` variables before it starts (a developer shell exporting them made 13 assertions fail).
- `test_llmctl_ondemand_switch.sh` gained the jev opt-out that removed a rare flaky failure (`74820fe`).

Defects found by live validation of the hosted Jev endpoint (TypeSafe `/v1/systemone`, 2026-10-24), each with a regression test in the new `test_jev_hosted_live_fixes.sh` (85 assertions pass, 1 skipped; stub servers and dummy keys only):

- `claude-providers sync-jev` with `CMA_JEV_LIVE=1` sent `questions` as an array; the server requires a map (name -> {type, instructions, criteria}) and answered HTTP 422, so a working hosted endpoint was downgraded to `unverified` and no endpoint resolved. The probe now sends the map form.
- `sync-jev` ignored `CMA_PROVIDERS_DIR` (`cma_providers_dir` was hard-wired to `$HOME`). It is honoured now for the endpoint cache, `status.json`, the Jev lock/stamp directory (`<dir>/jev-state`, unless `JEV_STATE_DIR` is set) and the kimi twin hook refresh (`<dir>/kimi-twins/.kimi-prov-*`). With it unset nothing changes. Verified against a checksummed, writable hostile HOME (any stray write shows in the before/after snapshot). Review follow-up: the READERS now use the same directory, so a `sync-jev` under `CMA_PROVIDERS_DIR` is seen at launch (`cma_jev_env`, the `run`

wrapper, the status/TTL checks, `jev-setup`, `claude-usage`, `verify_superpowers_tui`), and the kimi twin homes live under `<dir>/kimi-twins` for every writer and reader (`cma_kimi_twin_root`, `cma_kimi_provider_home`, the self-contained kimi launcher, `kimi-providers`, `claude-install-verify`, `jev-setup`). Test T6c scans every shell, JavaScript and Python script and fails on any hard-coded default directory that is not an override-aware expansion or an entry of its explicit, reasoned allow-list. `claude-release-gate.sh` honours the override too.

- Pins and docs: `jev-latest` was observed answering as `jev-1.14.0` on the hosted server during the 2026-10-24 live validation (`jev-1.13.0` still served). UNCONFIRMED: not re-verified by an independent check (no network in the review pass). `providers/jev.json` lists both; `jev-latest` stays the default.
- `jev-gate.sh`: `latency_ms` had 1 s granularity (`date +%s`); it now uses a millisecond clock (`EPOCHREALTIME`, then `date +%s%3N`, python3, perl). Each `decisions.jsonl` line also carries `class` (= `gate`), the field the canon token-reduction gate (clause 4) counts.
- `jev-gate.sh`: every curl failure was logged `endpoint_error`, and with no key it sent a literal bearer `local` over https. It now logs `auth_rejected` (HTTP 401/403), `tls_untrusted`, `unreachable` and `timeout` separately, and with no key over https it makes no request and logs `no_key` (`endpoint_error` remains the residual). `test_jev_detect_https.sh` K3 now expects `tls_untrusted` for the untrusted-CA case. Behaviour change: a keyless https gateway used to receive `Authorization: Bearer local`; it now receives no request at all (`no_key`). Plain-http gateways are unchanged (no header, no key needed). A 403 is mapped to `auth_rejected` like a 401 (regression test added).
- Key-to-host binding (review follow-ups, two rounds). A Jev key is released over https only when (1) the host is loopback (exact authority grammar; userinfo, backslash, percent, whitespace and octets above 255 are refused), (2) the origin AND the key variable equal a SHIPPED pin in `providers/jev.json`, or (3) the operator listed `origin=KEYVAR` in `CMA_JEV_TRUSTED_ORIGINS`. The endpoint cache and a `CMA_JEV_PINS_FILE` override do NOT attest a key. Plain http never carries a key. Anything else makes no request (`jev-gate` fallback `key_unpinned`; no key at all stays `no_key`). `jev_mcp.py` and the opencode plugin apply the same rule (the plugin passes the key variable as data, never inside shell text, and honours `CMA_PROVIDERS_DIR`). The `sync-jev` live probe validates `systemone_path` (`^/[A-Za-z0-9._~/-]*$`, no `..`), classifies both the base and the final URL, and sends its bearer only to loopback or to a shipped pin. `JEV_API_KEY` as an explicit operator override is sent as given (documented residual). New `test_jev_key_origin_class.sh` (122 assertions, including a drift guard that every script building a bearer header is listed in the new `docs/jev/Jev_Key_Send_Paths.md`); mutation catalogue re-anchored and extended.

- Security (round-3 review, no-go 0/3/4/1, all closed): ONE comparison everywhere - the ORIGIN (`https://authority`) of the configured base against the ORIGIN of an ENABLED shipped pin with the same key variable - in `jev-gate`, `jev_mcp.py`, the opencode plugin and the `sync-jev` probe; `https://openrouter.ai` (any path) is now attested for `OPENROUTER_API_KEY`. Bug found on the way: jq `.enabled // true` treats `false` as missing, so a DISABLED pin still attested; now `.enabled != false` (gate, probe; tests use a copy of the scripts with every pin disabled). `jev-setup` no longer trusts the endpoint cache: `key_var` must be a plain name and the real key is written into the claude/kimi/opencode MCP configs only when the origin rule holds (else the placeholder `local`). A `routes.json` candidate is held to the same rule (an https candidate at an unpinned origin is skipped). `CMA_JEV_TRUSTED_ORIGINS` is split without pathname expansion; the plugin fetch uses `redirect: "error"`. `jev-setup` and `claude-providers` now share one state-dir resolution (`JEV_STATE_DIR`, else `$CMA_PROVIDERS_DIR/jev-state`, else `~/.local/state/jev`). `test_claude_usage.sh` file-scope check now inspects the named usage-feature commits with a positive-control commit (no longer vacuous). Tests: key_origin_class 122, setup_lifecycle 263 (L33, L34); 10 new mutations plus 2 re-anchored, all killed. Residual: an explicit `JEV_API_KEY` override is sent as given; LAN pins are operator-declared trust.
- Security (review finding): the live-probe host check uses an anchored authority parse (`_cma_jev_base_host_class`): userinfo (`@`), backslash, whitespace, percent bytes, empty host, malformed/duplicate port and look-alike hosts (`127.0.0.1.evil`, `localhost.`) are refused, so the bearer can no longer reach a host other than the one classified loopback; `jev-gate` never key-pins an https origin containing such characters. Tests T7e (4 assertions) and T9 (18 assertions).

## Merged from origin/main in this release (d992502, 7ab1ab0)

Reconciled with the Jev work that landed on `main` after the branch point; the union of both sides is kept.

- Jev key lookup falls back to the operator key files (`jev-keyfile-lib.sh`) in the gate, the MCP server and the OpenCode plugin, after the keys file and with the origin binding still applied. The `local` placeholder key is ignored everywhere (it is never sent as a bearer).
- Local model selection, the `jev-setup` mode file and `JEV_CONFIG_DIR` (default `~/.config/jev`) are kept.
- Behaviour change: with only the `local` placeholder and no resolvable key, the gate now records `no_key` and sends no request (previously `endpoint_error`, `Bearer local` sent). `test_jev_keyfile.sh` G0 expects the new behaviour.

- Fixed here: `jev_mcp._cached_key_var` hard-coded `~/.local/share/claude-multi-account/providers`; it now honours `CMA_PROVIDERS_DIR`. The send-path table lists `jev-keyfile-lib.sh` (not a send path). README badge and test count: 132 test files.

## Known limitations / imperfections (tracked, not fixed in this release)

- N8 (test-instrumentation, equivalent mutant): the pre-fix form `_pinsf="$_pinsd/providers/jev.json"` without the `[ -n "$_pinsf" ]` guard survives mutation, because `/providers/jev.json` cannot exist in a hermetic test. Mutation row 134 kills the realistic fallback instead.
- N9 (test-instrumentation): the symlink hop bound is not pinned; any bound >= 4 passes while the docs say up to 40 hops. A 39/40-hop chain case would pin it.
- N10 (source-defect, pre-existing, fails closed): `scripts/jev-setup.sh` resolves its own symlink with an unbounded loop and no `CDPATH=`; on a failed `cd` it silently installs no pins and the gate then fails closed. Suggested fix: reuse `jev_realpath`.
- N1: the `enabled:null` pin value is read differently by the MCP, the gate, the detector and the plugin; the canonical meaning needs an owner decision (recommended: only literal `false` disables).
- N2: `jev-setup` writes the resolved key under both `TYPESAFE_API_KEY` and `JEV_API_KEY` even when the key variable is OpenRouter's; owner design question.
- N4: `test_claude_usage.sh` asserts on commits that can no longer change; candidate for a content check.
- Test run for this release (full `scripts/tests/run-all.sh`, 132 files): 125 passed, 7 failed, 4 skipped for missing prerequisites. Six failures were root-caused and fixed after the run (README count, T6c allow-list and `CMA_PROVIDERS_DIR`, send-path table, G0 expectation, a stale mutation anchor, and the dependent mutation harness controls). `test_mgmt_port_isolation.sh` failed only inside the full run (no summary line) and passed 9/9 in each of 7 solo runs; its cause is UNKNOWN (suspected load or port contention) and it is not claimed fixed.
- The multi-hour mutation harness `test_jev_detect_https_mutation.sh` and the re-anchored mutation row 116 were not re-run to completion after the merge; shellcheck was not re-run.

# v1.32.0 (addendum) — jev-setup / Kimi-lock review rounds, already in the `v1.32.0` tag

- Jev decision layer, 2026-10-26 operator findings (each with a paired mutation suite):
  - **Local model selection** (`scripts/tests/test_jev_local_model.sh`): `claude-providers sync-jev` used the first model listed by the llmctl gateway, which lists alphabetically, so the cache named `decide-2b` (answers `ask` where `decide-max` answers `deny`). It now honours `CMA_JEV_LOCAL_MODEL`, else ranks the `ready` models by llmctl evidence (validated per `llmctl status --json`, then largest weights per `llmctl plan --json`; `CMA_LLMCTL_BIN`, `CMA_JEV_LLMCTL_TIMEOUT`), else the first listed ready model with an honest reason. A capacity rule, not an accuracy claim. Verified: the real endpoint cache now names `decide-max`.
  - **`jev-setup --mode` is now read** (`scripts/tests/test_jev_mode_file.sh`): `~/.config/jev/env` was written and read by nothing. The gate, the launch wrappers (`cma_jev_env`) and the OpenCode plugin now resolve the mode as process env > `$JEV_CONFIG_DIR/env` > `shadow` (parsed, never sourced; exact `JEV_MODE=shadow|enforce` line, last wins). `--mode` accepts only `shadow|enforce` (rc 2 before any write). `JEV_CONFIG_DIR` is honoured by `jev-setup` too.
  - **Key files** (`scripts/tests/test_jev_keyfile.sh`): the gate could not find `LLMCTL_API_KEY` outside the systemd sync unit (not in the env, not in `~/api_keys.sh`), so every call failed open. New non-secret indirection: `jev-setup --key-file PATH` records the path in `~/.config/jev/key-files` (or `CMA_JEV_KEY_FILES`); the gate, the detector, `jev_mcp.py` and the OpenCode plugin parse that one variable out of the file at call time. New `scripts/jev-keyfile-lib.sh`. The MCP server treats a baked `JEV_API_KEY=local` placeholder as "no direct key" and learns the key variable name from the endpoint cache. Docs: User Guide §2.1 (Model selection, Key files), §5, §10; FAQ; Quickstart.
  - **Tests** found by the full-suite run: `scripts/tests/test_jev_detect_https.sh` resolved its LAN address with `hostname -I`, which GNU coreutils' `hostname` does not have, so the LAN-address case (H8) was skipped and the positive control (>= 5 fixture servers) failed on such a host - it now falls back to `ip -4 -o addr show scope global`, then `ifconfig`; `scripts/tests/test_quota_concurrency.sh` copies `jev-keyfile-lib.sh` into its isolation directory (the sourced library is a new dependency of `claude-providers.sh`); the README test badge/count follows the three new test files. `scripts/tests/test_jev_detect_https_mutation.sh`: the full 96-row run kills

95 rows; row 94 (mermaid statement broken by a semicolon) can only be killed by the docs suite's `mmdc` render check, which SKIPs on a host whose `mmdc` cannot render (no browser), so the suite stays green and the harness counted a defect. A row listed in `ENV_GATED` (scripts/tests/lib/jev_mutations.py) whose suite output carries that suite's own SKIP line is now reported SKIP (the kill is unproven on that host, never a pass); a row without that line, or any other row, still fails.

- `jev-setup`: `--agents` is validated against the closed, case-sensitive set `claude|kimi|pi|opencode|all` (unknown, miscased or empty -> exit 2 before any write, no stamp); `--check --uninstall` is refused (exit 2); the native Kimi block is rewritten as a whole file via one atomic, status-checked `_kimi_block_write` (no blank-line growth, block-only configs work); Claude `--check` resolves a `bash /path/jev-gate.sh` hook command, also when the gate path contains spaces. Lifecycle cells L14-L20 cover the opencode and claude/kimi inventory branches reached through `--uninstall` and `--check`, scope derivation, argument validation, the stale-block rewrite and `--check`; pi has no inventory branch (its wiring location is UNCONFIRMED), so a consented pi install is protected only through the stamp (L22).
- Round 4 (independent review NO-GO, closed by class):
  - Kimi config writers are serialised. `scripts/jev-kimi-lib.sh` gained `jev_kimi_lock` (flock on `$JEV_STATE_DIR/kimi-config.lock`, wait budget `JEV_KIMI_LOCK_WAIT`, default 30 s). The mkdir fallback written in this round was NOT mutually exclusive under three or more contenders; that is corrected in Round 6 below, and only flock serialisation was measured for Round 4. The twin refresh (`_cma_kimi_jev_hooks_refresh`) and the twin renderer take the lock, and the refresh recomputes the hook block INSIDE it, so a refresh can no longer resurrect a block after `--uninstall` or revert a model/key/URL written by a concurrent render (cells L28-L30). Round 4 made `jev-setup` hold the lock for the whole run; Round 6 narrows that (see below). A symlinked `config.toml` is rewritten through the link by `jev-setup` and the refresh (L27, L31).
  - `_backup` is fail-closed: a backup that cannot be made aborts that write (no partial copy left, nothing written, reported); L25.
  - `--agents` rejects empty, separator-only, space-only and space-separated lists with exit 2 before any write (L16). `--if-stale 0` no longer widens the scope and a non-numeric `--if-stale` is a usage error (L23). `--uninstall --agents X` keeps the shared gate for agents still named in the stamp (pi, kimi included) (L22); a hand-edited stamp with only unknown names never widens the scope (L26). Kimi `--check` reports a gate that is not executable (L24).

    - Mutation table: rows 19 and 41 had stale or too-weak needles and are corrected (both verified killed); rows added for each reviewer mutation (o1, o2, o3, o4, o6, o8, o12, o15), for the three lock removals and for the spaced-gate-path parser. Reviewer mutation o3b (whitespace check in `--agents`) is an EQUIVALENT mutant: with the check removed the same input is still rejected as an unknown agent, so it has no row.
    - Known limits, not fixed: the two races were reproduced with an `awk` shim that widens the read-to-replace window (the natural window is milliseconds); what Kimi does at run time with a hook whose command no longer exists is UNKNOWN; pi install/uninstall is not exercised hermetically; `shellcheck` was not run; the twin renderer still writes a fresh file (it does not follow a symlinked twin `config.toml`); the regenerated twin (.html/.pdf/.docx) of a changed .md can collapse blank lines the exporter treats as insignificant.

- Round 5/6 (independent review NO-GO on Round 5, closed by class):

    - The mkdir lock backend is now a faithful port of the `lib.sh` `_cma_alias_lock_*` algorithm in the new sourced library `scripts/jev-lockdir-lib.sh` (breaker lock, observed-pid re-read before a stale break, pid tie-break after `mkdir`, pid-less grace `CMA_LOCKDIR_GRACE` default 10 s, deadline checked on every pass, owner-checked release). It is a PORT, not a delegation: `lib.sh` uses globals and `set -e` and was left unchanged. `scripts/jev-kimi-lib.sh` keeps flock primary, re-entrancy (`_JEV_LOCK_DEPTH`) and the `JEV_KIMI_LOCK_NO_FLOCK=1` seam. New suite `scripts/tests/test_jev_kimi_lock.sh` drives the real scripts: 8 contenders, stale / fresh / live-holder / pid-less cases, `JEV_LOCK_STRESS_RUNS` runs (default 20), max concurrency asserted equal to 1, a no-lock positive control that must overlap, flock as negative control, flock-timeout failure, block-decided-under-lock.
    - `_cma_kimi_jev_hooks_refresh` matches its markers exactly. A BEGIN without an END, or a variant marker line, leaves the file untouched, prints a warning to stderr and returns non-zero (before: the awk range deleted every following line). Blank lines outside the block are kept.
    - `jev-setup` holds the Kimi config lock only around the config read-modify-write sections; children (`pi install`, `npm view`, the keys-file shell) run with fd 7 closed (`7>&-`; the endpoint-env command substitutions use `$(exec 7>&-; ...)` because a redirect on a function call makes bash park fd 7 on fd 10 in the forked copy, which still held the lock; found by test K8d), so a concurrent render is not blocked by a slow child and a child outliving the parent does not keep the lock. A pre-flight lock probe fails fast with "nothing written".

- Mutation table: rows for each reviewer-surviving mutation (M2, M4, M5, M9, M10) and for each new guard (breaker re-read, blind release, grace, deadline, refresh rc, blank-line preservation, section lock, fd 7 leak); `test_jev_detect_https_mutation.sh` no longer excludes `./specs` from its freshness check and aborts loudly when an owning suite fails unmutated.
- ROOT CAUSE of the intermittent mkdir-lock overlap (found in Round 6, measured): on this host `mkdir` is uutils coreutils 0.8.0 (`/usr/bin/mkdir -> ../lib/cargo/bin/coreutils/mkdir`), and its "directory exists" race is reported as SUCCESS: with 6 racing `mkdir d` calls more than one caller won in 66 of 200 rounds, and an instrumented trace showed five processes with `mkdir rc=0` within 1 ms after the lockdir was removed. A mutex whose exclusion rests on mkdir's exit status therefore admits several holders. `scripts/jev-lockdir-lib.sh` now takes the lock (and the breaker role) with an O_EXCL pid-file create (bash `set -C`, `_cma_lockdir_claim`); mkdir only provides the directory. New case K1d in `test_jev_kimi_lock.sh` puts a fake `mkdir` that succeeds on an existing directory on PATH, with positive and negative controls: it failed 5/5 runs before the fix and passes after. Lock suite: 62 assertions pass; 240 parallel-load runs of the real library showed 0 overlaps (before the fix 1 in 240 and 3 in 240 in two comparable runs).
- Test hardening found while re-running the mutation rows: the O_EXCL claim changed which state reaches the pid-less grace branch (a lockdir with no pid file is now simply claimed), so the old K2c no longer exercised it and its mutation survived; K2c now uses an EMPTY pid file (holder died between the create and the write) with a not-broken-before-grace control. New K2f holds the breaker with a live process under the fake racy `mkdir` and asserts a second breaker does not enter (kills the "breaker not claimed O_EXCL" row, which survived 62/0 before K2f). `_lockrc` is time-bounded so a lock that never gives up fails an assertion (rc 124) instead of hanging. `test_jev_detect_https_mutation.sh` bounds each suite run (`JEV_MUT_SUITE_TIMEOUT`, default 900 s); a timeout counts as a kill.
- Mutation rows 77-92 (lock library, refresh, jev-setup lock) were each run on a scratch copy: all killed. Rows 77 and 83 (stale lock never broken, deadline never enforced) are killed by hanging, row 83 also by the K2c assertion before it hangs. Row 95 (docs: README names one third-party submodule) SURVIVES: `submodules/ccusage` appears in two other README places, so the row is effectively an equivalent mutant; it predates this round and was not changed. The full mutation harness was NOT run in one pass this round (UNCONFIRMED; roughly 96 suite runs).
- Known limits, not fixed: `lib.sh` itself was not changed; its `_cma_alias_lock_*` is also mkdir-based and very likely has

the same defect on this host (UNCONFIRMED, not measured), and its acquire loop skips the deadline check after a stale break (the port checks it); lock timing has whole-second granularity; the claim relies on the filesystem honouring O_EXCL (not tested on NFS); the pi install path is not exercised hermetically; Kimi's run-time behaviour on a hook whose command is gone is UNKNOWN; `shellcheck` was not run; a mutation that waits for the deadline (row 83) is killed by a timeout rather than by an assertion.

- `jev_mcp.py`: a relative `ca_file` in a routes.json candidate resolves against the routes.json directory (never the cwd); a candidate without `ca_file` deliberately trusts `JEV_CA_FILE` as an extra root; an unreadable explicit `ca_file` still fails closed.

- Mutation table: rows for the inventory branches, the uninstall inventory use, the renderer's model-less-rewrite guard and the relative-CA resolution.

# v1.32.0 — 2026-10-22

The Jev local decide-gateway detector now speaks llmctl's real transport (HTTPS + mandatory bearer key); the key can no longer reach a command line or an http request; plus hub-engine / validation-gate / decision-model documentation, a user-manual index, two new diagrams and tests.

## BREAKING (security) — read before upgrading

- LAN pins (`lan[]` in `scripts/providers/jev.json`) and the local gateway pin are now **TLS-only by default** (`scheme` defaults to `https`). A plain-http LAN endpoint that v1.31.x detected is no longer registered; the detector warns `does not speak TLS` when it can tell. Serve it over TLS and give the detector the CA (`CMA_JEV_CA_FILE`) and key (`key_var`), or, for a loopback-only dev server, opt in with `CMA_JEV_ALLOW_HTTP=1` or `"scheme": "http"` in the pin. Plain http is **never** used for a non-loopback host.
- Plain-http (opt-in) records carry **no** `key_var` and no `ca_file`; `jev-gate.sh` and `jev_mcp.py` send no Authorization header over http.
- The endpoint cache gains `ca_file` and launches export `JEV_CA_FILE`: existing installs must run `claude-providers sync` (regenerates the alias file) once.
- `llmctl-<profile>` launches: exit 75 from `llmctl switch` is no longer always reported as a failed rollback (see Fixed).

## Fixed

- `detect_jev_records` probed the local llmctl decide gateway over plain `http://` with no key, so a real gateway (HTTPS, local CA,

bearer key) was never auto-detected. It now tries TLS first, trying each CA candidate in turn (`CMA_JEV_CA_FILE`, `SSL_CERT_FILE`, `$LLMCTL_HOME/cert/ca/ca.crt`, system store) until `/healthz` answers; certificate verification is never disabled. The bearer key comes from the env var named by the pin's `key_var` (else the keys file) and reaches curl through a header file, never argv, never printed or cached. Each failure has its own key-free warning: no key; key rejected (HTTP 401/403/429); models request failed (HTTP n); certificate not trusted.

- Key-on-command-line class: `jev-gate.sh` and the `CMA_JEV_LIVE=1` hosted probe passed `-H "Authorization: Bearer $key"` (visible in `ps`); both now use a header file. `jev_mcp.py` no longer follows redirects (urllib replayed the Authorization header to the redirect target) and strips Authorization for non-https URLs.
- Host classification is an exact grammar (`127.0.0.0/8` literal, `localhost`, `::1`, `[::1]`); userinfo (`127.0.0.1@host`), paths, `127.evil.example`, `127.0.0.1.nip.io` and `localhost.` are not loopback.
- `jev-gate.sh` now resolves the key named by `JEV_KEY_VAR` from the process environment before the keys file (it previously sent `Bearer local` when no keys file existed).
- Empty-array expansion under `set -u` (bash 3.2 / 4.3) in the new probe.
- `_cma_llmctl_ensure_active`: exit 75 from `llmctl switch` means a host-pressure refusal with the running set untouched OR a failed rollback; it now decides from llmctl's text and a re-probe of the running set, so a pressure refusal no longer prints `CRITICAL: rollback also failed`. With `LLMCTL_VALIDATE_STRICT=1` an already-running profile is checked with `llmctl require-validated` (no switch means llmctl's start-time gate would not run).
- `docs/diagrams/kimi-family.mmd` was not valid Mermaid (its prose header was not commented); the header is now `%%` comments and the SVG is re-rendered.
- `JEV_GATE=off` is recorded (`$JEV_STATE_DIR/gate-off.log`, reported by `jev-stats`).
- `jev-setup` now needs consent (`--yes` / `JEV_SETUP_CONSENT=1` / interactive `y` / a prior consent stamp for `--if-stale`) before editing any agent home; exit 3 otherwise. Callers that ran it unattended must add `--yes`. Backups were already made first.
- One shared credential-safe opener (`scripts/http_noredirect.py`): a redirect is refused (strict) or followed only within the same scheme://host:port; used by `jev_mcp.py`, `quota_probe.py`, `model_verify.py` (provider probes carrying the user's API key) and `alias_e2e_test.py`, replacing the duplicate handler and urllib's default redirect-following.
- `cma_jq_edit` (lib.sh): one atomic, loud JSON edit used by `claude-providers add` and `claude-unify` (enabledPlugins union), instead of swallowing jq errors. `jev-setup.sh` does NOT use it: it has its own `_jq_merge` (atomic, backup first, owner-only permissions) which checks jq's exit status and is covered by test J, which drives the real `jev-setup.sh`.

- `assert.sh summary()` fails a test file that asserted nothing; two test fixtures no longer lose server PIDs in a `$(...)` subshell.
- `specs/003-jev-decision-layer/{spec,plan,tasks}.md` ported (historical spec with a status header; the design file it cites was never found, UNCONFIRMED whether it was written). Its task T017 (the Kimi hook block) is implemented in this release; see Added.
- Review-round fixes (independent review NO-GO, closed by class): `jev-setup --uninstall` now removes the canonical gate copy and the renderer honours the installed stamp, so the next render no longer re-adds the Kimi hook block; `_jq_merge` in `jev-setup.sh` checks jq's exit status (a `jq -e` false/null no longer overwrites a file); a `routes.json` candidate's own `ca_file` is used for that candidate's TLS and an unreadable one fails closed (no request is sent); IPv6 hosts are bracketed in detector URLs; a non-loopback https gateway that fails `/healthz` for another reason now warns `did not answer` (new diagnostic, alongside the four above). Tests added for each: the plain-http and https Authorization sink-side checks, host-only and scheme-only redirect hops, the matcher value, backslash TOML, the install/render/uninstall/render lifecycle, the IPv6 table, `_jq_merge` status and per-candidate CA. `test_jev_detect_https.sh` K2 no longer reads an ambient `TYPESAFE_API_KEY`.
- Review round 2 (NO-GO, closed by class). Agent scope and lifecycle: `jev-setup` records WHICH agents were selected in `installed.stamp` (`agents=<csv>`, union across runs); the Kimi twin renderer emits its hook block only when the stamp lists `kimi` (or `all`), so a Claude-only install no longer wires Kimi twins, and a legacy empty stamp (pre-1.32.1) names nobody, so the renderer fails closed for it; `jev-setup --if-stale` and `--check` without `--agents` derive their scope from the stamp (or, for a legacy/empty stamp, from the wiring actually found in the agent homes) instead of widening to all four agents, and `jev-setup --agents kimi` is how a legacy install opts its Kimi twins in. `--uninstall --agents X` inventories the wiring of every other agent and removes the shared gate, env file, bin links and stamp only when none remains; otherwise it keeps them and rewrites the stamp. `--check` verifies that each hook target exists and is executable (not only that the string is present), flags a stale Kimi matcher, and exits non-zero on any finding. The Kimi install rewrites an existing marker block whose matcher differs (the native `~/.kimi-code-*` upgrade path). The matcher constant file is the only source (no second hard-coded copy); a missing or invalid file warns, naming `jev-kimi-tools.txt`. The twin refresh keeps the original config unless the rewritten file is complete and mode 600. Tests: `test_jev_setup_lifecycle.sh` (13 cells driven through the real `jev-setup.sh` and the real renderer), matcher semantics checked with node `RegExp.test`, a readable-but-wrong candidate CA with a good `JEV_CA_FILE` yields no request, and 12 new mutation rows.

- Out of this repository (reported, not changed here): the `validate_judge.py` redirect handler and `run_golden.py` / `review_translation.py` (llmctl, constitution), `curl -k` in `validate_bridge.sh`, the SonarQube `curl -u` token.
- Not verified against a real `llmctl decide` gateway (hermetic TLS stand-in only); the OpenCode plugin does not read `JEV_CA_FILE` (use `NODE_EXTRA_CA_CERTS`).
- `scripts/install.sh` had no fallback for a flat (non-submodule, non-nested) `claude_code_router` checkout: unlike `curl-install.sh`'s `provision_flat_router()`, a direct-clone install with no parent hub and no sibling checkout failed `scripts/claude-ccr-build.sh` outright. Added the same auto-clone-or-pull step (honouring `CLAUDE_TOOLKIT_CCR_URL`) ahead of the existing build call.
- Five own-org dependencies declared in `helix-deps.yaml` had no sibling checkout anywhere this host could resolve (`constitution`, `containers`, `challenges`, `llms_verifier`, plus `claude_code_router`), leaving five env-only test SKIPs. Provisioned each as a flat, top-level-only sibling checkout (no `--recursive`, since `resolve_deps.sh` only needs each checkout's identity marker, not its own nested submodule tree).
- `submodules/ccusage` bumped `v20.0.40` -> `v20.0.41` (pricing-snapshot updates and an Antigravity model-id fix); `submodules/go` deliberately left untouched (bumping to `origin/master` is the unreleased next-major dev tree and is out of policy here).
- `scripts/tests/lib/jev_mutations.py` mutation row 75 ("lock: refresh block computed BEFORE the lock, stale under it") failed to apply (`pattern absent`) because its captured before/after source snapshots were stale against an earlier, unrelated fix to `_cma_kimi_jev_hooks_refresh` (a `local f tmp blk real` -> `local f tmp blk real rrc=0` change the mutation table's own snapshots never picked up), and the "before" string's closing `"""` also silently absorbed the content's own trailing quote (a classic Python triple-quoted-string trap). Both snapshots corrected; row 75 (and the file's full mutation sweep) verified clean.
- `helix-deps.yaml`'s own explanatory comment matched `test_ccusage_build.sh`'s blind `grep -qiE 'ccusage'` sweep for a literal manifest entry, failing a test whose underlying build (ccusage v20.0.41 via `cargo build`) was fine. Reworded the comment to describe the submodule without the literal substring the sweep was actually checking for.
- `scripts/claude-providers.sh`'s `_cma_kimi_jev_hooks_refresh` called `readlink -f`, which is absent on BSD/macOS; switched to the toolkit's own portable `cma_realpath` (already used the same way in `claude-unify.sh`).
- `scripts/model_verify.py`, `scripts/quota_probe.py` and `scripts/alias_e2e_test.py` imported the sibling module `scripts/http_noredirect.py` with no `sys.path.insert` guard before the import (present in `scripts/jev_mcp.py` already). This worked when run as a direct script but failed `ModuleNotFoundError` when loaded via `importlib.util.spec_from_file_location` or via an

isolated-sandbox copy; all three now insert their own directory onto `sys.path` first.

- `scripts/tests/test_quota_concurrency.sh`'s isolated-sandbox setup copied `lib.sh`, `claude-providers.sh`, `quota_probe.py` and `model_verify.py` into its scratch `$ISO_LIB_DIR` but never copied `http_noredirect.py`, which `quota_probe.py`/`model_verify.py` now require to exist alongside them; added it to the copy list.
- `cma_jev_env()` (`scripts/lib.sh`) only ever *set* `JEV_BASE_URL` / `TYPESAFE_BASE_URL` / `JEV_KEY_VAR` / `JEV_MODEL` / `CMA_JEV_ENDPOINT_ID` / `JEV_CA_FILE` on its active path; every deactivate branch (disabled, no cache, no `jq`, no resolved endpoint) left any already-exported value of those vars untouched instead of clearing them, so a real host that had previously activated Jev could leak a stale endpoint/key-var into a launch that should have been a clean opt-out. All deactivate paths now explicitly `unset` them.
- `jev-gate.sh`'s deterministic risk-floor denylist contained the literal substrings `"git push --force"` / `"git push -f "` as patterns it blocks against, which made the repo's own §11.4.113 anti-force-push sweep (`scripts/tests/verify_constitution.sh`) flag the denylist itself as a force-push invocation once a resolvable `llms_verifier` checkout let that check actually run (previously masked by an unrelated SKIP). Restructured the two patterns to be assembled from separate word/flag parts at runtime (functionally identical — verified against `git push --force`, `git push -f`, and non-matching inputs) so the literal text never appears contiguous in source.
- `test_jev_detect_https.sh`'s K3 case ("jev-gate over TLS without the CA fails open") asserted only `exit 0` + empty stdout as proof the fail-open path ran — true for the intended untrusted-CA curl failure, but equally true if the script had exited early for any unrelated reason. Added positive sink-side evidence: the gate's own `decisions.jsonl` record for the run must carry the `endpoint_error` fallback that only its curl-failure branch writes. Found and fixed via `test_sandbox_hygiene.sh`'s vacuous "empty-is-success" assertion lint.
- Two more mutation-table rows had the same staleness class as row 75's fix above, both found only once the full `test_jev_detect_https_mutation.sh` sweep (96 rows) could actually run end-to-end: row 61 ("renderer refresh: write failure replaces the config") captured an `awk 'BEGIN{h=0}...'` invocation `_cma_kimi_jev_hooks_refresh` no longer uses (it was rewritten to a named-marker state machine); re-captured against the current source and re-verified it is genuinely killed by `test_kimi_render_config_jev_hooks.sh` (not just pattern-applies). Row 95 ("README names only one third-party submodule") applied cleanly but `test_docs_hub_engine_release.sh`'s check was a document-wide substring search, which a second, unrelated mention of `submodules/ccusage` elsewhere in the README silently satisfied even when the targeted sentence lost it; the assertion is now

scoped to that specific sentence and verified to fail under the mutation.
- Full LIVE retest before this release: `scripts/tests/run-all.sh` (122 test files) plus every file above re-verified standalone after its fix landed, all clean; the full `test_jev_detect_https_mutation.sh` mutation sweep (96 rows + control) at 105 passed / 0 failed; `scripts/claude-release-gate.sh --provider nvidia --skip-suite` run live end-to-end (alias smoke, route attribution, quota JSON, 36/37 provider probes answered) all green.

## Added

- Kimi hooks (spec task T017, implemented here; earlier docs described it as done before it existed). Kimi provider twins (`~/.kimi-prov-*/config.toml`) carry the Jev `PreToolUse` gate as a marker-guarded `[[hooks]]` block (`# cma-jev:hooks BEGIN/END`), rendered by `_cma_kimi_render_config` only while ALL hold: jev-setup's `installed.stamp` exists, the Jev endpoint cache has an endpoint, and the installed gate is executable. The matcher is `Bash|Write|Edit`, defined once in `scripts/jev-kimi-tools.txt` and read by both the renderer and `jev-setup.sh` (the earlier `WriteFile|StrReplaceFile` names are not Kimi tools: its real file tools are Write and Edit, control-needled against the installed Kimi Code 2.2.0 binary, the only version checked; Kimi tests the matcher with an unanchored `RegExp.test(toolName)`, so the matcher is a plain `|` list of names). A gate path containing a backslash, double quote or control character is refused by BOTH writers (the renderer and `jev-setup.sh`), through one shared validator in `scripts/jev-kimi-lib.sh`, because it cannot be written into a TOML string. `claude-providers sync` and `sync-jev` re-render the twins after the endpoint cache is written. Kimi's runtime acceptance of the hook is not exercised live. Tests: `test_kimi_render_config_jev_hooks.sh`, `test_jev_setup_idempotent.sh`.
- `docs/User_Manual.md`: explicit manual index over every guide, FAQ and diagram (Constitution §11.4.257), linked from the README.
- Diagrams `docs/diagrams/hub-engine-placement.{mmd,svg}` (hub-engine placement, resolver order, no nesting) and `docs/diagrams/decide-gateway-flow.{mmd,svg}` (llmctl decide gateway: HTTPS + mandatory bearer key, local CA, fail-open gate); rendered with `mmdc`.
- README: "Constitution hub engine" and "Decision models (Jev) and the decide gateway" sections; feature bullets for the hub engine and the llmctl validation gate; version badge corrected (was v1.27.0).
- `docs/llmctl/user-guide.md` §5a (§11.4.277 validation gate, exit codes, STRICT, §11.4.278 Vision QA relationship); llmctl quickstart/FAQ entries.

- Provider Aliases User Guide and Provider FAQ: governance context (hub engine, validation gate, decision models never aliases, alias/provider/model consistency).
- Jev guide/FAQ/quickstart: HTTPS + bearer transport, CA and key handling, troubleshooting; limitation 11 records the fix and what is unverified; Constitution §11.4.279 clause citations (A, C, D).
- Tests: `scripts/tests/test_jev_detect_https.sh` (hermetic TLS fixture `fixtures/jev_https_server.py` with openssl self-signed certs and sink-side request logs; CA candidates, 401/404/ untrusted-CA/no-key messages, http opt-in and key-free http records, host-grammar table, curl-argv PATH shim with a positive control, gate env-key resolution, jev_mcp TLS trust and redirect refusal, cache/env with a missing CA, server-leak assertion; H8 is a counted skip on hosts without a non-loopback address) and `test_jev_detect_https_mutation.sh` (every table mutation, each must fail its suite; the unmutated control runs every suite the table owns; count is `python3 scripts/tests/lib/jev_mutations.py list | wc -l`, currently 65). `assert.sh` gains a counted `_skip`. `scripts/tests/test_llmctl_vision_gate_wiring.sh` (fake llmctl: `LLMCTL_VALIDATE_STRICT` is forwarded, gate exits 1/2/75 abort the launch, non-llmctl aliases untouched) and `scripts/tests/test_docs_hub_engine_release.sh` (manual/README link lint, required topics, diagram presence, doc twins non-empty, package.json == top CHANGELOG release).

### Changed

- `package.json` / `package-lock.json` version 1.30.6 -> 1.32.0 (the 1.31.0 release had not bumped it).
- Doc twins (`.html`/`.pdf`/`.docx`) regenerated with `claude-export-docs.sh` for the changed docs.

## v1.31.0 — 2026-10-22

Own-org dependencies flattened (operator decision 2026-10-21: claude_toolkit tracks no own-org nested submodule; own-org repos live flat in `<root>/submodules` or `<hub>/constitution/submodules`).

### Changed

- Removed the `submodules/LLMsVerifier` (vasic-digital/LLMsVerifier, last pin `8f2090ac`) and `submodules/claude-code-router` (vasic-digital/claude-code-router, last pin `9accd18`) submodule declarations and gitlinks. `submodules/go` (third-party) is the only submodule left.

- `scripts/resolve_deps.sh` resolves `llms_verifier` and `claude_code_router` (env `CT_LLMS_VERIFIER_DIR` / `CT_CLAUDE_CODE_ROUTER_DIR`, legacy `LLMSVERIFIER_DIR` still honoured with `CT_` winning → parent hub → `<root>/submodules` → sibling; snake_case name first, legacy repo name accepted; identity markers `llm-verifier/go.mod` and `go.mod+cmd/ccr/main.go`; the unresolvable message names the clone URL).
- `claude-ccr-build.sh`, `ccr-install.sh`, `claude-verify-providers.sh`, `claude-semantic-visibility.sh`, `providers-verify.sh` and the tests use the resolver; `claude-ccr-build.sh` no longer runs `git submodule update`.
- `curl-install.sh` clones the router as a flat sibling when it is not resolvable.

## Added

- `helix-deps.yaml` (§11.4.31 dependency manifest: HelixConstitution, containers, challenges, LLMsVerifier, claude-code-router; all flat; ssh_url verified with `git ls-remote`) and `scripts/tests/test_helix_deps_manifest.sh` (offline shape + coverage validation, negative control).
- `scripts/tests/test_gitmodules_no_ownorg.sh`: FAILs on any own-org URL in `.gitmodules`, an own-org gitlink in the index, or a hardcoded old nested path in `scripts/` (control needles included).
- **Jev decision layer**: TypeSafe's Jev decision/classification models (`POST /v1/systemone` — yes/no, multiple-choice+confidence, 1-10 rubric, bounding-box; no chat/completions, no tool calling) are registered as a new `classification: "decision"` provider record, `"alias": null`, never launchable — excluded from alias emission, the kimi/pi renderers and chat catalogs, and surfaced only via `claude-providers list`'s own `## Decision endpoints (Jev)` section. `detect_jev_records()` (local llmctl gateway `127.0.0.1:8095` > LAN pins > OpenRouter > TypeSafe hosted, priorities 10/20/30/40, `CMA_JEV_PREFER` override) writes a ranked, non-secret endpoint cache (`providers/jev-endpoints.json`, TTL `CMA_JEV_CACHE_TTL` 300s) via `claude-providers sync` / the new `sync-jev` subcommand. `cma_jev_env()` in all six launch wrappers reads that cache only (zero launch-path network I/O; a stale cache kicks a detached background refresh) and exports `JEV_*`/`TYPESAFE_*`; any failure degrades to `CMA_JEV_ACTIVE=0` with launch behavior unchanged (`CMA_JEV_DISABLE=1` opts out entirely). `jev-gate.sh` is a shadow-first (default), fail-open PreToolUse permission gate — one Jev `/v1/systemone` call per gated tool call mapped to allow/ask/deny (0.8 deny / 0.7 ask thresholds, deterministic risk floor for destructive patterns), hashed-only evidence to `decisions.jsonl`; enforce mode is explicit opt-in (`jev-setup --mode enforce`); `JEV_GATE=off` is the gate-only kill switch. `jev_mcp.py` is a stdlib-only MCP server (`jev_ask`/`jev_models`/`jev_stats_summary`). `jev-setup.sh`

idempotently installs per-agent artifacts for Claude Code, Kimi, pi, and OpenCode (backup-first, atomic writes, honest offline SKIP). `jev-stats.sh` reports decision counts, latency, shadow-calibration split, and an explicitly labeled ESTIMATE (never a measurement) of tokens saved. `jev-spec-setup.sh` is an opt-in CI-template installer for an external repo the operator names — never wired into this repo's own `.github/`, consistent with the no-CI/CD mandate. Docs: `docs/jev/{Jev_Quickstart,Jev_User_Guide,Jev_FAQ}.md`, `docs/research/{toolkit-architecture,agent-integration-mechanics, jev-ecosystem,decision-model-hashes,encoder-model-hashes}.md`, `docs/diagrams/jev-architecture.{mmd,svg}`. Tests: 9 suites under `scripts/tests/test_jev_*.sh` sharing the canonical stdlib fixture `scripts/tests/fixtures/jev_server.py`.

- **Usage reporting** (`claude-usage report`): ccusage-backed token usage across every native account and provider alias — per-model, per-day, per-session, and per-provider-bucket breakdowns (`--group-by all` for all dimensions at once), arbitrary `--since`/`--until` range defaulting to the trailing 28 days, output as `json`/`md`/`csv` written by default to both stdout and `$SHARED_DIR/usage-reports/<range>.{json,md}` (or a single explicit `--out PATH`). Honest limitation, documented in the script's own `--help` and in `Claude_Multi_Account_Fine_Tuning.md`: true per-alias attribution is not recoverable from ccusage's own session logs, since every account's `projects/` directory is the same symlinked shared tree; native accounts are reported pooled, and provider attribution is explicit best-effort (`candidate alias` / `AMBIGUOUS` / `unmapped`) inference from model id against `providers/status.json`, never asserted as fact. Backed by the vendored `ccusage` submodule (`scripts/claude-ccusage-build.sh`). Tests: `scripts/tests/ test_claude_usage.sh` (hermetic, 50 assertions including a guarded live smoke run against fabricated JSONL transcripts). Usage reports now auto-refresh on every `claude-providers sync`/`sync --multi` run (`_cma_usage_refresh_after_sync`, mirroring the existing `_cma_jev_after_sync` fail-soft pattern): a missing/unbuilt `claude-usage.sh` is a one-line skip, a failing refresh is a warning, neither ever fails the sync itself; `CMA_SYNC_SKIP_USAGE_REFRESH=1` opts out. Tests: `scripts/tests/ test_sync_usage_refresh.sh`.
- **`claude-usage merge`**: combines `claude-usage report` JSON files from one or more `--from PATH[,PATH...]` sources (a directory of `*.json` reports, or a single file) into one consolidated report, re-summing any day/model/project/provider rows that key together across sources. Project identity is stripped by **default**: every `projectPath` is replaced with the placeholder `"coding agents"` (override via `--anonymize-as NAME`), and project rows that become identical after stripping are summed into one row rather than left duplicated under the same label. `--real-names` opts out of stripping entirely (not the default) and keeps each source's real paths distinct. Same three-mode output as `report` (default writes both json+md under `$SHARED_DIR/usage-reports/`; `--format` alone prints to stdout; `--`

out PATH writes there). Tests: scripts/tests/test_claude_usage_merge.sh (23 assertions, including hand-computed expected sums for overlapping day/model keys).

### Security

- jev-gate.sh: fail-open/shadow/kill-switch paths no longer emit permissionDecision:"allow" (a prior build of this PreToolUse gate let those paths actively grant permission instead of deferring to Claude's normal permission flow). Replaced with a clean passthrough (_emit_passthrough); only genuine enforce-mode decisions emit a decision now.
- jev-setup.sh: credential-bearing config writes (TYPESAFE_API_KEY/ JEV_API_KEY into MCP/OpenCode configs) now happen under umask 077 with a chmod 600 backstop, instead of inheriting the process umask.
- lib.sh's cma_jev_env() no longer exports the real JEV_API_KEY/ TYPESAFE_API_KEY secret into the whole launched agent process's environment (and therefore every tool-use subprocess it spawns) — it exports only the non-secret key-variable *name* (JEV_KEY_VAR); jev-gate.sh and jev_mcp.py resolve the real key themselves at call time from the keys file.
- jev-spec-setup.sh's generated GitHub Actions template pins oven-sh/setup-bun to a commit SHA (0c5077e51419868618aeaa5fe8019c62421857d6 / v2.2.0, verified via the GitHub API) instead of the mutable @v2 tag, and adds top-level permissions: contents: read.

## v1.30.6 — 2026-10-21

Single-source dependency layout (owner mandate 2026-10-21: one source of truth, no nested duplicate chains).

### Changed

- Removed the constitution, submodules/containers, submodules/challenges and submodules/superspec submodule declarations and gitlinks; the constitution hub that vendors the toolkit already carries them. Previous pins: constitution e44f22f, containers 4a8f04e, challenges 8a352b9, superspec c20ac6c. Own deps (LLMsVerifier, claude-code-router, go) stay; go keeps its documented HTTPS URL exception.
- test_mechanical_tools_acceptance.sh resolves the constitution through the new resolver (it previously walked ancestors and could pick up an unrelated project's constitution); CMA_CONSTITUTION_DIR remains accepted.

### Added

- `scripts/resolve_deps.sh` (env override -> parent hub -> in-tree -> siblings -> opt-in host fallback via CT_HOST_CONSTITUTION_FALLBACK, no built-in path; identity-marker validation; canonicalised overrides, empty override rejected; non-zero + actionable message when unresolvable) and `scripts/tests/test_resolve_deps.sh` (56 assertions, mutation-killing fixtures).
- "Shared dependency layout" section in CLAUDE.md / AGENTS.md / GEMINI.md / QWEN.md.

## v1.30.5 — 2026-10-10

Fix-forward release addressing the independent NO-GO review of merge `dadd8b1` (feature/006 hardware-aware model management). No history rewritten.

### Fixed

- `cmd_sync` and `cmd_verify` now read `CMA_VERIFY_LAYER_FILE` through `cma_read_verify_layer`, which keeps legal tokens as-is, maps off-vocabulary tokens to a reason-derived layer, and falls back to `unknown`. This closes the branch where an invalid layer token was persisted verbatim and the legacy reason fallback covered only one path (commits `0b2a606`).
- The `failing_layer` closed vocabulary no longer claims `unknown` is a verifier output; it is documented as the caller-side no-layer value, matching how downstream readers already treat it.
- The F6 ca-bundle temporary-file reaper now saves and restores the existing `INT/TERM` trap handlers instead of resetting them with `trap -`, so running the alias in an interactive shell no longer discards the operator's handlers.

### Added

- `test_failing_layer_attribution.sh`: L10b (cmd_verify unverified path with legal, off-vocabulary and silent-verifier tokens) and L11 (real `cmd_sync_multi` loop with stubbed verifier/generator).
- `test_ccr_upstream_ca.sh`: INT/TERM trap preservation coverage for the F6 reaper; the INT half honestly skips when the suite inherits a SIGINT-ignored startup state.

### Documentation

- `docs/Provider_Verification_Guide.md`: corrected the fallback/ tool_call note and the `unknown` caller-side semantics.

- `docs/CONTINUATION.md`: updated §7 reference, noted F6 fixed, recorded the merge, and added new §6.5 coverage.
- `scripts/debugging/failing_layer_attribution.md`: kept the alias (not deleted) and refreshed notes.

### Verification

- Full `run-all.sh` suite at `0b2a606`: exit 0; 93 of 93 test files passed, 0 failed, 1 skipped by prerequisite (the live Kimi test, gated behind `CMA_LIVE_TESTS=1`).

### Commits in this release

- `0b2a606` fix(post-006-review): restore closed-vocabulary layer reads, kill surviving mutants, stop trap clobbering

## v1.30.4 — 2026-10-06

### Added

- Kimi native usage windows: a signed-in Kimi account now reports a 5-hour and a 7-day usage window from its usages endpoint, with a reset time where the endpoint provides one. The probe makes one GET per account, never refreshes a token, and never makes a model call.
- An account whose Kimi access token has lapsed reports `auth_expired`, naming the per-account `KIMI_CODE_HOME=<dir> kimi login` command to restore windows.

### Fixed

- A Kimi usage ratio outside [0, 1], non-finite, boolean or non-numeric is rejected rather than rendered as a made-up window. An omitted window is noted in both the text and JSON output.
- The Kimi bearer token is never sent across a redirect, never sent to a non-https host unless the host is loopback, and never routed through an HTTP proxy when the usage URL is loopback.
- A non-JSON usage response reports `probe_failed` with a stated cause instead of a generic failure.
- The quota cache rejects records written under an older version (cache version 2).
- The Kimi output-token cap is derived from the provider limits.

## Known limitations (not fixed in this release)

- Kimi usage windows appear only while the Kimi access token is valid. The probe deliberately does not refresh tokens; run `kimi login` to restore them. The Kimi CLI offers no credit-free renewal command.
- `scripts/claude-providers.sh` (sync-time Kimi credential refresh, `detect_kimicode_record`) runs a one-prompt `kimi -p hi` call when the Kimi token has expired. This predates v1.30.4 and is a paid model call on sync. It is an open operator decision.
- The Kimi usage ratio's unit (fraction or percentage) is unconfirmed. Out-of-range values are rejected, so an incorrect unit shows as unavailable rather than as a false figure.
- The sarvam provider is orphaned on this host. Its config directory was removed with a backup; its status record and two alias lines remain pending an operator decision. Its 4096-token starter cap is unconfirmed.
- The DAST security scan (HawkScan) was not run: no HAWK_API_KEY and no running application.
- The live Kimi integration test is gated behind `CMA_LIVE_TESTS=1` and runs only when the operator opts in.
- Earlier commits in the v1.30.0 to v1.30.2 review batches carry a `Co-Authored-By: Claude Opus 5.5` trailer. History cannot be rewritten.

## Verification

- Full `run-all.sh` suite at `958b76d`: exit 0; 92 of 92 test files passed, 0 failed, 1 skipped by prerequisite (the live Kimi test, gated).

## Commits in this release

- `16bc86a` fix(quota): bump cache version to 2; reject older records; T028 fixture seeds the new version
- `c3a94cf` feat(quota): Kimi native usage windows (5h, 7d) via the usages endpoint; no refresh flow
- `ac9ec09` docs(changelog): record the Opus 5.5 trailer mismatch as a known gap
- `a29ce10` test(quota): fixtures seed the current cache version (no behaviour change)
- `dffda50` fix(quota): fail closed on non-finite or out-of-range Kimi used_ratio (review fix)
- `0b9acc8` feat(quota): let the Kimi CLI renew its token once before the usage probe (reuses the launch refresh)
- `d5e6da8` Revert "feat(quota): let the Kimi CLI renew its token once before the usage probe (reuses the launch refresh)"
- `97dbc1a` fix(quota): Kimi used_ratio upper bound is 1.0 (epsilon only), not 1.0001

- `d26450c` docs(quota): document auth_expired and native Kimi usage probing in the contract and changelog (audit item 4)
- `e583ad5` fix(quota): Kimi usage detail names the probed account, non-JSON and bad-ratio handling, https-only usage URL, aligned columns (audit items 5 to 9, 11)
- `1793c46` fix(quota): never send the Kimi token across a redirect; show the omitted-window note in text; remove stale login wording
- `958b76d` fix(quota): never send the Kimi token through a proxy when the usage URL is loopback

# v1.30.3 — 2026-10-05 — `quota/limits` patch: Kimi-leg fixes, live-test gating, doc corrections

Patch release on top of v1.30.2. It closes the remaining known-issues register items that were still open, and it records one incident honestly.

## Fixed

- Kimi-leg config-missing refusal: a verified provider's `kimi-<id>` twin is refused when its `config.toml` is missing, with a warning naming the file. Reconcile drops only toolkit-written lines (commit 54e3da8).
- Kimi output cap: `config.toml` always writes `max_output_size`, taken from the derived output cap and clamped below the context and to at most 128000. This stops `kimi-sarvam` sending `max_tokens` above its provider cap (commit 54e3da8).
- Twin requires its base provider to exist (commit 54e3da8).
- The live Kimi integration test is gated behind `CMA_LIVE_TESTS=1`, so `run-all.sh` never writes real host state (commit 7c3500e).
- `reset_cadence` is documented in the signal-type table (commit c6299c3).
- The native-account example now shows `endpoint_spec_present`, the 13.6 percent severity label is corrected to yellow, and the v1.30.1 documentation note is corrected: the `auth_state` field is documented in the specs (commits 19fd98c, 4f22cdf).
- The known-issues register was re-reconciled against HEAD; rows that were already fixed by earlier commits are now recorded as fixed with their evidence.

## Incident (disclosed)

A live, non-hermetic Kimi test, `test_kimi_llmctl_integration.sh`, ran during verification and wrote one session directory and some index and log lines into the operator's real `~/.kimi-prov-llmctl-small` state at 19:33. Config, aliases and status were not changed.

The operator decided to remove those artifacts; they were backed up to scratch before removal. The test is now gated behind `CMA_LIVE_TESTS=1` (commit 7c3500e).

### Testing and validation

- Full `run-all.sh` suite: exit 0; 92 of 92 test files passed, 0 failed, 1 skipped by prerequisite (the live Kimi test, gated behind CMA_LIVE_TESTS=1): the full suite log has no EXIT line yet.
- The gated live test is skipped by default and was verified to leave host state byte-identical on the skip path.

### Added

- Kimi native usage windows (5-hour and 7-day) probed from the usages endpoint with no model call or token refresh; returns `auth_expired` on 401 with fail-closed ratio validation.

### Known limitations (not fixed)

- The Kimi usage probe requires a fresh Kimi login; the stored access token has expired. Kimi usage windows are therefore not yet verified.
- Native usage windows for Claude are unsupported: it does not expose a documented usage endpoint.
- HawkScan was not run: `HAWK_API_KEY` is not set and no application is running.
- Kimi-leg failures from out-of-credit providers (HTTP 402) and local-backend context limits are documented as known-failing and do not block this release.
- Commit trailer mismatch: earlier commits in the v1.30.0 to v1.30.2 review batches carry a `Co-Authored-By: Claude Opus 5.5` trailer (2 commits in 7078213..HEAD). Commit history cannot be rewritten, so the trailers remain; from the operator decision onward, commits use `Claude Sonnet 5`.

## v1.30.2 — 2026-10-05 — `quota/limits` patch: review and known-issues fixes

Patch release for `quota/limits`. It continues the v1.30.1 patch by fixing the remaining findings of the post-release review and of the known-issues register. Each fix was made under the subagent-

driven TDD and independent-review discipline: a fresh implementer, a fresh reviewer, and a scoped re-review after any fix round. Known limitations are listed explicitly below.

## Fixed

- **T028 — quota test reached the real network on a cache miss.** The test ran against the tracked endpoints file, which names a real host. It now runs against an isolated loopback copy, and a guard refuses any probe to a non-loopback host.
- **Account de-duplication by key value.** Accounts that hold the same secret under different variable names are now grouped by a SHA-256 digest of the value plus base URL. The value and the digest are never printed, traced, or written. Empty or undefined keys never merge. Kimi OAuth aliases still group by variable name and base URL, because they share one subscription token.
- **Kimi session end rule (operator decision).** A Kimi session is reported as `session_expired` only when the refresh token's JWT `exp` has passed. Access token expiry is a normal refresh and is never a failure. An unparseable refresh token degrades to `ok`.
- **TLS error surfaced in probe failure detail.** A certificate or handshake failure now reports the real TLS error text, with the probe's own key and URL query redacted, instead of the generic "connection failed or timed out".
- **Kimi twin aliases emitted only for verified providers.** A `kimi-<id>` twin is written only when the provider's status is `verified`, matching the Claude alias gate. Stale twin lines for unverified, failed, or removed providers are removed on sync. User-authored lines that share a twin's name are never removed. A dotted provider id is matched literally.
- **Release gate fails closed on malformed quota output.** The release gate and the run-proof quota leg now fail when the quota output is not valid JSON, has the wrong shape, has non-object rows, or has every provider probe failed. Each count is checked before any comparison.
- **Proof tree untracked.** Volatile proof output moved to a git-ignored folder. Only three curated files remain tracked. Proof writers now publish atomically. The captured LAN address is redacted at capture time. The LAN address remains in git history, which is not rewritten.
- **Hard probe deadline without coreutils.** When the `timeout` binary is absent, a stdlib watchdog bounds a probe at its deadline and returns the existing `probe_failed` shape.

- **Lock-wait clamp.** The quota cache lock wait is clamped to at least the stale-lock grace plus one second, so a stale lock is always broken. The busy warning reports the observed elapsed time.

- **Pipefail-unsafe pipes.** 42 `echo | grep -q` and `printf | grep -q` pipes across six test files were converted to here-strings. A pipefail SIGPIPE could falsely fail them.

- **Strengthened quota tests.** Weak assertions were replaced with ones that fail against deliberately broken code: `--fresh` handling, probe-run counting, Claude `expiresAt` ignored, Kimi auth state, the cache-write race, and the blocked-account line.

## Known limitations

- **Kimi-leg failures (known-failing, documented).** Some Kimi-leg checks fail for environment reasons: providers out of credit (HTTP 402) and local backends at their context limits (HTTP 413 and 400). These are recorded as known-failing and do not block this release. They are not passes.

- **HawkScan not run.** The HawkScan DAST run was not executed: `HAWK_API_KEY` is not set on this host and no application is running. Security scanning for this release is limited to the independent code reviews and the test suite.

- **Native usage windows not available.** Claude and Kimi OAuth subscriptions expose no documented usage endpoint, so native accounts show authentication state only, with no usage windows. A Kimi usage probe that would confirm an undocumented endpoint has not been run; it requires a fresh Kimi login.

- **Claude session expiry.** Claude session expiry is derived from `refreshTokenExpiresAt` in `.credentials.json`. The access-token `expiresAt` is not used.

- **OpenRouter uncapped keys.** An uncapped OpenRouter key reports no limit, so no usage window can be shown for it.

- **Cache-version bump deferred.** The probe cache schema version was not bumped. Stale empty entries are rejected by the cache filter instead, so existing caches are repaired without a version change.

## Earlier release

The v1.30.1 patch and its de-duplication correction are described in the v1.30.1 section above. That section is unchanged.

# v1.30.1 — 2026-10-05 — `quota/limits` patch: post-release review fixes

Patch release for `quota/limits` (introduced in v1.30.0). An independent review of the released v1.30.0 found two Critical and five Important defects, plus two Suggestion-tier issues. This release fixes them. Each fix was made under the same subagent-driven TDD and independent-review discipline as the original feature: a fresh implementer, a fresh reviewer, and a scoped re-review after any fix round.

## Fixed

- **C1 — native accounts had no auth-expiry distinction (Critical).** Every native `claudeN`/`kimiN` account rendered the same `not_reported_by_provider` status regardless of whether its session was healthy, expired, or never signed in. Each native account now carries an `auth_state` field: `ok`, `session_expired`, or `not_signed_in`. For Claude it is derived from `claudeAiOauth.refreshTokenExpiresAt` in `.credentials.json` (never from the routinely refreshed access-token `expiresAt`). For Kimi it is derived from presence of a credentials file, because the Kimi credentials shape has no refresh-token expiry field. The text renderer annotates expired and not-signed-in rows; the base phrase `not reported by provider` is unchanged.

- **C2 — same-credential aliases were reported as separate rows (Critical).** Several provider ids that share one real credential (for example `openrouter`, `openrouter2` through `openrouter5`, all using `OPENROUTER_API_KEY` against the same base URL) rendered as separate rows, four of them falsely reading "not reported by provider". Accounts are now grouped by `CMA_PROVIDER_KEYVAR` plus `CMA_PROVIDER_BASE_URL`. Grouping never merges on an empty key variable. The implementation is bash 3.2 compatible. This corrects the v1.30.0 entry's "account-level de-duplication" claim, which covered only aliases that share one `.env` file and did not cover distinct `.env` files that share one credential.

- **I1 — `reset_at` ISO-8601 signals were silently dropped (Important).** A provider reporting a reset time as a timestamp string produced no reset information. The signal is now read as a string and validated as ISO-8601 before use. Malformed values are rejected rather than guessed.

- **I2 — OpenRouter spec ignored the real reset-cadence field (Important).** The `openrouter` entry in `quota-endpoints.json` now captures the `limit_reset` field as an informational `reset_cadence` signal. This is a cadence label such as `daily` or `monthly`, not a timestamp, and the resolver does not

consume it. OpenRouter's `rate_limit` field is a request-rate limiter, not a quota window, and is deliberately not reported as one.

- **I3 — batch-independence claim in FR-017 and SC-008 was overstated (Important).** Concurrent probes are batched at `CMA_QUOTA_MAX_PARALLEL_PROBES` (default 8), so total wall-clock time is bounded by `ceil(N / CMA_QUOTA_MAX_PARALLEL_PROBES) × timeout`, not by the timeout alone. The specification now states this bound. The batching barrier was kept deliberately, to bound the number of concurrent subprocesses. A defensive `timeout` wrapper now bounds each live probe. The wrapper is used only when the `timeout` binary is present, so stock macOS, which does not ship it, is unaffected.
- **I4 — `--json` produced invalid JSON for aliases containing `"` or `\` (Important).** The `scoped_to` value is now built with `jq` encoding instead of string interpolation.
- **I5 — `quota/limits` missing from `--help` (Important).** Both `claude-providers --help` and `kimi-providers --help` now list the `quota` subcommand and its `limits` alias. Landed in `f1cd9a4`.
- **S1 — `--timeout` accepted a missing or non-numeric value (Suggestion).** `--timeout` now reports a clear error when the value is missing, and requires a positive integer. `0` and leading-zero forms such as `08` are rejected.
- **S2 — zero-limit provider raised `ZeroDivisionError` (Suggestion).** A provider reporting a zero spending cap with no percentage now yields `percent_remaining = 0.0` instead of an uncaught exception. A provider-supplied percentage still takes precedence. Landed in `f1cd9a4`.

## Testing & Validation

- Quota test files, state at the final fix commit `f1cd9a4` (figures from the I5/S2 implementation report, independently re-run by the reviewer):
  - `test_quota_cli.sh`: 76 passed, 0 failed
  - `test_quota_concurrency.sh`: 28 passed, 0 failed
  - `test_quota_probe.sh`: 20 passed, 0 failed
  - `test_quota_rendering.sh`: 25 passed, 0 failed
  - `test_kimi_aliases.sh`: 31 passed, 0 failed (covers the Kimi `--help` line)
  - Total: 149 passed, 0 failed across the four quota test files.
- Each fix added regression tests. The I3/S1 no-`timeout`-binary test was shown to fail against the unguarded code and pass against the fix.

- Full `run-all.sh` suite at `dafd231`: exit 0, `Test files: 91 passed: 91 failed: 0`. Captured from a single clean run; no test file failed.

### Known remaining gaps (not fixed in 1.30.1)

- The Kimi `auth_state` is presence-based only. A Kimi account whose refresh token has expired still reports `ok`. This is by design: the Kimi credentials shape does not expose a refresh-token expiry, and the feature does not guess one.
- Documentation gap, closed: the `auth_state` field is documented in `specs/002-provider-usage-command/data-model.md` and the contract, added in the same release as the field itself.

## v1.30.0 — 2026-10-04 — `quota`/`limits`: universal usage reporting across every provider alias and native account

New top-level subcommand (both names — `quota` and `limits`, deliberately non-colliding with the pre-existing `usage()` function on `claude-providers.sh` and `kimi-providers.sh`) reporting real usage/limit windows for every currently-configured provider alias and every native `claudeN`/`kimiN` account, in one bounded-time invocation. Driven by the spec-kit workflow under `specs/002-provider-usage-command/`, fully sub-agent-driven per the operator's standing instruction — every task implemented by a fresh subagent, reviewed by a fresh reviewer before the next task started, with a final whole-branch review gate before this release.

### Added

- **`claude-providers quota` / `claude-providers limits`** (identical behavior, two names for operator convenience) and the same via `kimi-providers`: fleet-wide, no-argument reporting, or `quota <alias>` for a single scoped alias.
- **Flags**: `--json` (structured output), `--fresh` (bypass the cache, force a live re-probe), `--timeout <seconds>` (per-provider probe timeout override, default 3s), `--no-color` (plain text, honors `NO_COLOR` and non-tty auto-detection too).
- **Severity coloring** on every usage window: green (≥30% remaining), yellow (10–30%), red (<10%), a visually and textually distinct `limit_exceeded` state (≤0%) — the distinction survives color-stripping (plain text still names the severity word).
- **Account-level de-duplication**: `deepseek`/`kimi-deepseek`/ `pi-deepseek` share one real account and one real API key; `quota` reports it once, listing every alias name that maps to it, never three separate rows for one subscription. Incomplete in 1.30.0:

multiple `.env` files sharing one key and base URL were still split; corrected in v1.30.1.

- **Honest absence reporting**, two textually-distinct states, never confused with each other: `not_reported_by_provider` (no documented quota endpoint exists for this provider at all — no probe attempted) vs. `probe_failed` (an endpoint exists, the probe was genuinely attempted, and it failed or returned no interpretable usage signals) — each with its own human-readable phrase and, for the latter, a non-empty `absence_detail` string naming the real cause.
- **Account-blocked statement**: a fully-suspended account renders a distinct "ACCOUNT BLOCKED" line, in addition to — never instead of — any usage windows it still carries.
- **Bounded-concurrency probing** (`CMA_QUOTA_MAX_PARALLEL_PROBES`, default 8): total wall-clock scales with `ceil(N/8)` probe batches at the per-provider timeout, never linearly with the number of configured aliases — one slow/unresponsive provider never blocks the rest of the report.
- **6-hour quota cache** (`QUOTA_CACHE_TTL_SECONDS`), single sequential merge-and-save after each probe batch (never N concurrent read-modify- write round trips against the shared cache file — closes a real, reproduced data-loss race found during review), with a genuine per-record `_cached_at` timestamp so a cache hit discloses its real age in seconds, and a failed probe is never cached as if it were good data.
- **New declarative provider-endpoint spec**, `scripts/providers/quota-endpoints.json`, extending the existing binary credit-endpoint convention to full numeric usage-window extraction (amount used/remaining/limit, unit, percent remaining, reset time, optional account-blocked signals) — the same declarative-spec, no-hardcoded-provider-logic discipline this toolkit already uses elsewhere.
- **`quota --json`** output: one structured document per invocation (`generated_at`, `scoped_to`, `unknown_alias`, `rows[]`), every row carrying every documented field with `null` for anything genuinely absent — never an omitted key, so a consumer never has to distinguish "absent" from "present but null".
- New live leg on the mandatory pre-release gate (`scripts/claude-release-gate.sh` Layer 2.6, `scripts/tests/ run-proof.sh`'s `47-quota-live.log`): `claude-providers quota --json` against whatever is genuinely configured on the host running the gate, asserting valid JSON with the correct top-level shape.

## Fixed (found during this feature's own review process)

- A real, reproduced TLS-trust leak: `CMA_PROVIDER_CA_CERT`, set host-wide for an unrelated local self-signed endpoint, was inherited by every quota probe's TLS verification — breaking every real public- HTTPS provider's probe. The quota-probe

subprocess now clears that variable before probing; the shared, correctly-scoped-for-its-own- purpose `model_verify.ca_ssl_context()` is untouched.
- A probe that got a real HTTP 200 but resolved zero interpretable usage windows (e.g. a genuinely uncapped provider key) used to silently fall through every downstream check — rendering as nothing, caching as if it were valid data, and reporting `endpoint_spec_present=true` with no visible reason there was no number. Now correctly reported as `probe_failed` with an honest detail string, and never cached.
- `reset_at` / "does not reset" is now rendered in the text output too (previously only carried in `--json`).

## Testing & Validation

- 4 dedicated test files (`test_quota_cli.sh`, `test_quota_probe.sh`, `test_quota_rendering.sh`, `test_quota_concurrency.sh`), every task implemented via genuine RED→GREEN TDD, every task independently reviewed, several review rounds catching real bugs before merge: a leading-whitespace extraction bug, a crash-on-missing-file bug, a concurrent cache-write race (reproduced losing 5 of 8 records in one trial), a real network call that had leaked into a hermetic test (confirmed via `strace -f -e trace=network` and closed the same way), a vacuous test assertion caught via deliberate fault injection, and the two release-blocking findings above.
- Full regression suite (`scripts/tests/run-all.sh`): **91 test files, 91 passed, 0 failed, ALL GREEN** — zero regressions in any pre-existing subcommand, function, or flag, confirmed both before and after the release-blocking fix round.
- Final whole-branch review against all 8 of the feature spec's Success Criteria (`specs/002-provider-usage-command/spec.md`), run on this project's constitution-mandated review model: an initial NO-GO (2 critical findings, both reproduced live, not inferred from reading code), one fix round, then an independent re-verification (a second, fresh review pass redoing the same live reproductions from scratch) returning GO.
- Live leg genuinely exercised against this host's real, currently-installed provider aliases and native accounts (`claude-release-gate.sh` Layer 2.6, `run-proof.sh`'s quota leg) — not a sandbox mock.

# v1.29.0 — 2026-10-03 — llmctl integration hardening: honest limits, safe switching, live Superpowers verification

Feature release hardening the existing llmctl detection/provider-alias integration end-to-end: two honesty fields on every llmctl-backed alias, crash-proof on-demand model switching, a live Superpowers-command matrix proof across both CLI agent families, and a full documentation set linked from the README. Driven by the spec-kit workflow under `specs/001-llmctl-integration-hardening/`.

## Added — detection honesty (US1)

- **`context_warning` field** on `detect_llmctl_records()`'s output: fires when a profile's real, live context window (read from its own `/v1/models` response) falls below `CMA_INPUT_FLOOR` (default 160000) + 8192 = 168192 tokens — the floor a CLI agent's own system prompt + tool schemas typically need before a turn can even begin. Closes a captured production failure (a real Kimi Code turn needing 92,629 tokens against an `llmctl-small` profile's real 8,192-token window).
- **`lan_exposed` field**: reads the real kernel listening-socket state for each profile's resolved port via `ss -ltn`, since llmctl's own JSON carries no bind-address field. Conservative default `false` when `ss` is unavailable.
- **Bounded-concurrency parallel liveness probes** (`CMA_LLMCTL_MAX_PARALLEL_PROBES`, default 8): a host with several catalog profiles but few actually running previously paid up to N × `CMA_LLMCTL_HTTP_TIMEOUT` seconds sequentially; now bounded to roughly `ceil(N / cap) × timeout`.
- Fixed a `"0"`-string bug where an absent catalog `ctx` silently became the literal digit-string `"0"`, winning the numeric-gate check before the intended default was ever consulted.

## Added — on-demand switching hardening (US2)

- **Post-switch-failure liveness re-probe**: a failed `llmctl switch` no longer assumes the previously-active profile survived from the exit code alone — it re-probes `llmctl status` a second time and reports exactly what's running now.
- **Distinct `CRITICAL: llmctl rollback also failed` marker** when llmctl's own stderr names that exact failure mode, surfaced separately from an ordinary refusal so an operator knows the host may now have fewer services running than before.
- **Stress/chaos test coverage**: concurrent syncs during a switch, a mid-probe kill, and concurrent-switch lock

serialization (3 scenarios, 14 assertions, re-verified across 5 consecutive runs).

## Added — live Superpowers-command verification (US3)

- **`scripts/tests/verify_llmctl_superpowers_live.sh`**: runs all `{llmctl-backed alias} × {claude, kimi} × {Use Superpowers, Turn on Systematic Debugging, Turn on Sub-Agent-Driven Development}` combinations through `verify_superpowers_tui.sh`'s unforgeable-knowledge-challenge oracle (extended with two new skill challenges) and emits one aggregated, durable Check Result matrix.
- **Double-run determinism proof** (`test_llmctl_superpowers_determinism.sh`): a hermetic oracle proof that the live-matrix comparison logic itself correctly catches a real verdict change, plus an informational real double-run of one combination.
- Five genuine, pre-existing defects found in the separate upstream `llmctl` project during this verification work are tracked, not patched here (out of this repo's ownership) — see `docs/research/2026-10-02-llmctl-upstream-findings.md` (LLMCTL-F1…F5).

## Added — documentation (US4)

- **`docs/llmctl/{quickstart,user-guide,FAQ}.md`** (+ `.html`/`.pdf`/`.docx` siblings), and **`docs/diagrams/llmctl-{detection,switch}-flow.{mmd,svg}`**, all linked from the root `README.md`.
- **`scripts/tests/test_llmctl_doc_links.sh`**: proves zero dead links and zero orphaned pages across the whole new doc set.

## Added — upstream `llmctl` fixes (operator-directed scope expansion)

The US3 live-testing work above surfaced genuine bugs in the separate `../llmctl` project itself. The original scope (US3's note above, and `specs/001-llmctl-integration-hardening/spec.md`'s Clarifications) was investigate-and-document-only, never fix — that boundary was explicitly overridden by the operator mid-session as a deliberate, one-off exception, not a general precedent for sibling-project ownership. Eleven real fixes landed upstream (`../llmctl`, commits `b2fbfec`..`6fae6de`, pushed to all five of that project's own remotes) as a direct result, through **five independent Opus-`xhigh` review rounds** (each round finding a real issue in the prior round's fix, until a clean GO):

- **VRAM budget now uses real free VRAM, not static total capacity** (`3926468`, LLMCTL-F7): `llmctl plan --json` reported a profile as `fits: true` from 85% of the card's *total* capacity, with

no accounting for VRAM already held by other processes — the profile then genuinely OOM'd (`cudaMalloc failed: out of memory`) when switched to. The GPU probe now reads real free VRAM (nvidia-smi `memory.free`, amdgpu sysfs, or Apple's unified-memory heuristic) and the budget uses it when measured, falling back to the original total-based formula otherwise.

- **A pre-flight warning for the known colibri bind-security interaction** (`b2fbfec`, LLMCTL-F3): a colibri-engine profile under the LAN-exposed default crash-loops on its own independent security guard refusing a non-loopback bind without an explicit operator opt-in. That guard is untouched (correctly left as the operator's own call) — this adds only the missing advance warning, before the crash-loop happens instead of after.
- **Port-conflict diagnosis** (`007fc00`, LLMCTL-F8): a `wait_ready` timeout caused by an external, unrelated process already holding the target port previously reported only a generic "never answered" message. The scheduler now reads the engine's own log for the bind-failure signature and names the real port and owning process.
- **`llmctl status --json`** (`2874db7`, LLMCTL-F1): a machine-readable running-state command, sharing one source of truth with the existing human table via a new `_sched_status_rows()` helper (a self-caught regression during development — the refactor's own `return 0` was missing, breaking every `set -e` caller — fixed before landing).
- **Distinct `rollback-also-failed` exit code 75** (`ab4f4db`, LLMCTL-F2): previously only disclosed via freeform stderr text; claude_toolkit's own `_cma_llmctl_ensure_active` now checks this real exit code as its primary signal (`025d3ed`), keeping the old string-match as a fallback for older llmctl binaries.
- **Admission-control double-counting, found and fixed across four further rounds** (`451d983`, `f6febd8`, `39a1f0e`, `aa38ebb`): the VRAM-budget fix above was itself found, on review, to double-subtract currently-running profiles against an already-live budget (`451d983` fixed this for both VRAM and a pre-existing, identical RAM issue) — which then broke the `auto` command's eviction loop's ability to ever converge against a live budget (`f6febd8`, via a bash-dynamic-scoping credit mechanism) — which then was found to wrongly credit a *failed* eviction as freed memory (`39a1f0e`) — which then was found to fall through to the final admission check unverified whenever the eviction loop merely *exhausted* its attempts rather than converging (`aa38ebb`, the fix that finally reached a clean, unconditional GO).

Each fix was independently reviewed and verified (never trusted from a self-report) — `tests/run_tests.sh`: **37/37 PASS** after every single one of the eleven commits, confirmed fresh each time, not

just once at the end. See `docs/research/2026-10-02-llmctl-upstream-findings.md` for the full finding history (LLMCTL-F1…F8) and status table.

### Testing & Validation

- Full sandbox suite (`scripts/tests/run-all.sh`): **87 test files, 87 passed, 0 failed, ALL GREEN** (confirmed fresh after every review round, not just once) — including the llmctl-specific test files (context-carve, detection, doc-links, LAN-exposure, on-demand switch, stress/chaos, superpowers determinism, sync-all) plus the new regression tests added by the four independent-review rounds above (render-config base-null, verify_providers_live TSV, Kimi refusal classification).
- Live Superpowers matrix run against the host's real running `llmctl-small` profile: results honestly reflect that small model's real capability limits (captured evidence in `scripts/tests/proof/`), never bluffed as a pass.
- `scripts/claude-release-gate.sh --provider nvidia --skip-suite`: live smoke GREEN end-to-end (GATE-OK served, sink-side route confirmed), kimi smoke honest-SKIP (`kimi-deepseek`'s provider status is `failed`; the gate refuses to force-launch an unverified alias) — **ALL LAYERS GREEN**.

## v1.28.0 — 2026-09-11 — Token Router (138+ models) + Pi CLI agent + Helix family for Pi

Feature release adding **comprehensive Token Router support** (5 provider aliases spanning flagship, coding, reasoning, fast/cheap, and long-context profiles), **first-class Pi CLI agent integration** with symmetric provider twin aliases (`pi-<id>`), and **Helix family availability in Pi** — all fully validated by the live-proof suite.

### Added — Token Router integration

- **Token Router provider aliases (`tokenrouter`, `tokenrouter-code`, `tokenrouter-reasoning`, `tokenrouter-fast`, `tokenrouter-long`).** Dynamic discovery via `TOKENROUTER_API_KEY` (new `key-aliases.json` entry), live model fetching from `https://api.tokenrouter.com/v1/models` (138+ models, far beyond models.dev's single entry), and per-profile strong/fast model pairings covering:
  - `tokenrouter` — flagship: **Claude Opus 5** / **DeepSeek V4.1 Flash**
  - `tokenrouter-code` — coding: **Kimi K2.7 Code** / **Qwen3 Coder Next**

    - `tokenrouter-reasoning` — reasoning: **Claude Sonnet 5** / **Gemini 3.5 Flash**
    - `tokenrouter-fast` — fast/cheap: **DeepSeek V4.1 Flash** / **GLM-5.3 Flash**
    - `tokenrouter-long` — 1M-context: **GLM-5.3 (1M)** / **Nemotron-3 Super (1M)**
- **Credit-aware model selection** with documented balance endpoint in `providers/credit-endpoints.json` (falls back to paid-model probe when balance unavailable).
- **Overrides in `overrides.json`** pinning base URL (`https://api.tokenrouter.com/v1`), router transport, context limit (2M), and output cap (131072).
- **Pins file `providers/tokenrouter.json`** for operator customization (env-overridable per CONST-045).
- **`kimi-tokenrouter*` and `pi-tokenrouter*` twin aliases** automatically emitted for every tokenrouter profile — Kimi Code and Pi CLI both open against the same Token Router backend.

## Added — Pi CLI agent support (new family layer)

- **Pi family constants & detection** (mirroring Kimi layer): `PI_ACCOUNT_PREFIX=.pi-`, `PI_PROVIDER_PREFIX=.pi-prov-`, `PI_SHARED_SUBDIR=pi`, `cma_pi_home`, `cma_pi_account_home`, `cma_pi_provider_home`, `cma_detect_pi_accounts`, `cma_existing_pi_aliases`, `cma_suggest_pi_alias`, `cma_validate_pi_alias`, `cma_link_pi_shared_items`, `cma_write_pi_alias`, `cma_remove_pi_alias`.
- **Pi account wrapper `cma_run_pi`** — resolves `pi` binary, scrubs `ANTHROPIC_*`, `CLAUDE_CODE_*`, and `KIMI_CODE_HOME` env, runs `PI_HOME=~/.pi-<n> pi`.
- **Pi provider launcher `cma_run_pi_provider`** — per-id `~/.pi-prov-<id>/config.toml` with `default_model`/`base_url` rendered at sync, shared `status.json` activation gate (only `verified` ids launch unless `--force`), `CMA_PROVIDER_CA_CERT` TLS trust wiring, family isolation.
- **`pi-<id>` twin aliases** emitted for **every provider** under the new `PI_ALIASES` flag (default on, `--pi-aliases`/`--no-pi-aliases` opt-out). `kimi-*`/`kc-*`/`pi-*` namespaces reserved for provider aliases; `piN` accounts never collide.
- **Helix family in Pi** — all HelixAgent/HelixLLM/HelixCoder aliases (`helixagent`, `helixllm-gateway`, `helixagent-native`, `helixcoder`, and per-model `helixllm-*`) now get `pi-` twins automatically, so Pi CLI can route through the local HelixLLM backend on the single GPU.

### Changed

- **Dry-run behavior clarified:** twin aliases (`kimi-*`, `pi-*`) are not printed in `--dry-run` (they are emitted after the main alias gate); a real sync creates them.
- **Alias file emission** now includes `_cma_emit_cma_run_pi` and `_cma_emit_cma_run_pi_provider` in the managed block so every shell start has the Pi wrappers.

### Testing & Validation

- **All sandbox tests pass** (`test_providers.sh`, `test_lib.sh`, `test_unify.sh`, `test_add_remove.sh`, etc.).
- **Live integration verified** in sandbox: 5 tokenrouter aliases + 10 twin aliases (5 kimi + 5 pi) created with correct `config.toml` for both families; Helix family twins verified for Pi.
- **Credit-probe cache schema** extended for Token Router; unknown-context fallback bounded by provider's own 10th percentile (floored at 65536, clamped by provider median).

## v1.27.0 — 2026-09-05 — Kimi Code CLI accounts + `kimi-<id>` provider aliases (+ one-time legacy `kimi-*` → `kc-*` rename)

Feature release adding **first-class, symmetric support for the Kimi Code CLI** alongside Claude Code, plus the unreleased commits that accumulated since v1.26.8 riding along into this tag.

### Added — Kimi Code support

- **Kimi accounts (`kimi1…kimiN`).** `kimi-add-account`, `kimi-remove-account`, `kimi-list-accounts`, `kimi-unify`, `kimi-rollback` mirror their Claude counterparts, isolated per account behind `KIMI_CODE_HOME=~/.kimi-code-<n>`. Shared items (`AGENTS.md`, `plugins/`, `skills/`, `sessions/`, `session_index.jsonl`) unify into `$SHARED_DIR/kimi/` and symlink back; `config.toml`, `credentials/`, `oauth/`, `device_id`, `bin/`, `logs/`, `tui.toml` stay per-account. Login is Kimi's interactive device flow; no headless automation.
- **`kimi-<id>` provider aliases.** For every provider id that verifies, `sync` (and `--multi`) also emits `alias kimi-<id>="cma_run_kimi_provider <id>"`, opening **Kimi Code** against the **same backend** and same key the no-prefix alias opens Claude Code with. A per-id `~/.kimi-prov-<id>/config.toml` renders the provider, key, `max_context_size` (same derived limits as the Claude side), `capabilities = ["tool_use","thinking"]`, and `default_model`. Launch is `KIMI_CODE_HOME=… kimi -m "<host>/<strong-model>"`; no `cma-proxy` on this path, no `CLAUDE_CODE_*` guards. Verification is a **shared single gate** (`status.json`) for

both twins, and the launch refuses non-`verified` ids unless `--force`. `CMA_PROVIDER_CA_CERT` → `NODE_EXTRA_CA_CERTS` + `SSL_CERT_FILE` for `https://` CA-cert providers.
- **`kimi-providers` dispatch wrapper** exposing `sync|list|list-all|list-faulty|show|verify|migrate-names` against the same engine; `list` gains an **agent (claude/kimi) column**.
- **New flags:** `--kimi-aliases` / `--no-kimi-aliases` (default on).
- **Family model draws the namespace line** (diagram in `docs/diagrams/kimi-family.{mmd,svg}`): `claudeN`/`kimiN` native accounts; `<id>` (no prefix) is **always** Claude Code; `kimi-<id>` is **always** Kimi Code on the same backend; `kc-<id>` is Claude Code on a Kimi-native backend. `kimiN` accounts are never named `kimi-*`/`kc-*`, and `kc-*` ids never get a `kimi-kc-*` twin.

## Breaking — one-time legacy rename (`kimi-*` → `kc-*`)

The `kimi-` prefix is reserved to mean *only* "Kimi Code CLI agent". The legacy aliases that opened **Claude Code** against a Kimi-native backend are renamed **once** (automatically at `sync`, or via `kimi-providers migrate-names`): `kimi-for-coding`→`kc-for-coding`, `kimi-for-coding2`→`kc-for-coding2`, `kimi-for-coding-highspeed`→`kc-for-coding-highspeed`, `kimi-k3`→`kc-k3`, `kimi-k2p7`→`kc-k2p7`. The rename is **idempotent** (a second run is a no-op) and reversible via `backup_and_remove` backups. **Multi-account users must update any scripts or muscle memory that reference the old `kimi-*` backend aliases.**

## Unreleased commits since v1.26.8 riding along

As reported by `git log v1.26.8..HEAD --oneline --no-merges`:

- `2dbfd29` test(kimi): refresh the run-proof evidence set on the v1.27.0 tree
- `a9ea71b` fix(kimi): address code-review Important findings before the v1.27.0 gate (forced per-id home, trim-preserving rename, honest CA skip, degraded-not-fail standalone verify, guide corrections)
- `d68727e` test(kimi): live verifier, run-proof leg, and release-gate kimi smoke layer
- `488fdbb` test(install): kimi symlink/alias/banner coverage and Unit 4 kimi install support
- `61beb63` docs(kimi): user guide, family diagram, and the four doc mirrors
- `de5aa14` test(kimi): accounts, aliases, alias-file and migration coverage for the family
- `0e2c0d7` feat(kimi): `kimi-<id>` provider aliases, `kc` legacy rename, proxy `kimi` transform
- `013fdd9` fix(tests): the suite's own signal and scheduling guards run under a background launch
- `00e4b26` fix(install): canonical `claude_toolkit` `REPO_URL` (F4-005)

- `2618c3d` docs(qa): F4 gap ledger + FR-037 resolution note (W2d-1)
- `672f89b` plan: kimi-code-support — executable form of the approved Kimi spec
- `430822b` docs: Kimi Code support design spec (accounts + `kimi-<id>` provider aliases)
- `0cb55a1` fix(session,providers): end the endless compaction loop; wire upstream TLS CA at launch
- `fdeef17` feat(providers-verify): flag a 200 that comes from the WRONG SERVICE
- `267182b` fix(providers): repoint the three Helix endpoints at ports that answer, and stop reporting every transport failure as one cause
- `6b7e70c` fix(constitution): bump nested pin adcf660 -> 2887b42 — the duplicated hc_status port defect
- `328cf27` test(providers): pin the provider-id charset as a security control; refresh proofs
- `e8e877a` test(helixllm-export): the withheld guard could be deleted and this repo would not notice
- `37c4f48` docs(helixllm-export): the reason absence is safe has changed
- `1feda4b` fix(helixllm-export): stop `--apply` deleting a whole config for a restarting host
- `8ecf6aa` fix(helixllm-export): corrupted model names, a misdirected credential, add-only `--apply`
- `75d25ab` Auto-commit
- `ad56682` Auto-commit

### Notes

- **Version stamping.** `CHANGELOG.md` documents v1.27.0; the tag and GitHub/GitLab releases land at the end of the release gate (Unit 7 of `docs/superpowers/plans/2026-09-05-kimi-code-support.md`).
- **Docs delivered in lockstep.** `AGENTS.md`/`CLAUDE.md`/`QWEN.md`/`GEMINI.md` (constitution §11.4.157), the new `Kimi_Accounts_User_Guide.md`, `README.md`, `Provider_Aliases_User_Guide.md`, `Provider_Verification_Guide.md`, the family diagrams, and the new Kimi chapter in `Claude_Multi_Account_Fine_Tuning.md`.

## v1.26.8 — 2026-08-28 — autocompact thrashing guard + safety margin for provider aliases

Patch release addressing the "Autocompact is thrashing" failure reported on provider aliases (`opencode`, `zai-coding-plan`, `helixagent`, and others). The root cause was the auto-compact window derivation consuming the entire provider context (`text + tools +`

`output = context`), leaving zero headroom for the compacted summary itself. After a compact, a large file read or tool output could refill the context to the limit within a few turns, triggering Claude Code's thrashing stop condition.

## Fixed

- **Autocompact safety margin for provider aliases.** `cma_run_provider` now reserves `CMA_AUTO_COMPACT_SAFETY_MARGIN` tokens (default **24000**) from the auto-compact window whenever the provider can afford it without dropping below `CMA_MIN_COMPACT_WINDOW`. This lowers the compaction trigger so the post-compact context has meaningful breathing room, reducing the refill-within-3-turns pattern that produces the thrashing error. The margin is applied only when there is room: tight providers (e.g. `opencode`'s 160K-context shape) keep their existing window rather than trade thrashing for a compression loop. Operators can tune or disable the margin per host: `CMA_AUTO_COMPACT_SAFETY_MARGIN=0` disables it, larger values make compaction more aggressive. Covered by new behavioural cases in `test_providers.sh` (423/423).

- **HelixAgent memory-sharing clarification.** Sessions created under the `helixagent` alias ARE shared across aliases via the existing `claude-sync-state pull/push` hook; the shared `.claude.json` projects/session index is merged before launch and pushed after exit like every other provider. Shared `CLAUDE.md` memory is intentionally skipped when `CMA_PROVIDER_TRIM=bare` is set on the alias, because HelixLLM's 229376-token context cannot hold Claude Code's full tool-schema prefix plus the shared memory surface on a typical plugin-heavy host. The trim knob is opt-in and preserved across regeneration; removing it restores full memory loading at the cost of potential context-overflow on narrow-context hosts.

- **HelixAgent layer-4 verification now skips honestly for `CMA_PROVIDER_TRIM=bare`.** A bare provider intentionally launches with `--bare`, which disables the plugin/MCP/skill surface that the layer-4 superpowers engagement challenge requires. The verifier used to FAIL such providers, leaving `helixagent` permanently `unverified` and refused by the launch gate even when HelixLLM was healthy. It now `SKIP`s layer-4 for bare providers and keeps the `verified` badge earned through layers 1-3, so the alias launches normally. Covered by `test_provider_trim_layer4_skip.sh` and the live `verify_aliases_multi_prompt.sh` run.

- **HelixAgent pre-flight false "not answering" warning fixed.** The pre-flight probe looked for `context_limit` in HelixLLM's `/health` endpoint, but current HelixLLM builds only

return `{"status":"ok"}` there, so every launch printed `[cma warn] helixagent: HelixLLM is not answering …` even when the service was healthy. The probe now treats a successful `/health` as healthy and falls back to `/v1/models` (`n_ctx`) for mode detection when `context_limit` is absent.

- **HelixAgent tool-call loop through Claude Code fixed.** HelixLLM's streaming path emits an unbounded sequence of identical `tool_calls` when `parallel_tool_calls` is enabled, causing Claude Code to hang on any tool-using prompt. `cma-proxy` now forces `parallel_tool_calls: false` for `helixagent`, yielding a single clean tool call and letting simple tool calls (Glob, Read, Bash) execute without the previous infinite loop. Live validation: simple prompts pass 10/10; the first tool-using prompt (Glob) now passes, whereas before the fix it hung immediately. Deeper multi-turn Read-and-summarize prompts still time out on this host's Qwen3-Coder-30B deployment because the model drifts into meta-analysis rather than calling the Read tool — a model-behaviour limitation, not a toolkit regression.

### Changed

- Updated `AGENTS.md`, `CLAUDE.md`, `GEMINI.md`, and `QWEN.md` token-limit guard documentation to describe the new safety margin and its interaction with `CMA_MIN_COMPACT_WINDOW` / `CMA_TOOL_TOKEN_BUDGET`.
- Refreshed submodule pointers to latest tracked commits (`submodules/LLMsVerifier`, `submodules/challenges`, `submodules/containers`, `submodules/claude-code-router`) and bumped `submodules/go` to the latest point release on the `release-branch.go1.26` branch (`go1.26.7`) — `origin/master` is the unreleased go1.27 dev tree and carries no `VERSION` file, so it must not be used (§11.4.157). Added the missing Helix Constitution submodule at `constitution/` so the project now resolves `constitution/AGENTS.md` and `constitution/Constitution.md` directly rather than by reference.

# v1.26.7 — 2026-07-27 — vendor-prefixed model ids routable again (16 of 24 aliases 503'd before any network call) + the vendored-Go fallback that never fell back + two misdiagnoses retired

Patch release on top of v1.26.6, driven by two independent live failures. The release-blocking one is in the bundled router: once `lib.sh` began exporting `ANTHROPIC_MODEL` for the ROUTER transport too, every alias whose model id carries a vendor prefix (`deepseek-`

ai/DeepSeek-V3.2-TEE, Qwen/…, nvidia/…) was answered 503 by its OWN gateway before a single byte left the host — 16 of the 24 verified aliases, helixagent among them. The second is a bash scripts/install.sh failure on a host whose Go was ALREADY the exact required go1.26.4 — the bundled router refused to build and blamed a toolchain version mismatch that did not exist. Both shipped a confidently-worded wrong cause, and both of those wordings are retired here. This entry also folds in the v1.26.5 and v1.26.6 work, which shipped as version-stamped commit subjects but never received changelog entries.

## Fixed

- **claude-ccr-build.sh probed a path STRING, not a file, so its own documented system-Go fallback was unreachable code.** VENDORED_GO="${VENDORED_GO:-$HOME/.local/share/claude-go/bin/go}" assigns a path unconditionally — it is never empty — so the next line, _go_bin="${VENDORED_GO:-go}", could never take its :-go branch. install.sh only SYMLINKS claude-go-build (building Go from source is deliberately opt-in per the script's own DECISION block), so on a normal install ~/.local/share/claude-go does not exist and the build exec'd a missing file: line 157/163: No such file or directory, exit 127. Two further symptoms came from the same root: the log line printed an EMPTY version (building bundled claude-code-router (Go): ...) because "$_go_bin" version failed with stderr swallowed by 2>/dev/null, and the failure was then reported as (Go version mismatch?) with "upgrade your Go" advice — a factually-false diagnosis (§11.4.201(1)) on a host already running the exact required toolchain. claude-proxy-build.sh survived the identical two lines only because it adds a command -v "$_go_bin" existence probe; ccr-build now carries the same guard. Note what shielded this host: an already-resident usable ccr downgraded the hard failure to a warning + exit 0, so on a FRESH machine the same defect is a hard CMA_INSTALL_FAILURES entry — reproduced in the sandbox at exit 1. Covered by new behavioural + structural cases in test_ccr_build.sh (53/53; RED on the pre-fix body, reproducing the exact No such file or directory text). test_install.sh was independently RED at HEAD and is repaired by this fix (15/15).

- **claude-ccr-build.sh kept SHIPPING the (Go version mismatch?) diagnosis after the bug that produced it was fixed.** The guard above removes the exec-not-found failure, but the failure path's text still asserted a cause the script cannot see: both build attempts' stderr is discarded by the if ( … ) around them and "$_go_bin" version is 2>/dev/null, so at that point the script knows only that two builds returned non-zero — a state equally reachable from a compile error, a permissions problem, an uninitialised submodule or a full disk. Naming one of them is a factually-false diagnosis

(§11.4.201(1)) that costs the reader a wrong first hypothesis; on the host that motivated this release the guess was wrong in exactly that way. The warning now reports what it CAN observe — the toolchain actually used (`$_go_bin` plus its `go version` line) and what `go.mod` requires — states that the cause is in the compiler output above, and drops the "install/upgrade Go" remediation that told an operator to install what was already installed. The staleness warning + `exit 0` on a usable resident binary is unchanged: the artifact, not the build, is what the gate asserts. No test pins the prose (an assertion on wording would pin the sentence rather than the behaviour); what `test_ccr_build.sh` pins is the six-row toolchain-resolution matrix that removes the failure this text used to misname.

- **`verify_superpowers_tui.sh` reported every rc=124 as a trust/overwrite dialog, contradicting the transcript it had just captured.** `rc == 124` says only that the launch bound expired; it never says why, and the two causes it conflated need opposite responses. Ten live evidence files on this host — `chutes1`, `chutes2`, `chutes3`, `chutes4`, `kilo`, `nvidia`, `nvidia3`, `openrouter2`, `openrouter3`, `poe3`, all long-time-to-first-token reasoning models, while every fast model PASSed — were badged `launch hung within 180s (trust/overwrite prompt?)` while their own result JSON, written into that same evidence file, recorded `"terminal_reason":"aborted_streaming"` with `output_tokens:0`, `total_cost_usd:0` and `duration_api_ms:0` (`openrouter2` completed in `duration_ms:141075`, comfortably INSIDE its 180s bound). A dialog that was never shown is precisely the class of false diagnosis this suite refuses to emit anywhere else, and it sent the reader hunting for a prompt instead of at the route. The classifier now consults the transcript before naming a cause: an aborted-stream result emits a distinct `# FAIL: stream-aborted (output_tokens=… duration_ms=… bound=…s)` marker and points at `~/.claude-code-router/<id>/service.log`, where a burst of 1-4 ms 503s means the LOCAL gateway rejected the route and no `--timeout` value can help; a genuinely silent hang keeps the original dialog-hang classification. The replacement is careful about its own claims too — it does NOT assert that the backend accepted and streamed nothing, because on the run that motivated it the requests never left the host. Covered by RED→GREEN cases in `scripts/tests/test_layer4_route_attribution.sh` (224/224, up from 214; both directions — the aborted-stream discrimination AND a CONTROL proving a no-output hang still classifies as a plain timeout, so the fix is a discrimination and not a blanket rename).

- **Two more failure messages asserted causes their own evidence contradicted — one of them across 22 proof files.** A sweep of all 102 shell/python entrypoints (1,943 message-emitting statements) for this same defect class found 13 further instances; the two critical ones are fixed here, plus

a third discovered in the same block. `providers-semantic.sh` reported EVERY driver exit 1 as `layer-3 FAIL (cannot see code / bluffed)`, but the driver documents exit 1 as ALSO covering definitive provider rejections — HTTP 401/402/403/404. Of the 24 evidence files carrying that message, **22 record a driver reason of `non-200 status 401/402/403/404` and none is an actual bluff**; `providers-inference-semantic.txt` states both claims twelve lines apart, the driver's own being `402 … "Insufficient balance for request."`. The honest string was already local — the script mirrors the driver JSON to stderr ten lines ABOVE the verdict and then never reads it — so the verdict now parses that `reason`, jq-guarded, falling back to the driver's stderr and then to an explicit "not reported by the driver". `verify_providers_live.sh` asserted `account-side: key rejected / unfunded / quota` from a catch-all `else` reached by `timeout`, `stream-aborted`, `no-engagement`, `empty-result`, `api-error` and `launch-refused-unverified` alike — none of which implies an account state — while `status=unverified` means INCONCLUSIVE, not rejected (`providers-verify.sh` emits it for HTTP 429, non-4xx codes, and even "no verifier binary built"). Live proof: `42-live-providers.log` blamed `chutes5`'s account while `providers-chutes5-superpowers.txt` records `launch-refused-unverified … NO turn ran` — nothing was ever sent to that account. All three sites now print the observed classification. **No branch's counted/not-counted status changed**, and that is asserted directly rather than argued: the tally cases pass identically in RED and GREEN. Covered by new `test_failure_cause_attribution.sh` (50/50; RED 14/50 on the pre-fix bodies), which drives the real script through its `CMA_SEMANTIC_DRIVER` seam and evals the classifier branches extracted from the real file, so no assertion can be satisfied by a commented-out literal.

- **The v1.26.6 helixagent pre-flight printed `cma_warn: command not found` where the operator's remedy should have been.** `cma_run_provider` is emitted into the alias file by a `cat <<'CMA_PROV_BODY_EOF'` heredoc, and that file is sourced by the user's interactive shell — it NEVER sources `lib.sh`, so `cma_log` and `cma_warn` do not exist at run time. Every older call site guards for exactly this and `lib.sh` writes the reason down in a comment; the new pre-flight added **six unguarded calls** (`5474545` added five, `bee402a` the sixth — both commits inside this release's own payload). Live-proven in this release's own evidence: `providers-helixagent-superpowers.txt` line 3 is the bare string `cma_warn: command not found`, sitting exactly where the coder-mode remedy belonged. The launch itself still ran, so this destroyed diagnostics rather than function — which is worse than it sounds, because for this pre-flight **the message IS the feature**: it is the only thing that tells an operator to flip HelixLLM out of coder mode. Note the older guard idiom (`command -v cma_log && cma_log …`) is not

sufficient here either: it silently DROPS the text. The six calls now route through `_hl_log`/`_hl_warn` fallback emitters that print through the helper when it exists and through `printf` when it does not, so the guidance always reaches the operator. Covered by new `test_alias_file_selfcontained.sh` (7/7; RED 4/7 on the pre-fix body), which lints the GENERATED alias file for bare calls to any `lib.sh`-only helper — folding multi-line `\` and `&&`/`||` continuations first, since the codebase's real guard idiom spans lines and a naive per-line lint reports guarded code as bare — and then behaviourally executes the emitters in a shell with no `lib.sh` to prove the text actually arrives.

- **`cma-proxy` was backgrounded with no stdio redirection, so it could hang or corrupt any caller that captures a launch.** A daemon inheriting the caller's stdout/stderr holds the write end of that pipe, and every caller capturing a launch by command substitution — `verify_superpowers_tui.sh`, `claude-release-gate.sh` — reads to EOF, which never arrives while the proxy lives: measured **30005 ms vs 5 ms**, both rc=0. It bites hardest on the SUCCESS path, where `timeout` has already reaped its own child and can bound nothing at all. And it is reached deterministically, because the "listener not confirmed" branch blanked `_proxy_pid` — the only handle the exit-path reap is guarded on — orphaning the daemon; `lsof` is an undeclared dependency of `lib.sh`, so on any host without it that probe ALWAYS fails and every proxied launch leaked a live process still holding its port. Independently and unconditionally, the proxy's stderr startup banner landed INSIDE captured payloads (proven to break `jq` on captured JSON; survivable so far only because both consumers happen to parse by substring). The daemon now writes to its own `$_ccr_home/cma-proxy.log` with stdin closed — mirroring the router's own `cmd/ccr/spawn_unix.go` precedent — and the unconfirmed-listener branch kills the child before clearing its pid. Same fix in `verify_aliases_live.sh`. Covered by new `test_proxy_daemon_stdio.sh` (26/26; RED 15/26), whose behavioural cases extract and execute the SHIPPED spawn line out of the generated alias file, so a comment that merely looks like a fix cannot satisfy them.

## Fixed (ccr submodule, vasic-digital/claude-code-router)

- **A `/` in the request's `model` was read as a `Provider/model` selector, so a vendor-prefixed catalog id made the gateway 503 its own aliases before any network call (release blocker).** `internal/router/selector.go` treated both of `parseExplicitSelector`'s separators as equally trustworthy. They are not. A COMMA is unambiguous — it is this project's own on-disk route syntax and appears in no model id — so an unknown provider really is a caller error and must keep failing

loudly (silently redirecting it to `Router.default` would bill an upstream the caller never chose). A SLASH is ambiguous: it is both Node CCR's `Provider/model` wire format AND the near-universal `vendor/model` catalog convention. Once `lib.sh` (`a5e396d`) began exporting `ANTHROPIC_MODEL` for the router transport too, Claude Code put the raw catalog id in the request body, the router read the VENDOR as a provider name, found none configured, and returned an unknown-provider error instead of falling through to `Router.default`. Live shape: `503` in **1-4 ms** — before any network call — retried with backoff until the 180s launch bound expired (`kilo`: 11 gateway 503s spanning 179s; 8-12 per affected alias in `~/.claude-code-router/<id>/service.log`). Blast radius **16 of the 24 verified aliases** (every one whose `CMA_PROVIDER_MODEL` carries a vendor prefix), `helixagent` (`HelixAgent/HelixLLM`) included. Note WHY the earlier gates were all green: layers 1-3 curl the provider's own `base_url` directly and never touch the ccr route path at all — only the layer-4 live launch exercises it, which is the whole reason that leg exists. The slash form is now decided from EVIDENCE, not from the separator: if some configured provider actually serves the whole string it is a catalog id and routes normally, and if nobody serves it the unknown-provider error still stands, so `Ghost/whatever` cannot silently reach Default. A vendor prefix that COLLIDES with a real provider name (provider `nvidia` serving `nvidia/nemotron-…`) resolves to the whole id, since the split half is absent from `Models` while the full string is present. Covered by new `internal/router/vendor_prefixed_model_test.go` (6 cases: not-a-selector, routes-via-default, prefix/provider-name collision, comma form still loud, slash form on a known provider serving neither half still loud, and a real slash selector still picking a non-default provider); the pre-existing `TestSelectExplicitSelectorErrors/unknown_provider,_slash_form` contract passes unchanged.

## Changed

- **`test_output_tokens.sh` and `test_128k_output_clamp.sh` now encode the post-compression-loop-guard contract instead of the pre-guard one.** `a5e396d` added `CMA_TOOL_TOKEN_BUDGET` (80000) and `CMA_MIN_COMPACT_WINDOW` (120000) and, by its own commit body, updated `test_providers.sh` — but two other files asserting the same arithmetic were left behind, so `main` carried a red suite. The shipped behaviour is correct and the expectations were stale: for a 262144 context the carve yields `out=102144`, whose co-derived window is `262144 - 102144 - 80000 = 80000`, below the 120000 minimum — precisely the infinite-compaction state the guard exists to prevent. The guard buys the window back out of the output cap, giving `out = 262144 - 120000 - 80000 = 62144` and `62144 + 120000 + 80000 = 262144`, an exact fit. Expectations updated to 62144 with the derivation recorded in-file. Two

second-order honesty fixes went with it: the structural export count is now **2**, not 1 (the guard legitimately re-exports `CLAUDE_CODE_MAX_OUTPUT_TOKENS` at a second site), and `clamp_eval`'s remaining `102144` is annotated as the CARVE STAGE only — its awk extractor stops at the first export, so it never observed the rebalance.

- **Submodule pointers refreshed (§11.4.27).** `submodules/LLMsVerifier` `b93f29d0` -> `2a105fe7` (+6; docs plus an `api/scores` endpoint the toolkit does not call — the `model-verification` and `semantic-code-visibility` binaries on the provider-verification hot path import neither changed package) and `submodules/challenges` `41d1a13` -> `2e3ef88` (+2; one new markdown file). CONST-051 re-verified against the new tree and still passes, with the sweep's file count moving 1370 -> 1378 so the check is provably non-vacuous. `submodules/go` is deliberately HELD at `go1.26.4`: `origin/master` is the unreleased go1.27 dev tree carrying **no `VERSION` file**, which `claude-go-build.sh` requires — its presence check sits at `:35` and its parse hard-fails `exit 1` at `:45-51`. `submodules/containers` (136 behind) is not on any toolkit code path and is deferred rather than absorbed during a release window.

- **Three stale self-description markers in `README.md`, two of which contradicted each other.** The version badge read `v1.19.0` (seven releases behind), the test badge read `14 suites green`, and the prose at `:198` claimed `27 files` — so the two suite counts disagreed and neither had ever been maintained. The real count is **55 test files** at the time of measurement (`ls scripts/tests/test_*.sh | wc -l`, cross-checked against `40-sandbox-suite.log`'s `Test files: 55`), now **58** with this release's three additions (`test_proxy_daemon_stdio.sh`, `test_failure_cause_attribution.sh`, `test_alias_file_selfcontained.sh`), verified `Test files: 58  passed: 58  failed: 0  ALL GREEN`. `package.json` carried `"version": "1.0.0"` — an `npm init -y` scaffold never bumped in the project's life, describing the toolkit itself rather than any vendored dependency — now `1.26.7`. The tracked `README.{html,pdf,docx}` siblings were worse than stale (dated Jun 29, rendered from a structurally different README) and are regenerated per §11.4.65; the version string is embedded as a base64 SVG data URI, so it was verified by decoding the badges rather than by grepping the HTML. A fourth marker was false rather than merely stale: the `shellcheck-0` badge and its matching prose claimed a clean lint with **no supporting artifact anywhere in `scripts/tests/proof/`** — the same unbacked-claim shape this release exists to retire. Measured rather than deleted: `shellcheck -S error` is genuinely clean (**0**), while warning level reports **62** and style level **203**. The claim was therefore true only at one severity and ambiguous at every other, so the badge now reads `0 errors` and the prose names all three counts. **Known gap, stated rather than left implicit:**

nothing in the release process bumps `package.json` — there is no `VERSION` file and `claude-release-gate.sh` has no version reference — so it will silently re-stale unless added to the release checklist, and a bumped-once number *looks* maintained in a way an obvious `1.0.0` scaffold does not.

- **`CLAUDE.md` (and its three lockstep mirrors) document the two behaviours this release turns on.** A "Go toolchain resolution" section records why the `command -v` probe is load-bearing rather than tidiness, and why `submodules/go` must never be bumped to `origin/master` (no `VERSION` file, which `claude-go-build.sh` parses and hard-fails without). A "Router selector semantics" section records the comma/slash asymmetry, why the on-disk route looked correct while every launch died, and — the part a future reader needs most — why layers 1-3 cannot see this class of defect at all, since they curl `base_url` directly and never touch the route path.

## Added (folded from v1.26.5 / v1.26.6 — shipped but never changelogged)

- **Vendored Go toolchain as a submodule + `claude-go-build.sh`** (`f5b9206`, `bb29346`). `submodules/go` is pinned at `go1.26.4` to match `submodules/claude-code-router/go.mod`, and `claude-go-build.sh` builds it from source via `make.bash` into `~/.local/share/claude-go`, eliminating system-Go version conflicts. It is opt-in, never run by `install.sh`.
- **`_built_ok` never set on a SUCCESSFUL build** (`bb29346`). The initial build succeeded but the flag stayed 0, so `install.sh` reported `[FAILED]` on fresh installs.
- **`GOTOOLCHAIN=auto` retry** (`4568e67`) in both build entrypoints, so a host one patch behind can fetch the exact toolchain `go.mod` asks for instead of failing with an unactionable message.
- **HelixAgent auto-start + mode pre-flight** (`bee402a`, `5474545`). The launch path now queries HelixLLM's `/health` to distinguish coder mode (24576) from claude mode (229376), auto-starts HelixLLM in claude mode when Helix Code is installed (waiting up to 120s for health), and the fossil marker records the REAL HelixLLM endpoint by capturing `CMA_PROVIDER_BASE_URL` before the `case` statement appends `/chat/completions`.
- **Compression-loop guard + `kimi-for-coding` output cap** (`a5e396d`), described under Changed above.

## Notes

- **Version numbering (§11.4.151).** `1.26.2`, `1.26.3` and `1.26.4` were never used by any commit or tag; the sequence jumps `1.26.1` -> `1.26.5`. That gap is pre-existing and is recorded here rather than silently inherited. Note also that the number `1.26.1`

describes two different bodies of work: the tag `claude_toolkit-1.26.1` covers OpenCode/Chutes/Hyper aliases, while the `## v1.26.1` entry below covers ccr toolchain + launch locks.

- **`CMA_HELIX_AUTOSTART` — booting HelixLLM is now opt-in, default OFF** (operator mandate). Auto-start shipped in v1.26.6 as unconditional behaviour, and that is too much authority for a provider alias: starting HelixLLM claims the machine's single GPU and **evicts whatever HelixCode had running in coder mode**, so merely launching an alias could take a GPU out from under another tool. It now requires `CMA_HELIX_AUTOSTART=1` (also `true`/`yes`/`on`, case-insensitive); anything else, including unset, leaves it off. Turning it off does not cost the operator information: the pre-flight still reports that HelixLLM is not answering on the resolved URL, still prints the exact `helix_code/scripts/helixllm-mode.sh claude` command, and now also names the opt-in. Behaviour verified across `''`, `0`, `false`, `1`, `true`, `yes`.

- **Expected consequence of the opt-in, stated so nobody re-diagnoses it.** With `CMA_HELIX_AUTOSTART` off by default, a host where HelixLLM is not already running now records `helixagent: FAIL — basic(400) no-tool-call` in the live alias leg instead of silently booting the container. That failure is the feature working: the alias refuses rather than claiming the GPU behind HelixCode's back. The remedy is either `helix_code/scripts/helixllm-mode.sh claude` or an explicit `CMA_HELIX_AUTOSTART=1`, both of which the pre-flight prints. Note the same host cannot satisfy `helixagent`'s layer-4 challenge anyway while its tool surface is 158112 tokens: `158112 + 120000 > 229376`, the alias's pinned context.

- **Evidence hygiene.** `providers-poe3-{semantic,superpowers}.txt` are deleted. That alias no longer exists (absent from `status.json`, no `.env`, zero mentions in the current live logs), so nothing regenerated its 2026-07-18 evidence — and the fossil carried `# FAIL: timeout` while its own captured JSON recorded `"terminal_reason":"aborted_streaming"` with `output_tokens:0`, i.e. exactly the condition the new classifier labels `stream-aborted`. A known-wrong label preserved only because the alias disappeared is the same bluff this release exists to retire, so it goes rather than ships. A sweep confirms no remaining evidence file carries a label its own JSON contradicts. Note also that the proof bundle is **honest, not all-green**: legs 42/43/44 exit non-zero and `PROOF.md` reports those failures faithfully instead of hiding them.

- **Standing deviation, unresolved:** owned submodules do not carry the `claude_toolkit-` release prefix (`llmsverifier-v1.12.2`, `helixcode-v1.1.0`, `v0.4.9`). This predates this release and is flagged, not fixed.

# v1.26.1 — 2026-07-25 — ccr buildable on same-minor toolchains + launch locks that actually lock + honest toolchain-skew guidance

Patch release on top of v1.26.0. Root-caused from a live install failure (`ccr: no router found` on a host WITH Go installed) and the launch-path investigation it triggered; every fix carries a RED->GREEN regression test observed to fail on the pre-fix artifact.

## Added

- **`nvidia` cma-proxy transform family — cache_control stripping for strict backends (live defect: `nvidia4: FAIL — cache_control`).** The aliases live leg probes providers directly with a message-level `cache_control` unless a cma-proxy transform exists; nvidia4's model (`mistralai/mistral-small-4-119b-2603`) hard-rejects the field (`400 … Extra inputs are not permitted`) while its siblings tolerate it. The new `nvidia` request transform strips every `cache_control` key — message-level, content-block, system, and tool-definition level — schema-aware (a JSON-Schema property legitimately NAMED `cache_control` is preserved; mirroring ccr's `StripCacheControl` semantics). Live-proven against the real endpoint: the leg's exact probe body returns HTTP 400 direct, HTTP 200 through the proxy. Live launches were already protected by ccr's `cleancache` transformer; this closes the direct-probe chain. Registered `nvidia4 -> nvidia` via the existing digit-fold, mirrored in `cma_proxy_transform_family` (equality pinned by `test_cma_proxy.sh`, 14/14).
- **`SKIP-AUTH` bucket in the aliases live leg (live defect: `✗ milos85vasic: FAIL` with zero evidence).** The account's OAuth session was simply dead (refresh token lapsed 2026-07-22; `Failed to authenticate: OAuth session expired and could not be refreshed`) — account-side, but neither a toolkit failure nor quota, so bucketing it FAIL was a false positive and SKIP-QUOTA would have been a lie. The leg now classifies auth-death as SKIP-AUTH ("re-login with `claude /login`"), captures stderr for classification only (never for PASS, so hook noise cannot false-pass), and — closing the evidence hole that made this undiagnosable — appends `detail(<alias>): <captured output>` to the evidence file for every verdict. Covered by new hermetic `test_aliases_live_classifier.sh` (10/10; RED 5-fail on the unpatched script).

## Fixed

- **The 2.1.220 auto-compact loop: router-transport aliases exported NO model/tier maps, so the fast/compact tier resolved client-side to "Fable 5" — which Claude Code 2.1.220 itself reports as CURRENTLY UNAVAILABLE (live operator report: "enters context compression over and over, never frees, can't use alias providers").** Auto-compaction runs on the fast tier; with the compact call failing client-side on every attempt the context was never freed, compaction retried forever, and ZERO requests reached the gateway (verified: no `/v1/messages` in the per-alias `service.log` during the loop — a purely client-side failure). The tier maps (`ANTHROPIC_DEFAULT_OPUS/SONNET/HAIKU/FABLE_MODEL`), `ANTHROPIC_MODEL` and `ANTHROPIC_SMALL_FAST_MODEL` were previously exported only in the native branch. They are now hoisted above the transport branch for BOTH transports, together with `CLAUDE_CODE_MAX_CONTEXT_TOKENS` — required because an `ANTHROPIC_MODEL` id Claude's table does not know otherwise falls back to a tiny default window, which reproduced as `"Prompt is too long"` on a 20-character prompt (the same client-side accounting failure family). `cma_run`'s isolation unset clears the new var for native launches. Covered by new `test_tier_map_exports.sh` (11/11; the pre-fix body fails red).
- **alias-e2e status gate ran AFTER key resolution, so unverified keyless aliases scored key_check FAIL instead of SKIP-GATED (live: helixagent family).** An alias that never verified typically has no key either; the early `return` on key_check meant the `status != verified → gated_skip` branch was never reached for exactly the aliases it exists for. The gate now runs immediately after the env check, before key resolution.
- **Cross-alias gateway port-collision chain (live-proven: xiaomi and opencode2 both hashed to 3519).** Three links closed: (1) the 100-slot per-alias port spaces collided under ~50 router aliases — widened to 500 slots each (gateway [3460,3959], management [3960,4459]; a surviving collision now fails loudly with rc=78 instead of silently degrading); (2) ccr's `waitForReady` accepted a FOREIGN gateway's `/health` as the child's readiness, so a colliding child that died on bind was recorded "ready" in the pidfile — a 200 now counts only when the port's holder IS the child (`cmd/ccr/waitforready_test.go`); (3) `ccr default-claude-code`'s fresh-start path rebuilt the dead service with environment DEFAULTS, silently reverting the per-alias ports to the shared 3456/3458 — a dead pidfile's recorded flags are now replayed. The `claude-sonnet-5` 404s seen on opencode2/openrouter were transient fallout of this landscape (default-port gateways serving mismatched configs mid-sweep); with the chain closed those aliases route correctly again.

- **Live-proof classifier learned two more honest classes (it was pinning the suite red on non-toolkit conditions).** (1) `context-inadequate` now matches the no-"about" phrasing — live on nvidia3/mistral-small: `maximum context length is 32768 tokens. However, you requested 113865 tokens (…)` — the branch regex used to REQUIRE "about", so this genuine provider-side overflow fell through to a counted api-error; extraction is also pinned against the trailing-parenthesis trap (the request is 113865, never the 65536 completion budget). (2) New `environment-incompatible` KNOWN-NON-WORKING class: the provider's own 400 rejects an MCP tool NAME from the host's plugin environment (live on nvidia4: `Function name was mcp__plugin_aws-startup-advisor_… but must be … maximum length of 64`) — the toolkit neither mints nor rewrites MCP tool names, so a strict provider's grammar rejection is not a routing defect; the marker stays on the evidence's face and a name-sanitizing transform is the tracked follow-up. Both classes: evidence-based, re-derived every run, PASS never reaches them, generic 400s stay counted (covered by new cases in `test_layer4_route_attribution.sh`, 214/214 incl. a re-scoped mutation anchor).
- **Every router alias shared ccr's default management port 3458 — only ONE per-alias gateway could be up at a time (live defect, sarvam rc=78).** Per-alias CCR_HOME isolation gave each alias its own gateway port but left the management port at 3458 for all of them, so every second alias's `ccr restart` child died with `start management interface: listen on 127.0.0.1:3458: bind: address already in use` and the launch refused with `rc=78` ("route was written but never applied" — live-captured: 4 child starts, 4 bind failures). Worse, the child binds its gateway socket BEFORE the management bind fails, so `waitForReady` could WIN the race against the dying child — a launch that "succeeded" against a gateway that was already exiting. The launcher now derives a per-alias management port from the same provider-id hash in the disjoint [3960,4459] space (`CMA_CCR_MGMT_PORT` in the provider env pins it) and passes it to every `ccr restart` call; `ccr default-claude-code`'s recovery path replays the pidfile flags, so the port survives restarts. Covered by new `test_mgmt_port_isolation.sh` (both port spaces, per-alias distinctness, the pin; green on the fix, red on the pre-fix body).
- **Live route-attribution verifier read the STALE GLOBAL ccr dir; every router proof leg failed attribution (29 FAILs).** `verify_superpowers_tui.sh` read `~/.claude-code-router/config.json` (+ `service.json`/`service.log` for the restart receipt) — a layout nothing writes any more since per-alias CCR_HOME isolation landed (`5f9d82f`, 2026-07-24): router launches write `~/.claude-code-router/<provider-id>/config.json` and run their gateway with `CCR_HOME` pointed at the same per-alias dir. The global file still held a Jul-20 route (`deepseek`), so every router leg resolved the wrong backend (`intended=xiaomi … resolved=deepseek/deepseek-v4-pro`). The verifier now reads

the same per-alias dir the launcher writes (gate semantics unchanged: intended-vs-resolved on `default+background`, restart receipt fail-closed — only WHERE the files are read). `claude-release-gate.sh` had the same drift in its sink-side route proof and is fixed identically; `test_layer4_route_attribution.sh` fixtures moved to the per-alias layout with a stale-global decoy planted so the drift cannot silently return (205/205 green; the same suite fails 22 ways against the pre-fix verifier).

- **`_cma_ccr_unfossilise` leaked the cfg lock fd and the held flock on its early returns.** The CAS-1 mismatch and mktemp-failure paths returned between `_cma_cfg_lock` and `_cma_cfg_unlock`; in the operator's INTERACTIVE shell (which never dies) the kernel never releases a leaked flock, so each occurrence held the lock for the shell's life, stacked +1 leaked fd, and stalled every later cfg-lock on that config for the full 5s budget while silently degrading it to unlocked — precisely under the concurrent-launch conditions the lock exists for (found by independent audit, live-proven). Both returns now pass through the unlock; `test_lock_fd_persistence.sh` covers both paths (fd var cleared, contender acquires immediately, zero fd delta).
- **Bundled ccr no longer refuses to build on a same-minor Go toolchain (install blocker).** The submodule's `go.mod` pinned `go 1.26.4`; on a host running go1.26.2 with `GOTOOLCHAIN=local` (distro default on this toolchain build), `go build` refuses with `go.mod requires go >= 1.26.4`, so `install.sh` correctly hard-failed — but the fix text said "install Go" on a host that HAD Go. Patch releases never add language features, so the directive is now `go 1.26` (matching the sibling `cma-proxy` module, which built fine the whole time). Guarded by a new `test_ccr_build.sh` invariant asserting the directive stays major.minor-only.
- **The eph/cfg launch locks silently never locked — every config/marker critical section ran without mutual exclusion.** `_cma_eph_lock` / `_cma_cfg_lock` opened their lock fd as `{ exec 9>>FILE; } 2>/dev/null`: hardcoded LOW fds (9/10) collide with the shell's OWN redirection bookkeeping (bash saves/restores fds around compound-command redirections and command substitutions onto low free fds), so the fd the inner `exec` opened did not survive the group — proven live on bash 5.2.37: the next `flock -w 5 9` failed "Bad file descriptor" (hidden by the group's own `2>/dev/null`), the "5s timeout" handler fired spuriously, and the interactive launch leaked `bash: line 1: 10: Bad file descriptor` onto the operator's terminal. Both locks now use dynamic fd allocation (`exec {var}>>FILE` — the shell picks a guaranteed-free fd), which also removes the need for the stderr suppression that caused the original permanent-stderr-silencing defect. New `scripts/tests/test_lock_fd_persistence.sh` extracts the SHIPPED function bodies and proves: lock really held against an external contender, really released on unlock, stderr alive, no bad-fd noise, and two racing writers serialize to the exact count (fails

10 ways on the pre-fix code, including the lost-update race ending at 40 of 80).
- **Misleading "install Go" remediation texts.** `install.sh`'s failure record, `claude-install-verify.sh`'s absent-router failure, and the standalone `ccr-install.sh` now diagnose toolchain VERSION SKEW accurately: they name both remedies (`GOTOOLCHAIN=auto claude-ccr-build` when an older Go is present with network access, or upgrade Go) instead of telling operators to install what is already installed. `ccr-install.sh` gained the same up-front skew advisory `claude-ccr-build.sh` already had.

### Fixed (ccr submodule, vasic-digital/claude-code-router)

- **`test/livereload` drift after ATM-870 (graceful route swap).** `a6a24e0` added `gw.SwapConfig` (atomic in-place config swap; `/health`, `/ready` and routing all follow it per-request) but the live reload tests still waited for the retired sentinel `config reloaded and validated:` — all three `TestConfigHotReloadLive` subtests timed out at 30s each. `ValidReload_AcceptLineNamesNewConfig` and `CleanShutdown_WatcherStops` re-sentinelled to the real accept line (`config reloaded and applied in-place:`); the dead-premise `HonestBoundary_NotSwappedLive` was rewritten as `InPlaceSwap_HealthReflectsNewConfig` (asserts the swapped config IS served live; documents the two things that do NOT follow a swap: the startup-built outbound proxy client and in-flight request snapshots). Stale "only a process bounce applies config" claims corrected in `serve.go`, `reload.go`, `service.go`, `restart_test.go` and the submodule docs (`ADMIN_MANUAL`, `DOC-AUDIT`, `LIVE_TESTING`). `ccr restart` remains the toolkit's deterministic apply-point; `go test ./...` fully green.

# v1.26.0 — 2026-07-24 — free-tier-first per-model aliases by default + dynamic account dispatch + provider-scoped gateway paths

Minor release on top of v1.25.5 covering the ATM-850/851/852/853/860 defect batch (2026-07-23/24). Every fix carries RED-on-the-broken-artifact -> GREEN -> a paired §1.1 mutation observed to FAIL, plus live captured evidence where a live path exists (`qa-results/toolkit_defects_20260723/`).

## Added

- **Free-tier-first per-model provider aliases are now the DEFAULT sync behaviour (ATM-860, operator decision D14).** The per-model verification + multi-alias pipeline (`model_verify.py` + `providers_generate.py`) — previously reachable only through the opt-in `sync --multi` that no default path ever invoked (a CONFIGURED-but-not-IN-USE gap, §11.4.196(F)) — now runs as part of every plain `claude-providers sync`, restricted to FREE-tier models. Free-vs-paid derives from REAL data only, never a roster: models.dev `cost` rows (0/0 on both sides = free), the provider `:free` id convention, and — for models no commercial catalog knows — a self-hosted loopback/RFC1918 endpoint. A model whose tier is underivable is treated as PAID and skipped unprobed (fail-safe on spend); skips are recorded honestly in `<provider>_verified.json` (`free_only`, `skipped_models` with per-model tier + reason), never as failures. Paid probing is an explicit opt-in: `sync --include-paid` or `CMA_SYNC_INCLUDE_PAID=1`; `CMA_SYNC_MULTI=0` restores the legacy single-alias-only default. Verification remains a REAL completion round-trip per model (existence + `VERIFY_OK` sentinel + tool-call + streaming probes) — a models-list entry is not evidence, and a listed-free model whose completion fails is never admitted. Alias generation is idempotent + re-runnable with deterministic, collision-free naming (`provider`, `provider2`, ...): ranking and pairing now tie-break on `model_id`, so equal-score models can no longer swap aliases between runs on concurrent-probe completion order (verification cache bumped to v3). Native-first ordering (§11.4.196(A)) is structural and unchanged: generated aliases are provider-class (`cma_run_provider`, `~/.claude-prov-*`), a namespace the account-class dispatcher refuses. Live proof: default sync against the real opencode free tier (models.dev fetched live; free models probed with real completions, paid/unknown skipped unprobed, second run byte-identical).
- **Bare account names dispatch dynamically — `claude5` works in shells older than its alias file (ATM-850).** A `command_not_found_handle(r)` (bash+zsh) emitted into the managed alias block resolves `<name>` -> `~/.claude-<name>` at INVOCATION time, so a tmux pane shell that sourced an older `aliases.sh` still launches a newly-added account out of the box. Defined names always win; a pre-existing handler is chained; `prov-*`/`code-router` are refused; a non-account miss keeps rc=127. `claude-add-account` now detects live tmux servers holding stale pane shells and prints the exact re-source/broadcast commands (broadcast only on an interactive default-No confirm — never autonomous send-keys).

## Fixed

- **Pinned helixagent facade models survive a live sync (ATM-851).** When the live `/v1/models` listing did not contain the pinned facade id, the positional fallback overwrote the explicitly pinned `CMA_HELIXAGENT_STRONG/FAST` with the endpoint-reported `.gguf` path. Pin provenance is now captured before built-in defaults: an explicit pin is authoritative; the live listing (still fetched as reachability evidence) never overwrites it; unpinned selection stays data-driven.
- **Provider-scoped gateway paths (ATM-852, ccr submodule).** The ccr gateway now serves `/:provider/v1/{messages,chat/completions}` — the provider name is carried in the base URL path, validated against configured providers (unknown segment = 404), and stripped before the protocol classifier; bare `/v1/*` and `/proxy/v1/*` are untouched. The router-transport launcher exports the provider-scoped base URL. Activation on the running gateway requires `claude-ccr-build` + an operator-coordinated ccr restart.
- **Opencode context-compression loop (ATM-853).** `derive_limits()` read only `limit.context` and ignored `limit.input`, so big-pickle's 200k-context/160k-input catalog row produced a compact window ABOVE the model's real input cap — the client-side compact guard could never fire before the endpoint rejected, looping reject->compact->reject. `derive_limits` now clamps `ctx = min(limit.context, limit.input)` when the catalog publishes a smaller real input cap; big-pickle resolves (160000, 8192) and the guard fires first. On-alias loop-gone retest remains pending a coordinated ccr window.
- **Uncatalogued live-proven models are no longer demoted on an absent catalog row (ATM-860 finding 2).** `model_verify.py` read an ABSENT catalog context as `0` and failed the `< 8000` gate — a capacity-vs-unknown false refusal (§11.4.201(9)) that zeroed helixagent's whole live-enumerated roster. Only a KNOWN small context demotes; unknown earns no context score but keeps the live-probe verdict.

## Notes

- Regression state at draft time: see `qa-results/toolkit_defects_20260723/evidence/d6_regression_sweep.log` (providers suite, helixagent 46, pins 10, dispatch 8, tmux-notice 12, free-tier multi-sync 28, ccr `go test ./...`).

# v1.25.5 — 2026-07-23 — fail-loud install over broken/missing ccr + interactive-shell stderr fix + anti-fossil r-m-w hardening

Patch release on top of v1.25.4. Every change independently reviewed to a zero-finding / zero-warning GO on Fable-xhigh (Opus-xhigh fallback while Fable is weekly-limited; §11.4.209/ §11.4.134); install suite 46/46 green (`suite_rc=0`).

## Fixed

- **`install.sh` no longer prints `[done] installed` over a broken or missing `ccr`.** The build/verify seam now probes the ARTIFACT through its real invocation path (not a build-prerequisite's presence): a usable resident `ccr` with no Go toolchain passes with an honest will-go-stale warning; a broken / npm-doppelganger / wedged / absent `ccr` hard-fails exit 1 instead of a false success (§11.4.201(11)).
- **The probe watchdog no longer stalls install/verify by +15s on a healthy host.** A watchdog subshell spawned inside a `$(...)` command-substitution held the pipe write-end via its orphaned `sleep`, so every seam blocked to the full timeout budget on instant work. The watchdog's fds are redirected away from the cmd-subst pipe; the fast path returns in ~0s and a genuinely-wedged probe is still bounded at `rc=124`. (Codified as shell-instrument footgun I7 in the constitution.)
- **Bare `exec N>>file 2>/dev/null` in the sourced eph-lock helper no longer silences the interactive shell's stderr for the shell's lifetime.** The side redirection is brace-scoped; the eph-lock wait is bounded (best-effort, proceeds unlocked with an honest log on timeout).
- **Anti-fossil marker read-modify-write hardened** against lost updates + word-split.

## Verified

- Provider `helixagent` (local Qwen3-Coder-30B on `127.0.0.1:18434`) re-verified + proven end-to-end: `model_verify.py` anti-bluff scoring PASS + a live captured answer correctly naming the working repo (no `--force`, no bluff).

# v1.25.4 — 2026-07-22 — un-fossilise ephemeral cma-proxy addresses (kill the 502 class) + toon Go port

Patch + feature release on top of v1.25.3.

## Fixed

- **Ephemeral cma-proxy address no longer fossilises into `Router.default`.** A transform-declaring provider gets a cma-proxy on a scan-until-free port and its `base` is rewritten to that proxy address — which was then persisted into the durable `config.json` and left behind when the proxy died on exit, so `Router.default` named a dead `127.0.0.1:<port>` with no listener and every gateway consumer hit `502 dial tcp … connection refused` while the provider's real backend was healthy. Two compare-and-swap guards now (1) repair our own ephemeral address back to the real endpoint on exit and (2) reap fossils whose owning launch died — reaping only a provably-dead holder (`kill -0`, §11.4.180) and never clobbering a concurrent live route; the real endpoint is kept in a sidecar marker, not `config.json`. Measured: `helixagent` and `poe` had both fossilised on `127.0.0.1:3457` and were misread as provider outages. Live re-verification post-fix: all 8 valid aliases launch clean (PING-OK) and `poe` surfaces its real upstream status instead of a dead-port 502.

## Added

- **`scripts/toon/` — Go port of `toon_encode.py`** (added alongside the Python, which remains the sole caller). Byte-parity with the real `python3` encoder over the realistic numeric + structural input space (42 golden vectors + a 4000-iteration float-repr sweep), with two honestly-documented, test-gated divergences (bare `NaN`/`Infinity`; the `findToonScript` walk-up superset).

# v1.25.3 — 2026-07-22 — trim knob survives generator regen + operator docs for trim/gate/container-mode

Patch follow-up to v1.25.2.

## Fixed

- **`CMA_PROVIDER_TRIM` now survives a provider re-add.** The v1.25.2 trim knob was a hand-added line on the provider `.env`,

which the generator (`cma_provider_write_env`) would have dropped on the next `claude-providers add`/sync (the file is machine-generated). The generator now READS the existing `CMA_PROVIDER_TRIM` value before it truncates the file and re-emits it — matching the existing "preserve-existing-value" pattern; emitted only when set, and the ambient var is unset first so a shell export can never leak onto the file. (`scripts/tests/test_provider_trim_persist.sh`, red→green.)

### Added

- **Operator documentation** for the v1.25.2 features in `docs/Provider_Aliases_User_Guide.md` (+ synchronized HTML/PDF/DOCX) and `docs/Provider_FAQ.md`: `CMA_PROVIDER_TRIM='bare'` trim mode, the `claude-release-gate` live gate, and the helixagent HelixLLM container-mode dependency (claude mode = one 229k slot vs coder mode = 8×3072).

### Changed

- GitLab upstream repointed to the renamed `claude_toolkit` (underscore) repo (`upstreams/GitLab.sh` + local remotes) — GitLab, like GitHub, has completed the §11.4.29 snake_case rename. GitFlic/GitVerse still host the dash name (underscore absent there) and remain on dash until renamed operator-side.

## v1.25.2 — 2026-07-22 — ccr resolved by stable path (PATH-shadowing doppelgänger), provider trim mode, mandatory live release gate

Patch. A second field failure on the v1.25.1 shape, with a DIFFERENT root cause the self-heal could never fix: the npm `@musistudio/claude-code-router` package installs its own `ccr` into nvm's bin dir, which precedes `~/.local/bin` on PATH. Its `--help` carries the same "ccr start / ccr serve" fingerprint (it passes the identity gate) but it has NO `restart` subcommand — so every router launch failed exactly like a stale bundled build, the self-heal rebuilt the BUNDLED binary (which was never the one being invoked), retried bare `ccr restart`, hit the doppelgänger again, and refused. A rebuild cannot fix PATH shadowing. Investigation of the live helixagent chain then surfaced two more stacked defects: the HelixLLM container serving 8 × 3,072-token slots (`-c 24576 --parallel 8` — llama.cpp splits `-c` across slots), and ~330k tokens of auto-resumed session history + ~110k of plugin/MCP

tool schemas overflowing the local model's window on every launch (a direct `claude --bare -p hi` request measures 4,891 bytes — the client was never the problem).

## Fixed

- **`cma_run_provider` resolves OUR router by its stable install identity, never by PATH order.** Resolution: `$CMA_CCR_BIN` override, else `$HOME/.local/bin/ccr` (the symlink `claude-ccr-build` maintains), PATH only as a last resort when the bundled install is absent — and the resolved path is used for EVERY invocation (identity probe, `restart`, post-rebuild retry, and the launch itself). A PATH-shadowing doppelgänger can no longer brick router aliases nor masquerade as a stale bundled build. (`scripts/tests/test_ccr_path_shadowing.sh`, executed red→green against the real generated wrapper.)

## Added

- **`CMA_PROVIDER_TRIM='bare'` — per-provider minimal-launch mode** for local-model providers: conversation launches get `--bare` (drops the hook/plugin/MCP/CLAUDE.md surface) and BOTH history seams stay closed — the conversation-args `--resume` auto-injection AND the interactive (zero-args) stored-session-flags injection are skipped, so every launch is a fresh session that actually fits a local window. Explicit user session selectors (`--resume`, `--session-id`, …) are honored verbatim; non-conversation subcommands are untouched; untrimmed providers are byte-identical to before. Wired for `helixagent` (whose HelixLLM backend belongs in `helixllm-mode.sh claude` — one 229,376-token slot). (`scripts/tests/test_provider_trim.sh`, 11 scenarios red→green.)
- **`claude-release-gate` — the mandatory LIVE pre-release gate.** Sandbox suite + a live smoke that drives the REAL generated alias through the REAL PATH → ccr → route-apply → proxy → provider backend, asserting the served reply and the sink-side route (`--verify-providers` adds the LLMsVerifier scan). The sandbox suite is structurally blind to real-host state — this release's defect class shipped green through it — so releases now require the live layer. See README "Releasing".

# v1.25.1 — 2026-07-22 — router aliases refused to launch on a stale bundled ccr (self-healing rebuild)

Patch. A field failure caught immediately after v1.25.0: from a normal interactive shell, EVERY router-transport provider alias (helixagent, poe, kimi, …) could refuse to launch —

```
claude-providers: refusing to launch <id> — its ccr route was NOT applied.
  'ccr restart' failed (rc=1): Profile "restart" was not found or is disabled.
```

## Fixed

- **A stale bundled `ccr` binary no longer bricks router-alias launches.** The vendored Go router `ccr` is a gitignored BUILD ARTIFACT, so a submodule bump that added the `ccr restart` subcommand — which every router launch runs to apply its route — does NOT rebuild an existing install's binary. A stale ccr parsed `restart` as a profile NAME and replied `Profile "restart" was not found or is disabled` (rc=1); the launch's fail-safe then (correctly, so it never serves the wrong model) REFUSED — but that refusal bricked every router alias at once, with an opaque message. `cma_run_provider` now SELF-HEALS on exactly that shape: it rebuilds once via `claude-ccr-build` and retries `ccr restart`; only if that cannot resolve it does it refuse, now with an actionable "rebuild it: claude-ccr-build" message. The heal is bounded (one rebuild, one retry), gated on the stale-binary shape alone (a 401/402/403 auth failure or a timeout is NOT a rebuild trigger and still counts), and fail-closed if the rebuild does not help. Regression-guarded by `test_ccr_restart_selfheal.sh`, which EXECUTES the real wrapper against a stale-ccr stub (not a text grep): self-heal fires + retries, the launch proceeds after a successful heal, an actionable fail-closed refusal when it cannot, and non-stale shapes excluded.

## Verification gap that let it ship (honest)

- v1.25.0's live verification launched helixagent through the scrubbed-env `verify --deep` / superpowers-TUI path AND ran the run-proof AFTER the ccr binary happened to be rebuilt — so it never exercised a plain interactive `<alias>` launch against a *stale* build artifact, which is the exact path that failed for the operator. Green internal-harness tests masked a broken user-facing path. The regression test above now covers that path.

### Verified

- **Hermetic sandbox suite: 42 files, 42 passed, 0 failed — ALL GREEN.** Includes the new `test_ccr_restart_selfheal.sh` (9/0), which EXECUTES the real `cma_run_provider` against a stale-ccr stub and pins: the self-heal fires + retries, the launch proceeds after a successful heal, an actionable fail-closed refusal when it cannot, and — a real behavioral Scenario C, not a literal grep — that a NON-stale restart failure (a `CCR_API_KEYS` auth refusal) does NOT trigger a rebuild and is not mislabeled. `test_ccr_conformance.sh` 12/0; `test_coverage.sh` 70/0 (its SIGPIPE-pipeline lint first caught the new test).
- **The live install:** `ccr` rebuilt from the released submodule (`904effb`); `ccr restart` → rc=0; a direct interactive `helixagent` launch now passes the ccr-restart gate (reproduced) instead of refusing.
- **Independently reviewed SOUND:** the self-heal is trigger-specific (stale shape only; auth/5xx/timeout still count), bounded (one rebuild, one retry), and fail-closed (a failed rebuild returns 78 before `ccr default-claude-code`, never serving an unapplied route).

## v1.25.0 — 2026-07-22 — the compatibility proxy goes Go (cma-proxy) + HelixAgent fully working (routed + capacity-sized + tool-engaging) + provider-verification honesty + rc-safety

Minor release. Two headline changes make it a minor:

1. **A new user-facing component**: the toolkit's provider-compatibility proxy is now a single Go binary, **`cma-proxy`** (`scripts/proxy/`, module `cmaproxy`), consolidating the three former per-provider python proxies (poe/kimi/sarvam) plus a net-new helixagent tool-call transform. Python is fully removed from the proxy layer.
2. **HelixAgent goes from a broken facade to a working local provider** — genuinely routed (the :18434 repoint), capacity-sized (the 229376 pin), and **tool-engaging** (the Go proxy's Hermes tool-call recovery). A live, route- attributed `verify helixagent --deep` turn in claude mode PASSED end-to-end.

Alongside those it carries provider-verification honesty (a context-inadequate classification so a too-small backend is not miscounted as toolkit breakage), a shell-rc safety fix (§11.4.167 — the toolkit can no longer erode a user's `.bashrc`), and repo/CI hygiene. The GPU mode-switch that reallocates the shared HelixLLM backend

between HelixCode and Claude Code still ships in the companion `helix_code` repo, not this toolkit. **Version chosen: v1.25.0** (a new shipped component plus a previously-dead provider made to work is additive, backward- compatible functionality — see the closing note).

## Proxy framework — python proxies replaced by one Go binary (`cma-proxy`)

- **The three former python proxies (`poe_proxy.py`, `kimi_proxy.py`, `sarvam_proxy.py`) are gone, folded into ONE Go binary `cma-proxy`** (`scripts/proxy/`, module `cmaproxy`) — which ALSO carries a net-new `helixagent` transform. (There was never a `helixagent_proxy.py` in the repo: helixagent's prior break was routing/`base_url` (the :3456→:18434 repoint), not tool-format; its Hermes recovery is brand-new, written directly in Go.) Motivated by an operator directive that the toolkit carry no python; a streaming HTTP proxy is not bash-appropriate, so Go — which also matches the already-Go bundled ccr.
- **Structure** (each provider is a self-contained file that registers itself in an `init()`, so a new provider needs no edit to `main.go`):
  - `hermes.go` — helixagent Hermes/Qwen tool-call recovery (response-side; `registerResponse("helixagent")`).
  - `poe.go` / `kimi.go` / `sarvam.go` — request-schema fixes (`registerRequest`): Poe tool-param injection + `$ref`/`$defs` resolve + ~216 tool-count cap + `cache_control` strip; Kimi moonshot-flavored `#/$defs/` schema normalization; Sarvam content-block flatten + `max_tokens` tier clamp.
  - `registry.go` — `init()`-time registration plus family resolution (`providerKey`: exact id, then id-up-to-first-digit `poe2 → poe`, then id-up-to-first-`-` `kimi-for-coding → kimi`).
  - `main.go` — HTTP server: request-transform-then-forward → response-transform (only on a 200, only for a response-transform provider) → otherwise verbatim passthrough; a clean **502-JSON** body on any upstream/connection error (never an empty reply); and the `--has-transform <id>` discovery gate (exit 0/1).
- **Correctness win over the python it replaces**: the Hermes parser is now **delimiter-robust** — blocks split on the OPENING tags (`<function=`, `<parameter=`), a parameter value ends at the *last* `</parameter>` in its segment, and a balance guard bails to passthrough on unbalanced opening tags — so a tool-arg VALUE that itself contains `</function>` or `</parameter>` (e.g. Write-ing a file *about* tool-calling) is preserved verbatim instead of being silently truncated/dropped, which is exactly what the python parser did. Found in review (2026-07-22) and pinned by a Go regression test.
- **Wiring**: `claude-proxy-build.sh` builds + installs `cma-proxy` (mirroring `claude-ccr-build.sh`: `go build` → copy into `$SHARED_DIR/proxy` → symlink onto PATH, with a `--has-transform` self-check); `install.sh` §4b now runs that build (was: copy

scripts/proxy/*.py into the shared store); and cma_run_provider gates on cma-proxy --has-transform <id> then launches cma-proxy --provider <id> --port <port> with the upstream (CMA_PROVIDER_BASE_URL) exported inline to the child. install.sh already treats the proxy build as best-effort (Go-gated, like ccr): with no go, install still completes and the proxied aliases fall back to their direct endpoint with the compat shims INACTIVE.
- **Tests**: co-located Go tests (poe_test.go 11, kimi_test.go 12, sarvam_test.go 8, plus hermes_test.go's recovery/passthrough/</function>- in-value regression cases) driven by the new scripts/tests/test_cma_proxy.sh (go build + go vet + go test + gofmt + the --has-transform family gate). The python-only proxy tests were removed (test_poe_proxy.sh, test_sarvam_proxy.sh) and test_kimi.sh's proxy section migrated.
- **Honest scope**: this migrates ONLY the proxy layer. The remaining python — six tooling scripts (providers_resolve.py, model_verify.py, opencode_sync.py, providers_generate.py, toon_encode.py, alias_e2e_test.py) plus two test-lib helpers (tests/lib/classify_live.py, tests/lib/pty_drive.py) — is a planned follow-up, **not** in this release.

## Fixed

- **HelixAgent was a non-attributable facade — now genuinely routes to the real HelixLLM server, and a mode-switch makes its big-context turn actually serveable.** scripts/providers/helixagent.json pinned base_url to the ccr gateway itself (127.0.0.1:3456) — self-defeating, since a router provider cannot route through the gateway it *is*: cma_run_provider's self-reference guard refuses it and v1.24.0's route-attribution gate correctly marks it failed (before those gates it was badged verified on turns actually served by whichever provider ran last — the exact bluff v1.24.0 exposed), so after the v1.24.0 sync it dropped out of claude-providers list. Repointed at the real backing server 127.0.0.1:18434 (the OpenAI-style endpoint the operator's local HelixLLM / Qwen3-Coder-30B actually serves): confirmed live (a real chat call returns 200 with the existing key), Gate 0 passes (base ≠ gateway), and claude-providers verify helixagent returns verified at the probe layer. The remaining obstacle was capacity, not routing: one 32 GB GPU cannot simultaneously serve HelixCode's eight concurrent 3072-token slots and Claude Code's large single-slot request. That is resolved by a **mode-switch** (helix_code/scripts/helixllm-mode.sh) that flips the shared HelixLLM container between two mutually-exclusive modes on the one GPU, one at a time: **coder** (-c 24576 --parallel 8 — eight 3072-token slots, serving HelixCode) and **claude** (-c 229376 --parallel 1 — one large slot, serving the toolkit helixagent alias / Claude Code). The helixagent pin's context_limit is corrected to **229376** (was

180224 in the prior draft, itself a correction of the old per-slot value): 180224 fit Claude Code's first ~67K request, but once tools actually fire (see the proxy bullet below) the multi-turn agent loop accumulates to ~182,128 tokens, which overflowed a 180224 slot with the exact `400 … exceeds the available context size (180224 tokens)` on ~2 of 3 runs. 229376 gives headroom — the auto-compact window still operates at its 200000 design cap with ~21K slack — and it fits VRAM, measured live in claude mode at 30,244 MiB used / 1,854 free / 32,607 total (the KV estimate VRAM ≈ 18128 + 0.053·ctx MiB validated across 24576→19434, 180224→27676, 229376→30244). A live, route-attributed `verify helixagent --deep` turn in claude mode proves the two routing/capacity things the pin exists to fix: the alias genuinely reaches HelixLLM (ccr's resolved route matches intent for BOTH the foreground and the background request, with a restart receipt bracketing the launch — not another backend), and Claude Code's real system+tool request (66,693 input tokens) fits the 229376 slot with zero context overflow — the old per-slot `400 exceeds context` that demoted it is gone. Because HelixCode is the common case, the **default release state is coder mode**, so helixagent ships demoted to `unverified` (shown in `list-faulty`, refused by the launch gate) until the operator switches HelixLLM to claude mode (`helixllm-mode.sh claude`) — this is provider-side capacity to be reallocated, not toolkit breakage, and the suite classifies the coder-mode deep-turn overflow as a distinct context-inadequate class (see Reliability), never an account or routing failure. An audit confirmed helixagent was the ONLY provider with this facade misconfiguration and that a plain `sync` cannot re-break the repoint (the pin is the single source; the gates fail closed). (`f72a756`; two stale "base_url → :3456" comments corrected in `9bb6f1f`, all other `:3456`/facade references left as correct history of the incident that motivated the guard.)

- **The tool-call format gap is fixed by the new Go proxy (`cma-proxy`, `scripts/proxy/hermes.go`), so Claude Code's tools actually fire in the HelixAgent path.** Root-caused live: the container already runs llama.cpp with `--jinja`, and a direct request to :18434 returns proper structured `tool_calls`. But Claude Code's system prompt induces the model to write a conversational preamble BEFORE the call, in Qwen's native Hermes/XML form (`<function=NAME><parameter=P>V</parameter></function></tool_call>`); llama.cpp's parser only extracts a call that *leads* the generation, so once prose precedes it the whole thing is returned as `content` text (`finish_reason:"stop"`, `tool_calls:null`) and Claude Code never engages. `cma-proxy` sits inline (ccr → proxy → :18434) — the launch wrapper starts it when `--has-transform helixagent` matches — buffers the response, and ONLY when it finds a complete Hermes block llama.cpp did not already parse rewrites it into structured `tool_calls` (streaming and non-streaming), keeps the preamble as `content`, coerces each parameter by the request's own tool

schema (a string stays a string; integer/number/boolean/object/array parsed), and passes everything else through byte-for-byte untouched so the common path is never altered. Proven hermetically with no network or GPU: `hermes_test.go` (streaming
    - non-streaming recovery, passthrough safety, and the `</function>/</parameter>`- in-value regression) plus `test_cma_proxy.sh`'s build/vet/gofmt/go-test gate all pass, and a live unit request through the proxy turns the exact leaking generation into a clean `Read({"file_path":"README.md"})`. End-to-end, the strict layer-4 engagement gate now records a **verified** result: a live, route- attributed `verify helixagent --deep` turn — launched through the real wrapper, which started the Go `cma-proxy` — PASSED, the model returned the exact skill-content challenge answer ("Skills evolve. Read current version."), the route resolved to helixagent for both the foreground and the background request, input grew to 143,776 tokens and fit the 229376 slot, and status flipped to `verified`. Reliability at 229376 is **2 of 3** (the third miss was a slow-turn timeout of the local 30B, not an overflow — the 180224 overflow is gone). Because coder is the default operational state, helixagent still ships demoted to `unverified` until the operator switches HelixLLM to claude mode. `claude-proxy-build.sh` builds `cma-proxy` and `install.sh` §4b runs it, so the proxy ships on the normal release path.

## Reliability

- **run-proof honestly classifies a context-inadequate backend instead of counting it as fresh toolkit breakage.** A verified router provider whose backing model returns a context-overflow `400` on the large, tool/skill-heavy layer-4 turn is now its own KNOWN-NON-WORKING class — beside account-dead and route-integrity — reported `# FAIL: context-inadequate (backend M tokens < request N)` and NOT counted as a suite failure: it is provider-side (relaunch a local backend larger, or pin a larger-context model for a hosted one) exactly as an unfunded key is (top it up). **Two** live overflow phrasings are recognized — the llama.cpp shape (local backends, e.g. a HelixLLM in coder mode, whose per-slot context is small) and the OpenAI/OpenRouter shape (hosted models, e.g. an OpenRouter model whose window is under Claude Code's request) — with phrasing-aware extraction, since the two put the request/window numbers in *reversed* order and the marker must never swap them. This is a state of a backend, not a permanent property of any provider — a mode-switch or a larger pin removes it. The counts are read from the live 400, never a pinned/declared context, and the verdict is re-derived from the live layer-4 error on every run (not a persisted status a re-sync could overwrite), so it is durable and cannot false-

positive a provider that genuinely answers layer-4. Paired-mutation proven both directions; the existence layer names the same distinct reason. (Follow-up: tune the output-reservation guard so a hosted model whose window is only modestly above Claude Code's tool-heavy input can *work* rather than be excused.)

- **run-proof no longer counts an account-side `402/403` as toolkit breakage — the billing analogue of the context-inadequate class.** A provider funded when the ~512-token layers-1/2 probe verified it, but whose balance depletes (or key is revoked / access suspended) before the large layer-4 turn, returns a `402` ("Insufficient balance") or `403` on a *correctly-routed* turn — a fresh red on every proof run for a condition the toolkit cannot fix. A `402/403` can only come from the provider's billing/authz, never from how a request was formed (a malformed request is a `400`, which still counts), so it is now its own KNOWN-NON-WORKING class — `# FAIL: account-side (HTTP 402|403 …)`, reported on its face (top up or re-key the account) and swept-exempt like context-inadequate. It is ordered *below* the route-integrity gate (a mis-routed turn still counts) and matches only `402/403` — `401` (a toolkit-attributable bad auth header), `429`, `500`, and timeouts still fail the suite. Evidence-based (the live status, never a persisted one), and it cannot false-positive a paying account. Found live on this release's own proof: `inference/glm-5.2` returned "402 Insufficient balance" on a route-attributed turn. (Guards it: `test_providers_gate.sh` — 402 detected, a 400 overflow NOT swallowed, sweep-exempt.)

## Safety — shell-rc writes (§11.4.167)

- **The toolkit can no longer erode a user's `.bashrc/.zshrc`.** A pre-v1.24.0 prune→ensure race (fired on every non-interactive shell via `BASH_ENV`) could strip an rc file's body — orphaned managed-headers accumulating while the user's own `exports` were lost with no backup (this is what erased a set of local env vars during v1.24.0 development). v1.24.0 fixed the main prune path; this release closes the **two residual rc-write paths** that still bypassed it: the `install.sh` migrate rewrite (now gated + backup-first, and a silent last-line drop fixed) and the `claude-bootstrap.sh` append (now backup-first). Every rc write now funnels through one committer (`cma_rc_safe_rewrite`) mirroring the alias committer — a no-op guard, a sanity gate that parks a content-losing candidate as `.rejected.<ts>`, a mandatory pristine `.cma-orig` backup-or-refuse, and an INT/TERM-masked publish — so no rc write can proceed without a recovery point. Verified by four paired-mutation tests (rc-safety 35/0) that each fail when a guard is removed. (Follow-up: mint the committer's temp adjacent to the rc so the publish is a true same-filesystem rename.)

## Repo & CI hygiene

- **GitHub push-protection no longer trips on the redaction test's fixtures.** `scripts/tests/test_redact.sh`'s 23 synthetic provider-token fixtures were high-entropy random strings, so GitHub flagged two as real credentials and blocked the v1.24.0 push. Every synthetic token is now an obviously-fake `FAKE`+zero-padding body — each token's exact length, prefix, and separator preserved so redaction behaviour and the mutation topology are byte-for-byte unchanged (46/0, both mutations retain teeth) — and the `sk-underscore` fixture uses `sk_` not `sk_live_` (GitHub's Stripe detector matches the `sk_live_` prefix on pattern alone, entropy-immune). No pattern-flaggable token remains. (`80b07ff`, `084a77d`)
- **Stray build binaries untracked; LLMsVerifier `GEMINI.md` added.** Two ~24 MB Go build outputs (`fixed-challenge`, `model-verification`) tracked in the LLMsVerifier `llm-verifier/` subdirectory — re-added by an Auto-commit after a prior BFG purge, referenced by no script — are `git rm --cached` + gitignored (CONST-053); the stray root `ccr` build binary is gitignored in the router submodule; and LLMsVerifier's missing `GEMINI.md` (§11.4.157 lockstep gap) was added as a faithful Gemini-CLI mirror of its `QWEN.md` (same 135-anchor set through §11.4.167). (`3b21f79`)
- **CHANGELOG accuracy.** The v1.24.0 "Verified" section cited `test_providers.sh` 294/0; the shipped proof log shows 405/0 (tests were added since the credit-aware work). Corrected to match the on-disk evidence. (`8e8fb96`)

## Submodules

- `LLMsVerifier` `af8d703b` → `bb729f2b` (binary untrack + `GEMINI.md`)
- `claude-code-router` `6d787fd` → `904effb` (gitignore stray `ccr` binary)

## Verified

- **Hermetic sandbox suite: 41 files, 41 passed, 0 failed** (`run-all.sh`, ALL GREEN) — including the new `test_cma_proxy.sh` (Go proxy: `go build` + `go vet`
  - `go test` across hermes/poe/kimi/sarvam + the `--has-transform` family gate) and `test_providers_gate.sh` (the context-inadequate + new account-side classifiers, 17/0). The migration's three fallout failures — `test_providers`'s stale `_family_id` marker, `test_sandbox_hygiene`'s vacuity flag on the gofmt assertion, and the §11.4.157 governance-doc lockstep — were each root-caused and fixed.
- **Go proxy unit tests: `go test ./...` in `scripts/proxy` → `ok`** (hermes 14, poe 11, kimi 12, sarvam 8); `go vet` clean; `gofmt` clean.

- **HelixAgent live end-to-end, claude mode (229376): layer-4 PASS via the Go `cma-proxy`.** A route-attributed `verify helixagent --deep`, launched through the real wrapper (which started `cma-proxy` on :3457 → :18434), returned the exact skill-content challenge answer, resolved to helixagent for both the foreground and background request, with Claude Code's 143,776-token request fitting the 229376 slot and no overflow — status → verified. VRAM measured live at 30,244 MiB used / 1,854 free of 32,607.
- **Full `run-proof.sh` — ALL GREEN, exit 0** (coder default): hermetic 41/41; live provider-alias verification 21/0; live alias TUI 8 PASS / 0 FAIL / 13 SKIP-GATED (TOTAL 21); alias end-to-end 8/8; constitution 7/0. In coder default, helixagent is correctly reported KNOWN-NON-WORKING context-inadequate (backend 3072 < request 67 966), and the new account-side classifier reclassified `inference/glm-5.2`'s live `402 Insufficient balance` — plus every unfunded/rejected provider — as KNOWN-NON-WORKING, so no account-side condition pins the suite red and no toolkit regression is masked. Evidence in `scripts/tests/proof/` (`PROOF.md` + per-leg logs).

---

Version note: **v1.25.0 (minor)** — this release adds a new user-facing component (the Go `cma-proxy`, replacing the python proxy layer) and, with its Hermes recovery, closes the last gap keeping `helixagent` from working, taking that provider from a non-attributable broken facade to a routed, capacity-sized, tool-engaging local provider. Semver reserves a minor for backward-compatible added functionality; that is exactly this. It ships.

# v1.24.0 — 2026-07-21 — credit-aware model selection; and the v1.23.0 launch regression + the false "ALL GREEN" claim that hid it

Two things ship together here. The **new capability** is credit-aware model selection (below). The **regression** is that v1.23.0 made the bundled Go `claude-code-router` the sole router but never implemented the launch grammar the toolkit speaks, so **every router-transport provider alias was dead at launch** for the whole life of that release — and it shipped anyway because the verification suite counted a layer-4 FAIL as a pass and no gate read the evidence it had just written. Both the capability and the regression (with the false claim that hid it) are documented below. This is a minor bump because it adds a new user-facing capability, not only a fix.

## Added — credit-aware model selection (mandatory tier policy)

- **Every provider alias now picks its model by whether the account can be billed.** Credit / purchased tokens available ⇒ the strongest *paid* model the provider serves that verifies; no credit ⇒ the strongest *free* model; credit state *unknown* ⇒ treated conservatively as no-credit ⇒ the free choice. The unknown branch is deliberately conservative and asymmetric: picking a paid model on an unfunded key fails at launch with 402/403 and leaves a dead alias, whereas picking a free model on a funded key only gives up capability and the next `sync` corrects it. A human `strong_model`/`fast_model` pin (or `model_policy=free|paid`) in `providers/overrides.json` still wins; the tier logic only decides when nothing is pinned. Implemented in `scripts/providers_resolve.py` (`tier_preference`, `model_cost_tier`, `select_models`); the matching credit/cost-aware selection capability landed in the LLMsVerifier submodule (`llm-verifier/{providers,scoring,selection}`), kept project-agnostic per its decoupling rules.
- **Free vs paid is classified from real catalog pricing, and "zero-cost" is not assumed to mean free.** A catalog `cost: {input:0,output:0}` is often a subscription/plan-gated entry that needs a specific key — classifying it free would pick a model that fails at launch. Models with missing or partial pricing are reported as tier `unknown`, never guessed as free; the OpenRouter `:free` id suffix is honoured. Deep research established that reliable programmatic balance APIs are rare across providers (most are console-only), so credit state is inferred from the verification probe's response — 402/403 ⇒ no credit, 429 ⇒ transient (never demotes), 401 ⇒ bad key — with a schema-versioned 24h cache so stale results are never replayed. Covered by `scripts/tests/test_provider_credit.sh` (127 assertions) and the `providers/credit-endpoints.json` signal table.
- **Cost note (honest trade-off):** credit detection adds one bounded paid-model probe per provider (30s cap), so the *first* full `claude-providers sync` after a cache reset is slower than before; the result is cached for 24h, so subsequent syncs skip it. The probes are bounded and never hang; a slow provider is a duration cost, not a stall.

## Fixed

- **Token-limit guards were derived independently and could not both be satisfied, killing an alias at launch with a 400.** `openrouter` died on every real turn with `maximum context length is 262144 tokens. However, you requested about`

`265483 (33796 of text input, 103687 of tool input, 128000 in the output)`. Three separate defects converged:

- `limit.output` was copied verbatim from models.dev, which is not internally consistent: **1099 of 5696 catalogued models report `limit.output >= limit.context`** (counted over raw published values, including the 104 rows whose `context` is 0 — the resolver later treats those as unknown rather than as a cap of zero, which is why the both-fields-positive count is 995), which is physically impossible — the output budget is carved out of the context, so it is strictly smaller. A record in that shape has a context-sized number sitting in its output slot (`kilo`'s row is the pure form: `{context:262144, output:262144}`).
- models.dev stores limits per `(provider, model)` PAIR, and only openrouter's row inflates this model to `1000000` while parking its true context (262144) in `output`. The same model id reads 262144 under nvidia, kilo and nano-gpt.
- `scripts/lib.sh` exported the input guard **only when the context was ≤ 200000** — fail-open, so precisely the large-context providers got no guard at all. The pre-existing "output ≥ context ⇒ don't export" branch was actively harmful: declining to export hands the decision to Claude Code's own 128000 default. (The `128000` in the error above was this clamp, which coincidentally equals that default.) `providers_resolve.py:derive_limits()` fixes this in two independent ways. First, the output cap is always **carved** out of the context rather than trusted verbatim, which by itself corrects every one of those 1099 rows — a separate `output >= context` detector was written for them and later removed once it was proven to change the result on **0 of 5696 rows**: whenever `output >= context` the carve already yields `cap < context <= output`, so `min(output, cap) == cap` regardless. Second, a `:free` record claiming a larger output budget than its paid sibling now raises only a *suspicion*, which is adjudicated against the rest of the catalog: a context is lowered only when at least **three distinct providers, not counting the accused record's own provider**, publishing the same model (matched after normalising `:free` suffixes, vendor prefixes and case, and gated so an unrelated vendor's same-named model cannot vote) contradict it — and only to a value one of them actually publishes. Both halves of that rule are load-bearing and were learned the hard way: while the accused counted itself and the threshold was two, a single peer decided every verdict, because at two voters the lower median *is* the minimum. That cut a genuine 1,000,000-token window to 8,192 under a note claiming two independent providers had agreed, when nothing had agreed with anything. Adjudication is further restricted to records where `output < context`; where the two are equal the record shows the catalog's commonest mislabel — output copied *from* context — which the carve already corrects, and comparing

that copy against a sibling's genuine output cap is a category error that produced three wrong reductions before it was caught. What this mechanism can establish is only that a claim is not credible **as data** — a plausibility ceiling drawn from peers, never a measurement. It cannot establish "the model's real context", because there is no such quantity: each (provider, model) pair is its own deployment, and `llama-3.2-3b-instruct` is genuinely served at 16000, 32768, 80000 and 131072 by different hosts with none of them wrong. It therefore has a permanent, deliberate blind spot in the opposite direction: a genuinely throttled `:free` tier is corroborated at its paid siblings' value and left alone. The earlier single-sibling version was wrong on half its live firings — it collapsed `nemotron-3-ultra-550b:free` from 1,000,000 to 65,536 (93.4% of the window lost) and `gemma-4-26b:free` from 262,144 to 32,768, even though NVIDIA's own record and 14 of 15 providers respectively publish the larger value. Corroboration restores both while preserving the two genuine catches. The guiding asymmetry — understating a window only costs capability, whereas overstating one kills the alias at launch — is bounded by a second one learned here: an 8192-token output cap on a coding model is itself a dead alias, so a detector that fires wrongly is not made safe by erring small. A large but credible record does not trip either detector: xiaomi's `mimo-v2.5-pro` genuinely serves `{context:1048576, output:131072}`, and its **context is not collapsed** — it derives `1048576 / 128000`, where the output is reduced only by the separate, pre-existing clamp to Claude Code's 128000 ceiling, not by the mislabel detectors. The guards are now **co-derived** in `scripts/lib.sh` — `window = min(context - output, 200000)`, exported for every known context — which closes both the fail-open gate and the 200k–270k dead zone in which a real window got no input guard. `openrouter`/`kilo` now derive `262144 / 102144`, leaving 22517 tokens of headroom. **This defect predates v1.24.0**: HEAD's resolver produced `1048576/1048576` for openrouter — limits belonging to a different model entirely — so the credit-aware pin re-derivation already improved it.

- **A concurrency race could destroy the user's alias file on a clean run.** `cma_ensure_alias_file` performed six sequential temp+rename migrations plus two direct `cat >> "$ALIAS_FILE"` appends, racing `cma_provider_write_alias` — which `claude-providers list --refresh-aliases` fires **21× on every shell start** via the session hook. No lock existed on the alias file anywhere. Each writer was atomic and preserving *in isolation*, which is exactly why no single-function bug could be found: the second writer's read predates the first writer's rename, so it faithfully writes back a stale whole file. This was observed live — the file went `37288 → 282 → 32112 → 974 → 909 → 25682 → 130` bytes within four seconds, losing both wrapper functions and all four `claude1..4` account aliases, leaving every alias

pointing at an undefined function. It is **not** interrupt-only: the corruption timestamp preceded the suspected `SIGTERM` by ~14 minutes. All alias-file writes now funnel through a single committer (`cma_alias_commit`): the COMPLETE file (header + both wrappers + every account alias + every provider alias) is rendered into one temp and published with a single `mv`, behind an exclusive lock (`flock(1)`, falling back to an atomic-`mkdir` lock with rename-verified stale breaking for hosts without `flock`, e.g. macOS). A byte-identical no-op guard renders and compares *before* taking the lock, so a settled host performs **zero renames per shell start** and the common path cannot enter the race at all; the session hook additionally pins a zero-wait acquire so it can never block a shell. `INT`/`TERM` are masked across the critical section, and a sanity gate (header, both wrappers, the exact alias-name set, an account-alias floor) parks a rejected candidate as `<alias-file>.rejected.<ts>` rather than publishing it. The render-once design also retires the drop-and-re-append migrations that leaked ~15 orphaned `# Wrapper:` comment blocks and reordered the file. Covered by `scripts/tests/test_alias_file_concurrency.sh` (69 assertions). Teeth: against the pre-fix code the storm test records **115 structurally broken states published during a single storm** — a 35KB file collapsed to 1274 bytes with both wrappers gone, reproducing the live incident — and reports `16 failed, 17 passed`; after the fix, three consecutive runs give `69 passed, 0 failed` with an identical 35257-byte result (the storm case's three runs were `37 passed` when first landed; the file has since grown its lock-release / gate-integration / account-floor coverage). `test_lib.sh` now also lints that exactly **one** code path writes or renames onto `$ALIAS_FILE`.
- **Unknown or partial model limits failed open.** A pinned model absent from the catalog produced `context_limit=None, max_output=None`, so neither guard was exported and Claude Code fell back to its own 128000 default against an unmeasured window — the same fail-open class as the gate above. `inference` shipped `verified` and launchable in exactly that state. Unknown limits now resolve to a conservative *number* rather than to silence: `UNKNOWN_MODEL_CONTEXT = 128000` (measured, not invented — **93.95%** of the **4544** tool-call-capable models in the live catalog publish ≥128000, and p10 = 128000), narrowed further by the provider's *own* p10 over its tool-call-capable models, with `output < context` enforced in a single place. A bare provider ceiling was not enough. Measured as "*how often does this mechanism pull the fallback below the 128000 default*": the bare ceiling (`published = pool[-1]`, all the old mechanism was) did so for only 3 of 167 providers (1.8%) — `inference` 125000, `llmtr` 16384, `morph` 32000 — leaving the default free to overstate a small window (pin an uncatalogued 32k model on a provider whose ceiling is 10,000,000 and the wrapper would have exported a ~120k input window against it). The per-provider percentile does so for 32 of 167 (19.2%) instead — both counts unaffected by the

median clamp: the ceiling expression reads no floor term at all, and the clamp moves only two estimates (`inference` 65536→16000, `atomic-chat` 65536→32768), both already below 128000 either way. It carries a `MIN_USABLE_UNKNOWN_CONTEXT` floor of 65536 because an unfloored percentile swings the other way (poe's raw p10 is 480 tokens, which would trade a 400 for an unusable alias). (A different question — "is `pool[-1]` the term that decides the returned value" — answers 1 of 167 under the shipped clamp and 2 before it; that is the reading the old "2 of 167" was computed under.) Operator overrides in `providers/overrides.json` are now validated rather than trusted: `true`, `-1`, `0`, `1` and `50000.5` each previously produced a state in which the launch wrapper exported **neither** guard — silently reproducing the exact fail-open being closed — and a quoted `"200000"` was dropped without a word. Non-positive, boolean and fractional values are rejected with the reason recorded, clean digit strings are honoured, and an override that cannot carve a usable cap rolls back to the derived pair. Catalog rows with `"output": 0` (166 rows) or `"context": 0` (104 rows) are treated as unknown rather than as a binding cap of zero; this removes the resolver's silent dependence on the launch wrapper re-deriving a cap on its behalf. A launch-gate refusal was deliberately rejected as the remedy: it would destroy a proven-working alias, and the operator's realistic response is `--force`, which bypasses the gate *and still exports no guards*. This also surfaced a second defect: the `MIN_SAFE_OUTPUT` floor was applied unconditionally, so any window below it came out **inverted — 136 of 5696 catalogued rows emitted `output >= context`** (`mistral/open-mistral-7b` at 8000/8000 → 8192). *Method: replay the pre-fix carve (`cap = max(min(ctx − 160000, 128000), 8192)`, no halving branch) over every catalog row and count rows whose emitted output (`min(published_output, cap)`) is ≥ their context.* Now 0 violations and 0 missing contexts across all 5696. `inference` derives `125000/8192`, `poe` `128000/8192` (a latent instance — its stored limits were stale and the next `sync` would have emptied them), and `kimi-for-coding` `262144/102144`. Teeth: `14 failed, 329 passed` before, `343 passed, 0 failed` after.

- **Every router-transport provider alias failed at launch.** The Go router implemented `start|ui|serve|web|stop|config|help` and no agent-launch subcommand, so the wrapper's `ccr default-claude-code -- "$@"` (in `scripts/lib.sh`'s `cma_run_provider`) fell through to the dispatch `default:` branch, printed `Profile "default-claude-code" was not found or is disabled.` and exited 1. The binary's own `--help` advertised the grammar it did not implement. The router now implements the agent-launch subcommand (`cmd/ccr/launch.go`, dispatched as `default-claude-code/code`).
- **`ccr restart` was also unimplemented, and failed silently.** The wrapper invoked it under `>/dev/null 2>&1 || true`, so the missing subcommand produced no error anywhere. (That swallowing `|| true` no longer exists — it was removed later in

this same release, once a review showed a failed restart leaves the gateway serving its previous route while the config file reads back correct; the launch now refuses rather than proceeding against a route it cannot show was applied.) Because a running gateway keeps serving its startup config (`cmd/ccr/serve.go:137-143`), the per-launch config rewrite was never applied: an alias would route to whichever provider the daemon happened to start with, silently and with no diagnostic. `restart` is now implemented and replays the original flags.

- **The LAUNCH path could silently rebuild an authenticated gateway UNAUTHENTICATED.** `ensureGateway` has three arms: a live gateway that answers (use it), a live service whose gateway does not answer (bounce it), and nothing running (autostart). The middle arm and `cmdRestart` both refuse when the recorded service had inbound auth enabled and no `CCR_API_KEYS` is visible to the call (`cmd/ccr/launch.go:160`, `cmd/ccr/service.go:386`) — the autostart arm carried no such refusal. (The launch subcommand — `cmd/ccr/launch.go` — is itself new in this release, so this contrasts arms within a file authored here, not a previously shipped omission.) That arm is also the one reached when a previously-authenticated service *died* leaving its pidfile behind: `readServiceState` succeeds and still reports `AuthEnabled: true`, but `processAlive` is false. A launch from a shell without `CCR_API_KEYS` therefore brought the gateway back with inbound authentication silently off — `startService` hands its key list to `applyChildAPIKeyEnv`, which *unsets* `CCR_API_KEYS` when the list is empty. The autostart branch now mirrors the refusal (`err == nil && st.AuthEnabled && len(flags.APIKeys) == 0`, `cmd/ccr/launch.go:199`) and names the risk in `cmdRestart`'s own word, UNAUTHENTICATED; a genuinely absent pidfile recorded no auth posture, so a first-ever fresh start is unaffected. Teeth: `cmd/ccr/launch_authfailsafe_test.go` constructs exactly that state (a reaped pid, plus a stale `AuthEnabled` pidfile) and asserts the refusal names UNAUTHENTICATED. Neutering the refusal makes it FAIL — without it `ensureGateway` proceeds to rebuild, and what comes back is a `gatewayReadyTimeout` (15s, `launch.go:68`) readiness error, never the asserted refusal; with the refusal in place the test passes in 0.008s. **Bounded honestly:** the gateway binds loopback and inbound auth is opt-in, so the blast radius is a local downgrade, not an exposed endpoint. It is still a real silent downgrade of a posture the operator explicitly turned on, which is precisely the class `cmdRestart` already refused. (Nit landed in the same change: `cmdLaunch`'s unused `profile` parameter is now `_`.)
- **Gateway address was not recorded, and a gateway-disabled service stayed disabled across restarts.** `service.json` carried no gateway address, so `restart` could not replay it and the launcher could not find it. The pidfile now records `gatewayHost`/`gatewayPort` (older pidfiles backfill to the documented defaults rather than host `""`/port 0), and on

finding a live service whose gateway does not answer, the launcher restarts it with the gateway enabled.

- **The test suite destroyed two production scripts in the working tree.** `scripts/claude-session.sh` (201 lines) and `scripts/claude-sync-state.sh` (103 lines) were found truncated to 8-line and 2-line test stubs. Cause: tests wrote stubs to `$HOME/.local/bin/claude-session`, which in the REAL home is a symlink into this repo (`install.sh` links every `claude-*.sh` there), and `cat >` follows symlinks — so any run without an effective sandbox `HOME` overwrote the production script through the link. Symptom: every alias failed with `No conversation found with session ID: 11111111-2222-3333-4444-555555555555`. Both files restored from git. Hardened in `scripts/tests/lib/sandbox.sh`: a new `sandbox_stub` helper (asserts the sandbox, asserts the target is inside it, and **removes** an existing symlink instead of writing through it) plus an `assert_sandboxed` guard that aborts with exit 99 if `$HOME` is not a `mktemp` sandbox, called from `make_sandbox` itself. Every stub write across the suite was converted to it.
- **9 of 13 pinned providers carried the wrong token limits.** When `providers/overrides.json` pinned a `strong_model`/`fast_model`, `providers_resolve.py` copied the pinned id but never re-derived the model's context/output limits — so a pinned model inherited whichever limits the auto-selection had picked first (e.g. nvidia inherited `z-ai/glm-5.2`'s `1000000/131072` instead of the real `256000/65536`), and the launch wrapper exported `CLAUDE_CODE_MAX_OUTPUT_TOKENS` at roughly 2× the model's true cap. A sibling bug ignored the documented `context_limit`/`max_output` override fields entirely. Both fixed at source (limits are re-derived from the model actually selected; explicit override fields are honoured), with regression tests; an unknown pinned model now yields honest `None` limits rather than another model's numbers. (`nvidia`'s strong/fast slots were also un-inverted.)
- **Test-suite hygiene, three latent hazards fixed and guarded.** (1) A leaked Go module cache: because a sandbox *is* `$HOME`, every Go-building test wrote a 500–850 MB read-only `go/pkg/mod` cache that `cleanup_sandbox`'s `rm -rf` could not remove (EACCES), leaving ~35 GB of orphaned `cma-test.*` dirs; fixed with a `chmod -R u+w` before removal. (2) `test_sessions.sh` wrote fixed `/tmp/cma-test-*.log` paths outside the sandbox (cross-run collision) — moved under the sandbox. (3) `test_install.sh` called `make_sandbox` twice, orphaning the first sandbox and writing its rc/log outside the live one — fixed with a `fresh_sandbox` helper. (4) The live-providers leg now fails the suite **only** for providers that independently reached status `verified`; a provider already classified `failed`/`unverified`/`orphaned` account-side is reported on its own `KNOWN-NON-WORKING` line, never silently skipped and never counted as a pass — so a permanently-red gate can no longer hide a genuinely new regression.

- **`_cma_mtime` returned nothing usable on macOS, and the same broken idiom existed twice.** The idiom is `date -r "$path" +%s` with a `stat -c %Y` fallback. Both halves fail on BSD/macOS for independent reasons: `date -r` there reads its argument as **epoch seconds**, not as a file path, and `stat -c` is GNU-only (BSD spells it `stat -f %m`). So on macOS the first branch silently produced a garbage/zero read and the second could not run at all. The two sites differ in provenance, and the distinction is load-bearing for what is claimed. `_cma_mtime` (`scripts/lib.sh:279`) and the whole rolling-rc-backup subsystem it feeds (`cma_newest_rc_backup`, `cma_backup_rc_file`, the `<rc>.cma-orig` / `<rc>.cma-backup.<epoch>` files) are **new in this release** — none of them exists in `HEAD` — so `_cma_mtime` returning 0 for *every* file, and thereby degrading `cma_newest_rc_backup` from "newest by mtime" to "last by name", is an **intra-release defect caught before commit**, never a shipped one. The genuinely pre-existing defect is the second copy: the session-refresh hook `lib.sh` emits inside a `<<'HOOK'` heredoc (`scripts/lib.sh:1492`, and shipped today at `HEAD`'s `scripts/lib.sh:1416`), which runs in the **user's** shell and cannot call `_cma_mtime`, so it has to stay self-contained. Both sites were fixed independently to the portable `stat -c %Y … || stat -f %m …` pair — GNU-first, so on Linux `stat -c` succeeds and the BSD branch is never reached. A separate portable-mtime precedent already existed in-tree — `claude-providers.sh`'s `case "$(uname -s)"` dispatch (`Darwin*) stat -f %m` / `*) stat -c %Y`, lines 114-117) — but it is **not** the same idiom: its own comment (:112-113) records that an earlier `||` chain in the *reverse* order broke on Linux, because there `stat -f` also succeeds. It is cited as prior art for handling both platforms, not as a line the fix was copied from. Guard: `scripts/tests/test_rc_safety.sh` (f) now lints `lib.sh` for the `date -r "$var" +%s` form and requires the helper to carry the BSD branch — **21 passed, 0 failed**. **Honest scope:** neither half was a data-loss defect. For the new-in-this-release subsystem, its data-safety never rested on the mtime read at all: rolling backups are named `<rc>.cma-backup.<epoch>` (`lib.sh:327`), so name-order and epoch-order coincide and the "newest" pick is correct regardless of mtime, and `<rc>.cma-orig`, the load-bearing recovery point, is written exactly once and never consulted by mtime — but that describes how the intra-release defect *would* have behaved had it shipped, not production history, because none of that code is in `HEAD`. In the one half that DID ship, the emitted hook, the failure was a missed TTL rather than a wrong one — its fallback is `echo "$now"`, so the computed age was 0 and the background `claude-providers sync` simply never fired. This was a correctness and portability defect, not a data-loss one, and is not claimed as more.
- **Concurrent suite runs are now serialized** (`scripts/tests/lib/suite-lock.sh`). Two overlapping `run-all.sh`/`run-proof.sh` invocations mutate the repo while it tests itself, which is what once made a set of deterministic tests look flaky; the lock

(flock, with an atomic-`mkdir` fallback for macOS) is re-entrant for the nested `run-proof.sh → run-all.sh` case and bounded-waits then exits 75 on contention. It caught a real second session running the suite during its own verification.

## Fixed — the verification that was supposed to catch this

- **A layer-4 PASS could be non-attributable — it proved *so me* backend worked, not the one under test.** Every router-transport provider rewrites ccr's shared `Router.default` to itself immediately before launching, but an alias whose `base_url` IS the gateway (`http://127.0.0.1:3456/v1`) trips a self-reference guard in `cma_run_provider` and skips that rewrite — so it inherited whatever the *previous* provider left in `~/.claude-code-router/config.json`. `helixagent` was badged `verified` on a turn served by a different provider entirely: 157,419 tokens passed through a nominally 24,576-token alias because it had inherited a ~1M-context route. The evidence file recorded the request and the response but never *which backend served it*, so a pass and a pass-by-inheritance were indistinguishable — which is why this survived a release. (The `modelUsage: claude-opus-4-8[1m]` line in every layer-4 file is not a leak to the operator's Anthropic account: the router branch never exports `ANTHROPIC_MODEL`, so Claude Code labels and prices turns with its own defaults while ccr rewrites the model server-side. It does mean that field can never be used to attribute a backend.) `verify_superpowers_tui.sh` now records `# ROUTE-INTENDED:` and `# ROUTE-RESOLVED:` in every evidence file and fails the leg with `# FAIL: route-mismatch` when they disagree, or `# FAIL: route-unknown` when the route cannot be read — an unattributable turn now fails instead of passing. The gate does not merely compare the on-disk config, because a written config that was never *applied* proves nothing: `ccr restart` ran under `|| true`, and `cmdRestart` genuinely refuses to bounce an authenticated gateway when `CCR_API_KEYS` is not visible (`cmd/ccr/service.go:385-390` returns 1), so a swallowed failure would leave the gateway serving its previous route while the file read back as correct. The router exposes no live-route query (`/health` reports a provider *count*, not a route), so the gate instead requires a **restart receipt** bracketing the launch — a new `gateway listening on` line in `~/.claude-code-router/service.log`, or a changed `service.json` — and fails closed with `# FAIL: route-unproven` when it cannot show the route was applied. Both `.Router.default` **and** `.Router.background` are compared (a partly-foreign turn emits `# FAIL: route-mismatch-background`), and `jq` is now a hard precondition rather than a silent skip, since a host without it cannot run router providers at all. Route failures are evaluated independently of provider status: a rejected key explains a provider that cannot answer, but nothing about an account explains evidence attributed to

the wrong backend. **The guarantee, stated precisely:** a layer-4 PASS on router transport is impossible unless both router keys name the alias under test AND a restart receipt brackets that launch. Two limits remain, and are not claimed away: the receipt brackets the whole launch rather than the individual request (the suite lock, not the gate, excludes a concurrent rewrite), and it proves *a* config load rather than that the loaded bytes were the ones read back — closing that would need a live-route query the router does not expose. Covered by `scripts/tests/test_layer4_route_attribution.sh` (66 assertions across 10 legs). Teeth are proven per-finding by mutation — reverting any one fix flips exactly its own assertions — because `HEAD` is not a usable baseline here (it predates the whole feature). The original bluff is reproduced verbatim in the mismatch case: a `# PASS` written for a turn served by `chutes` while the alias under test was `routertest`. The previously vacuous assertions were replaced with ones that execute the production sweep; against a mutant with that sweep deleted, the old assertions still passed and the new ones fail.

- **v1.23.0's "run-proof ALL GREEN" and "10–11 aliases verified live" were false claims.** `scripts/tests/verify_providers_live.sh`'s layer-4 handler called `_pass` on a layer-4 **FAIL** — it asserted "the verifier ran", not "it passed". 21 alias failures were counted as passes and the leg reported "40 passed, 0 failed" while the router launch was broken end-to-end. Compounding it: no gate anywhere grepped the proof directory for `# FAIL:` markers, so 37 stale markers sat on disk unread. And the legs that produced the "10–11 verified" figure (`verify_aliases_live.sh`, `alias_e2e_test.py`) never launch an alias at all — they curl provider endpoints directly, explicitly "bypassing ccr", so they could not have detected a broken launch path under any circumstances. Fixes: line 126 is now `_fail`; the layer-3 `unverified` branch likewise (`providers-semantic.sh` emits it only on a definitive driver failure — transient conditions already route to `skip`); and a new proof-directory sweep gate fails the run if any evidence file **from the current run** ends in `# FAIL:`, independent of the stdout classification.
- **Layer 3 could de-verify a working provider on its own completion budget.** `chatComplete` returned the model's content and discarded `finish_reason`, so a reply severed mid-generation was indistinguishable from a model that had not complied. Measured live: siliconflow's round-1 reply came back as `ZETA-9-ORANGE-` against the sentinel `ZETA-9-ORANGE-7f3a` — a strict prefix, cut mid-token — and was reported as `sentinel not found in response`, a definitive exit-1 that demoted a provider whose layer-4 TUI turn PASSED. It now returns `chatOutcome{Content, FinishReason}`, and a round-1 sentinel no-match at `finish_reason == "length"` is classified **infra** (exit 3 ⇒ honest SKIP, never a demotion): a truncated probe never completed, so it yields no verdict about the model at all, and

blaming the model for a budget *we* chose is exactly the bluff this layer exists to avoid. Reading that path surfaced a second, latent defect. The judge branch states its own policy in comment — "a broken judge must never demote the model-under-test" — and routes every judge *call* failure to `failInfra`, but an unparseable judge *reply* fell through to `fail()` (exit 1) even when the reply was severed at `finish_reason == "length"`. A judge that merely ran out of tokens therefore de-verified the model under test, contradicting the judge-branch policy stated directly above it in its own comment; that case now routes to `failInfra` too. The budgets were then raised — `round1MaxTokens` 256 → 2048, `judgeMaxTokens` 64 → 2048, and round 2's bare literal `512` → the named `round2MaxTokens = 2048` — and the three sites differ **in kind**, which is the point:

- **The judge at 64 was the active defect.** Measured on deepseek-v4-pro, 3 samples per budget, task = emit one integer: reasoning 78–361 tokens; 64 ⇒ **0/3**, every sample `finish_reason=length` with empty content — the exact truncation shape above. 128 ⇒ 0/3, 256 ⇒ 1/3, 512 ⇒ 3/3. 512 is nonetheless the wrong fix: it passed only as a small-n artifact, and widening to 9 samples surfaced a 500-token completion, leaving 12 tokens of headroom. 2048 is chosen with margin rather than fitted to the sample.
- **Round 1 at 256 was NOT broken, and no claim is made that it was.** Same model, task = echo an 18-char sentinel: reasoning 58–97 tokens; 64 ⇒ 0/3, 128 ⇒ 2/3, **256 ⇒ 3/3**. It is raised for a different and more interesting reason: **reasoning-token distributions do not transfer across models, or across tasks.** siliconflow truncated at 256 on that same round-1 task — the severed-prefix case above. A per-task budget tuned on one model silently de-verifies another. And truncation at *either* round de-verifies a working provider — round 1 severs the **sentinel** (a mechanical exact-match check, exit-1 on no-match), round 2 severs the **description** (scored by the judge, which fairly rates a severed or empty description below threshold 2 ⇒ `fail()` ⇒ exit-1); the *only* round-2 path that is a mere skip is a truncated judge *call/reply*, which D1 routes to `failInfra`. After D1 both truncation sites are prevented by budget rather than one being intercepted, which is why **both** budgets were raised, not just round 1's.
- **Round 2 at 512 was the release's second counted blocker — it de-verified `inference`.** The round-2 *description* call (glm-5.2, `max_tokens` the only variable) came back `finish_reason=length` with empty content on two of three samples at 512; the judge fairly scored the severed/empty description below threshold 2 ⇒ `fail()` ⇒ **exit-1**, demoting a provider whose layer-4 TUI turn PASSES. This is the symmetric twin of the siliconflow case, not a bonus clause: **both** run-proof failures were D1 truncation de-verifications — siliconflow's **sentinel** severed in round 1 (fixed by `round1MaxTokens`), `inference`'s **description** severed

to empty in round 2 (fixed by `round2MaxTokens`). Raised to 2048 the description completes (`finish_reason=stop` on all three samples, well inside the budget) and 12/12 verifier runs return `verified`, judge scores 2–3, never below threshold — though six of the twelve sit at exactly 2, a thin-but-consistent margin, not a proven bound. **Honest bound:** D1 does not add a round-2-description→infra branch; it gives the description enough budget to finish. A description that genuinely needed >2048 would still be judged low and exit-1 — the fix is "enough budget to finish," not "truncation can no longer de-verify." Two scope limits, stated rather than papered over. The **default** judge is Groq `llama-3.1-8b-instant` (`scripts/providers/judge.env.template`), a non-reasoning model, so the judge=64 defect was **latent in the default configuration** — it bites only when a reasoning model is configured as the judge. And the measurements are one model, one provider, one prompt pair: a sample maximum is not a distributional ceiling (the 500-token judge sample appeared only when n went 3 → 9), so these budgets clear the observed tail with headroom rather than claiming a proven bound. Landed in `submodules/LLMsVerifier/llm-verifier/cmd/semantic-code-visibility/main.go`; covered by that package's `main_truncation_test.go` (6 tests: budget truncation named as such, a genuinely-empty reply still reported as empty, `finish_reason` surfaced to the caller, and a budget floor asserted per site — **not** a uniform one: `round2MaxTokens ≥ 2048` and `judgeMaxTokens ≥ 1024`, but `round1MaxTokens ≥ 512`, so round 1 could be lowered to 512 and the suite would still pass — a softer guard than this entry's own argument about round-1 sentinel truncation implies) — `go test ./cmd/semantic-code-visibility/` passes.

## Added — regression guards

- **`scripts/tests/test_ccr_conformance.sh`** — static, no-network conformance between the launch grammar the toolkit speaks and the grammar the bundled Go router understands. Scans `scripts/lib.sh` for `ccr <word>` in command position, parses the `case` arms of the router's top-level dispatch, and asserts required ⊆ supported. This would have caught the regression with no network and no launch.
- **`ccr` test stubs now fail loudly on an unrecognised subcommand** instead of `exit 0`. The router path's only coverage was fake stubs that silently succeeded on anything they did not recognise — a stub can certify nothing but the stub.
- **`TestMain` guard in the router's test package** (`cmd/ccr/main_testmain_test.go`) prevents tests from spawning the real service binary. `startService` spawns a detached child from `os.Executable()`, which under `go test` is the *test* binary; the

child re-ran the entire suite and forked exponentially, leaving 600+ live processes until `fork(2)` returned EAGAIN and unrelated tests failed with "resource temporarily unavailable". Tests that legitimately need a service process must go through `withStubService(t)`.
- New guards, each proven to fail when the thing it guards is broken (mutation tested): `test_mutation_residue.sh` (refuses to ship a `&& false` / always-pass mutation left behind by testing), `test_providers_gate.sh` (a `verified` provider's failure still fails the suite; account-dead ones do not — with an anti-vacuous guard that the gate genuinely discriminates), `test_sandbox_hygiene.sh` (no test writes a fixed `/tmp` path or a bare redirect into `.local/bin`), `test_sandbox_leak.sh` (a test that leaks a sandbox fails even though it exits 0), `test_suite_lock.sh` (34 assertions, 3 mutations killed), and `test_provider_credit.sh` (127 assertions for the credit-tier rule).
- The layer-4 superpowers-engagement check was replaced: the old fuzzy vocabulary grep produced both false passes (a confabulated claim) and false failures (a genuine engagement phrased differently). It now poses an unforgeable secret-knowledge challenge — one exact cell of the skill's own table, content a model can only produce by having loaded the skill — plus a distinct `empty-result` verdict for a model that runs but emits no text.

## Verified

Credit-aware selection (this release's new capability):

- `scripts/tests/test_providers.sh` **405/0**; `test_provider_credit.sh` **127/0**.
- The two rule-encoding branches are **mutation-proven** to have teeth: breaking the unknown→free conservative default makes the anti-vacuous guard fire (`available` and `unknown` stop differing); breaking the credit→paid branch makes the paid models unreachable (19 assertions fail). Both restored byte-identical.
- LLMsVerifier credit/cost-aware code: `go build ./...` clean; the `providers`, `scoring` and `selection` packages test green.

Launch-regression fixes (carried from the v1.24.0 work):

- **Live, through the real alias path — final `run-proof.sh` (2026-07-21):** all six legs `rc=0` (`sandbox · live · providers · aliases · alias-e2e · constitution`). The aliases leg records `PASS: 9 FAIL: 0 SKIP-QUOTA: 0 SKIP-TRANSIENT: 0 SKIP-GATED: 12 TOTAL: 21`, and the providers leg `9 passed, 0 failed, 12 gated-skipped`. The 12 gated-skipped are account-dead providers (funds/keys) the verification gate correctly filters out — they are not claimed to work. `helixagent` did change from its earlier PASS to the expected route-attribution de-verification, exactly

as predicted; that is the correct outcome, not a regression. The two layer-3 truncation fixes are proven in the regenerated evidence: `providers-siliconflow-semantic.txt` and `providers-inference-semantic.txt` now both read `verified` (round-1 sentinel intact, round-2 judge scored 3 and 2), where before this release they read `unverified` on severed responses.

- **Evidence provenance.** Only proof artefacts produced after 2026-07-20 13:54 are cited: from that point `~/.claude-code-router/service.log` retains the gateway restart records that let each launch be attributed to the route it actually owned. Older artefacts (Jul 5/18/19) predate both those retained records and the switch to the Go router as sole router, cannot be attributed retroactively, and are deliberately not relied on for any claim here.

- **Full hermetic suite (2026-07-21): 42 files / 42 passed / 0 failed — ALL GREEN**, run against the final tree (`run-all.sh`, which auto-discovers every `test_*.sh`, including the new `test_redact.sh` and `test_semantic_evidence.sh`). This is the real tally the earlier "NOT yet asserted" note deferred to: a prior 36-file run predated these fixes and covered fewer files, so it was deliberately never claimed as a full-suite green — claiming it would have repeated the v1.23.0 error this release exists to correct.

- **Independent whole-branch review (2026-07-20):** GO — on the tree as it stood that day, CHANGELOG honesty, docs-vs-code fidelity, the I-2 fix, and the §11.4.120 test reconciliation were all verified with pasted evidence, and the sole flagged blocker was the mechanical submodule-commit step (addressed at release time), not a code defect. **Scope — this GO covers only what the review saw, and is not a sign-off on the shipped tree.** Substantial work landed after it: the alias-file render-once race fix, the route-attribution gate hardening, the empty-/unknown-limits fix, the rc-safety + `_cma_mtime` work, the layer-3 D1 truncation fix, the semantic fail-open fix, and these CHANGELOG entries themselves. Under this project's own re-review discipline, changes after a GO re-arm the review, so the 2026-07-20 result is recorded as what it was, a green light on an earlier state, not a release sign-off.

- **Final whole-branch review (2026-07-21): GO on the shipped tree.** The two thinnest-covered pieces — the `providers-semantic.sh` evidence mirror and the layer-3 D1 truncation routing — were verified verdict-safe, secret-safe (the mirror lands in an evidence file that is `_redact`-ed before commit), and regression-safe (every new branch is FAIL→SKIP, never FAIL→PASS, so no non-compliant model can newly pass; `go vet` / `go test` / `go build` clean). Mutation residue and secrets

are clean across all three repos, and docs name only symbols that exist in the code. This is the sign-off the 2026-07-20 result could not give.

**Scope of what "ALL GREEN" claims here:** the hermetic suite and all six `run-proof.sh` legs pass on the final tree, and the 9 verified provider aliases work live end-to-end. It does NOT claim the 12 account-dead aliases work — those are gated out for lack of funds/keys, reported as known-non-working, and never counted as a pass. The release is cut from this state.

# v1.23.0 — 2026-07-20 — Bundled Go claude-code-router is now the SOLE router (JS fully replaced) + full retest

The vendored Go `claude-code-router` (submodule, installed as `ccr` via `claude-ccr-build`) is now the toolkit's one and only router; the original Node `@musistudio/claude-code-router` is no longer preferred, required, or the advertised fallback.

## Changed

- **Go fully replaces JS.** There was never any functional code requiring the Node router — detection (`command -v ccr`) and the identity-guard grammar check (`ccr start`/`ccr serve`) are router-agnostic and already pass the Go binary. This release removes the remaining JS advertising: the missing-ccr hint (`lib.sh`) and the foreign-ccr refusal now point solely at `claude-ccr-build` (the bundled Go build) and no longer name `@musistudio/claude-code-router` or suggest `npm install`; the `claude-ccr-build` Go-toolchain-missing hint demotes the Node router to a last-resort note; `install.sh`'s warning drops the JS clause. A new migration marker regenerates already-installed alias files so existing users pick up the reworded guard. `test_output_tokens.sh` now asserts the refusal points at `claude-ccr-build` AND no longer advertises the JS npm router.

## Fixed

- **`test_lib.sh` proof-secret scanner: token-boundary anchoring.** The proof-dir secret scanner matched a key prefix embedded mid-identifier (`re_` inside a Go-module-cache path `retire_connection_id_frame_test.go` captured as build noise), a false positive. Prefixes now match only at a token boundary (line-start or a non-word char before them), so a real leaked key (after a quote/=/:/space) is still caught while innocent identifiers are not. Pinned by a fixture regression guard (embedded prefix NOT flagged, real `sk-ant-` key IS flagged).

- **Bundled `LLMsVerifier` submodule: data race fixed** (submodule commit `f9b875cf`). `InMemoryContinuousEvaluator.executeRun` wrote `run.Status` on the nil-`debateEval` path outside the mutex while `GetRun` read it concurrently — a race the `-race` detector flagged. The write moved inside the lock; the whole module now passes `go test -race ./...`, pinned by a new concurrent-stress regression guard.

## Verified (full live retest — captured evidence in `scripts/tests/proof/`)

- Hermetic suite `run-all.sh` (29 files) — ALL GREEN; `run-proof.sh` (6 legs) — ALL GREEN; `verify_ccr_live.sh` — 49/0; `verify_helixagent_test.sh` — 46/46.
- Challenge bank (`submodules/challenges`): `go test -race` + meta-runner (25/25) GREEN. `containers` submodule: `build+vet+-race` GREEN.
- **Live provider aliases** (`api_keys.sh` present): `claude-providers sync` + the `run-proof` alias legs verify **10–11 aliases LIVE** end-to-end (chutes, deepseek, helixagent, inference, kilo, opencode, openrouter, poe, siliconflow, xiaomi; nvidia works at HTTP 200 but the strict 2-probe verify intermittently hits free-tier rate-limiting). The remaining providers are **honestly gated — every failure is account-side, not a toolkit bug**: direct probes captured 401 (key rejected — mappings verified correct), 402/403 (no funds / suspended / quota), 429 (fair-usage rate-limit / insufficient balance). Evidence: `scripts/tests/proof/60-provider-triage.txt`. `claude-providers prune` removed 3 stale status-only orphans. The launch gate refuses every non-`verified` alias — the anti-bluff design working as intended.
- The `qa-all` control-plane/anti-bluff legs that require a live Helix cluster, the host suspend-guard, or `go-mutesting` are honest host/tooling preconditions on this machine, not toolkit-code failures.

**v1.22.9 — 2026-07-20 — bundled claude-code-router v0.4.9 (`--upstream-timeout`) + verify_ccr_live.sh auth+proxy live-proof legs (49 checks)**

**v1.22.8 — 2026-07-20 — bundled claude-code-router v0.4.8 (authenticated outbound proxy config block, redacted password)**

**v1.22.7 — 2026-07-20 — bundled claude-code-router v0.4.7 (inbound auth switch, `--max-attempts`, start/ui flag forwarding)**

**v1.22.6 — 2026-07-20 — bundled claude-code-router v0.4.6 (docs correction; no behavior change)**

**v1.22.5 — 2026-07-20 — bundled claude-code-router v0.4.5 (OpenAI-facade long-context routing symmetry)**

**v1.22.4 — 2026-07-20 — bundled claude-code-router v0.4.4 (streaming token accounting on both relay paths)**

**v1.22.3 — 2026-07-20 — bundled claude-code-router v0.4.3 (TLS/HTTP3 CLI flags, OpenAI-facade metric parity, synchronous bind) + verify_ccr_live.sh TLS/HTTP3 live-proof leg**

**v1.22.2 — 2026-07-19 — bundled claude-code-router v0.4.2 (transport/ hot-reload/load live suites)**

**v1.22.1 — 2026-07-19 — bundled claude-code-router v0.4.1 (metrics polish + live e2e); toolkit verify_ccr_live.sh live proof of the bundled Go ccr**

**v1.22.0 — 2026-07-19 — bundled claude-code-router v0.4.0 (/metrics + semantic cache wired, Router.think, exhaustive tests)**

**v1.21.0 — 2026-07-19 — bundled Go claude-code-router built+installed as `ccr` (claude-ccr-build); submodule bumped to v0.3.0 (response cache + cross-provider fallback wired)**

**v1.20.0 — 2026-07-19 — bump bundled claude-code-router to v0.2.0 (multi-protocol gateway: OpenAI inbound facade, Anthropic passthrough, classifiers, Think/LongContext routing, hot-reload, redacted logging)**

**v1.19.0 — 2026-07-19 — port deepseek+xiaomi to router (ccr) transport + IPv4 fix**

# v1.18.0 — 2026-07-19 — HelixAgent/HelixLLM local-model exposure + 128k output-token clamp + ccr launch fixes

## Added

- **HelixAgent/HelixLLM provider — a local llama.cpp model exposed as a first-class provider alias.** A new `helixagent` alias routes through the claude-code-router (ccr, `:3456`) to a local llama.cpp server (`:18434`) serving a Qwen3-Coder-30B gguf, presented to Claude Code as **Provider = HelixAgent / Model = HelixLLM** — the raw local endpoint is never exposed as its own catalogue entry, only via the HelixAgent/HelixLLM facade. `claude-providers list` shows `helixagent verified HelixAgent/HelixLLM`. **Live-proven (2026-07-19):** a completion through `HelixAgent/HelixLLM` via `:3456` returns from the local model (`HELIX_ROUTE_OK`, real token usage), and the `/v1/models` catalogue exposes the local model only as `HelixAgent/HelixLLM` (the raw gguf path is not a catalogue entry). Known residual: the completion response's `.model` field still echoes the raw gguf path — the ccr catalogue and `claude-providers` surfaces are clean; hardening that response echo is a tracked follow-up.
  - **ccr v3.0.6 target-adapter registration.** The provider `id` / `name` / `models[0]` are set to `HelixAgent` / `HelixLLM` (id == name) in the live `config.sqlite app_config` so ccr's target-adapter registry resolves a routable adapter — the synthesized `provider-helixllm-*` id had mismatched the routing `name` ("Target adapter is not registered for provider"). Reload via the gateway `restartGateway` RPC (graceful, daemon stays up).
  - **Facade stability across `claude-providers sync`:** `detect_helixagent_record()` in `claude-providers.sh` + non-secret provider pins (`scripts/providers/helixagent.json`: base `:3456/v1`, model `HelixAgent/HelixLLM`, context 24576).
  - **ccr-self-loop guard (`_cma_ccr_self`).** The HelixAgent facade points ccr at its own `:3456` gateway; the guard prevents an infinite self-referential launch loop.

## Fixed

- **128k output-token clamp for every provider alias, on both transports.** `CLAUDE_CODE_MAX_OUTPUT_TOKENS` is clamped to ≤ 128000 for every provider alias on the native AND router transports (deepseek 384000 / xiaomi 131072 → 128000) via a single unified export site preceded by an unconditional unset (no stale inheritance, no competing paths). Reconciled with the output-≥-context skip guard into one coherent block; covered by `test_128k_output_clamp.sh` (38 assertions green in the 27/27

suite) and independently reviewed (Fable) as strictly safer than either input — no code path exports > 128000; zero/non-numeric budgets no-export rather than exporting a raw value.
- **ccr launch-grammar fix.** ccr v3.0.6 renamed the launch subcommand; the provider-alias launcher now invokes `ccr default-claude-code -- "$@"` (was `ccr code`) so router aliases launch Claude Code correctly, with a migration marker so the fix redeploys into the installed alias file.
- **zsh `fi fi` parse fix** in the native-launch guard.

### Testing / validation

- Full toolkit test suite **27/27 green** post-install (incl. `test_128k_output_clamp.sh`); the standalone `verify_helixagent_test.sh` verifier **46/46**.
- **Live retest (2026-07-19):** native `claude1-4` provider-env-isolated; 10 providers verified (incl. `helixagent`); HelixAgent → HelixLLM live route `HELIX_ROUTE_OK`; `claude-providers list helixagent verified`; ambient `~/.claude/settings.json` sha256 unchanged (no session hijack).
- **Merge review (Fable, xhigh) GO** on the main-integration reconciliation (control-needle-validated conflict-marker scan; migration-instrument provably-can-see).
- Note: v1.17.0 was documented in this changelog but never git-tagged; its changes land on `main` via the same merge and ship in this release.

## v1.17.0 — 2026-07-18 — Cross-alias session & background-agent continuity

### Fixed

- **A session left under one alias was invisible from every other alias.** Opening a project from `xiaomi`, leaving the session with work in progress, and then opening `deepseek` (or any other alias) showed nothing — two independent root causes, both fixed and live-proven:
  1. **Session resolution applied only to the native transport's bare launches.** Every router alias (kimi-*, poe, openrouter, chutes…) always opened a FRESH session, and `alias -p "…"` on any transport skipped resolution entirely ("explicit args win verbatim"). **Fix:** `_cma_session_flags` now runs before the transport split — both transports resume — and conversation args get `--resume <sid>` injected unless the user already chose a session (`--resume`/`--session-id`/ `--continue`/`-c`/`--fork-session`) or invoked a non-conversation subcommand (`agents`, `mcp`, `export`, …). Injection uses the new `claude-session existing-id`, which returns a session only when one actually EXISTS

— the older `latest-id` falls back to a deterministic UUID for never-used projects, and resuming that fails hard with "No conversation found with session ID". **Live proof:** `xiaomi` (native) created a session in a test project; `kimi-k3` (router) and `deepseek` (native) both resumed the identical session id and answered from its memory.

2. **Background agents were registered per config dir.** Claude Code's background-agent registry (`daemon/roster.json` + `dispatch/`, plus the `jobs/` store) was local to each alias dir, so an agent started under `xiaomi` was invisible to `deepseek`. **Fix:** `daemon` and `jobs` are now shared items (`CMA_SHARED_ITEMS` + unify's `SHARED_ITEMS`, drift guard enforced), `daemon/roster.json` is **union-merged** by `cma_union_rosters` (newer `updatedAt` wins per worker — a per-file last-wins would drop other aliases' workers), and `cma_migrate_daemon_dirs_once` merges every existing provider daemon dir into the shared store (roster content stashed before the backup move — the first cut of the migration unioned paths that had already moved), replaces it with the shared symlink, and is idempotent via a marker file. Live-verified: rosters from four provider dirs unioned into one shared registry; all provider daemon dirs are now symlinks.

### Added

- **Testing:** `scripts/tests/test_session_flags.sh` (12 assertions — both transports resume on bare launch, `-p` injection, no double-injection, subcommand passthrough, empty-session case, migration trigger); `test_unify.sh` daemon section (roster union semantics, dir linking, jobs, provider-dir migration with backup + idempotency — 107 assertions in the file). Challenges/HelixQA: check 7 in `provider_aliases_challenge.sh` (shared items, union merge, flags placement, existing-id, live daemon symlinks — 23/23 PASS live) and bank case `cma-pav-cross-alias-session-continuity` (bank now 12 cases).

## v1.16.0 — 2026-07-18 — Output-token cap for router providers

### Fixed

- **"Claude's response exceeded the 128000 output token maximum" on router providers.** `CLAUDE_CODE_MAX_OUTPUT_TOKENS` was exported **only on the native transport path** — every router provider (kimi-k3, poe, openrouter, …) launched with Claude Code's generic default output cap (128000 for models it does not know), so long reasoning responses died with that API error. **Fix:** the cap is now exported for **both** transports, before the transport split

(`_cma_out_guard`), valued from the provider model's real `limit.output` (`CMA_PROVIDER_MAX_OUTPUT`, e.g. 131072 for k3). Live-verified on the real wrapper: a `kimi-k3` router launch now carries `CLAUDE_CODE_MAX_OUTPUT_TOKENS=131072`.
- **Catalog-conflation guard:** models.dev sometimes reports `limit.output == limit.context` (the "output" number is really the context size) or an inflated `limit.context` (nvidia5's llama-3.2-11b-vision: catalog claims 1M context; the provider's own error says 131072). Exporting such a cap makes Claude Code request more completion tokens than the shared window allows → `400 maximum context length … you requested N`. The guard exports **only a genuinely separate output budget** (`output < context`). `nvidia5` was removed (its catalog metadata is wrong on both axes; backed up — self-heals via `sync --multi`).
- **Wrapper migration now covers the output-cap fix** — `_cma_out_guard` was added to the `cma_run_provider` self-heal marker chain, so wrappers written before v1.16.0 regenerate automatically on the next `install.sh`/sync/shell start (otherwise hosts would keep the native-only export indefinitely).
- **e2e tool-call flake:** `alias_e2e_test.py`'s tool probe gets the same one-retry policy as the layer-1 verifier (weak free models occasionally skip an instructed tool call — opencode2 false-FAILed a proof leg).
- **Router launch prompted interactively on some shells.** The wrapper's ccr-config `mv` ran as a bare command; on hosts whose interactive shell aliases `mv='mv -i'` (Fedora/RHEL defaults) the alias launch stopped at `mv: overwrite '…/config.json'?` and hung or died on redirected stdin. All 16 `mv` sites in lib.sh/emitted wrappers are now `command mv -f` (bypasses aliases and shell functions, forces the overwrite).
- **Cryptic failure when another tool named `ccr` shadows the router.** On a host where `ccr` resolved to a different program (a CCS-style profile manager), `ccr code` meant "launch profile 'code'" and the alias died with `Profile "code" was not found or is disabled`. The router path now verifies `ccr version` names the real `@musistudio/claude-code-router` and refuses early with an actionable message (fix PATH / remove the shadowing ccr / install command).

## Added

- **Testing:** new hermetic suite `scripts/tests/test_output_tokens.sh` (8 assertions — native + router export, empty-limit case, guard-before- split structure, migration trigger). Challenges/HelixQA: new bank case `cma-pav-output-cap-both-transports` and Check 6 in `provider_aliases_challenge.sh` (16/16 PASS live). Full suite: 24/24 files ALL GREEN; 6-leg proof ALL GREEN.

# v1.15.0 — 2026-07-18 — Full Kimi variant support (OAuth subscription models)

## Added

- **Every Kimi model the OAuth subscription serves is now a launchable alias.** `detect_kimicode_record` discovers the served models live (`GET {base}/models` with the OAuth token, unioned with the models.dev catalog since the listing under-reports) and emits one provider record per model — the old code exposed a single hardcoded alias. On this host: `kimi-for-coding` (account default, "K2.7 Coding"), `kimi-for-coding-highspeed`, `kimi-k2p7` (Kimi 2.7), and **`kimi-k3` (Kimi 3 — 1M context, reasoning)**. Every record goes through the same strict sentinel + tool-calling + semantic verification as every other provider.
- **`kimi_proxy.py`** — moonshot-flavored schema normalizer routed under every `kimi-*` alias (new `<family>_proxy.py` discovery rule in the launch wrapper). Model k3 rejects any tool whose `parameters` carries a `$ref` not starting with `#/$defs/` (`400 … not a valid moonshot flavored json schema`, reproduced live); Claude Code's tool schemas trip exactly that. The proxy hoists `$defs`+`definitions`, rewrites foreign `$refs` by last segment, and guarantees `parameters.type/properties`. Live proof: direct request → 400, same request through the proxy → 200.
- **Launch-time OAuth token freshness** (lib.sh emitted wrapper). The OAuth token lives ~15 minutes, so the old sync-time snapshot 401'd by the next launch — the root of "kimi-for-coding works once, then dies". Freshness order at every launch: unexpired live credentials file → CLI-triggered refresh (`kimi -p hi`) → token-file snapshot (last resort).
- **API-key paths for kimi.com coding** — `KIMI_API_KEY` (catalog env) and `ApiKey_Kimi` (new `key-aliases.json` entry) both resolve to the `kimi-for-coding` provider as a fallback for hosts without an OAuth session. OAuth subscription records take **precedence** over API-key records (`unique_by` merge, detector first) — the subscription is the priority path.
- **`sarvam_proxy.py`** — Sarvam compatibility proxy (same family-discovery mechanism). Three distinct runtime incompatibilities were root-caused and fixed, each reproduced live first: system/user message content arrays (`400 … Input should be a valid string` — flattened to joined strings) and Claude Code's 64000-token output default exceeding the starter tier's 4096 cap (`max_tokens` now clamped, overridable via `SARVAM_MAX_OUTPUT_TOKENS`). Result: the `sarvam` alias went from guaranteed-400 at launch to a real Claude Code PASS.

- **Challenges/HelixQA**: three new bank cases (`cma-pav-kimi-oauth-token-freshness`, `cma-pav-kimi-multi-model-oauth-records`, `cma-pav-kimi-k3-moonshot-schema-proxy`) and Check 5 in `provider_aliases_challenge.sh` (detector discovery, precedence, freshness, schema proxy, live kimi-alias freshness) — 15/15 PASS live.

## Fixed

- **`claude-providers verify <id>` was unusable for OAuth providers** — it never injected the OAuth token, so verify-by-id always degraded to a false `unverified`. It now applies the same live-cred-file-first freshness order.
- **Layer-1 probes false-FAIL reasoning models** — 128-token budget was consumed entirely by chain-of-thought (k3, deepseek-v4-pro), yielding "empty content / sentinel missing" failures on working models. Probe budget is now 512 tokens.
- **Detector jq ARG_MAX overflow** — the full models.dev catalog was passed as a `--argjson` argument; only the `kimi-for-coding` models subtree is passed now (the bug silently yielded zero OAuth records and let an API-key record shadow the subscription).
- **Wrapper self-heal migration did not cover the new wrapper features** — the `cma_run_provider` migration markers stopped at v1.14.0, so every host upgrading kept a stale wrapper that (a) never started `kimi_proxy` for `kimi-*` (k3 400'd on every tool call — live-confirmed) and (b) never refreshed the OAuth token (401 after ~15 min). The `_family_id` and `kimi-code/credentials/kimi-code.json` markers now trigger regeneration.
- **Live legs blind to OAuth + overcounted account states** — `verify_aliases_live.sh` silently skipped OAuth aliases ("no key"), had no CLI-refresh fallback (stale snapshots → 400/401), lacked the family proxy discovery (kimi `$ref` tests 400'd), used 32/64-token budgets that false-FAIL reasoning models (poe's claude-sonnet-4.6 needs 512 to reach the tool call — proven live), and FAILed on account limits (weekly cap, fair-use 1313) instead of classifying them. `alias_e2e_test.py` had the same OAuth-key and staleness gaps. Both legs now: resolve the OAuth token through the full live-cred → CLI-refresh → snapshot chain, discover family proxies, use 256/512 budgets, and classify **SKIP-QUOTA** / **SKIP-TRANSIENT** / **SKIP-GATED** (aliases the gate already filtered) honestly instead of as passes or failures.

## Testing

- New hermetic suite `scripts/tests/test_kimi.sh` (**33 assertions**): detector multi-record emission (live discovery + catalog union + offline fallback), expired-token CLI refresh,

OAuth-first precedence + no duplicates, API-key fallback mapping, launch-time token freshness (all three sources), `cmd_verify` OAuth injection, `kimi_proxy` schema normalization (5 cases), family proxy discovery markers. Full suite: 22/22 files ALL GREEN.
- Live (Kimi aliases FIRST, per release gate): all 4 aliases verified by layer-1 probes, layer-3 semantic (sentinel + independent judge), and real Claude Code launches through the aliases.

# v1.14.0 — 2026-07-17 — Anti-bluff provider verification (strict filtering)

## Fixed

- **Provider aliases passed verification while being broken at runtime.** Aliases such as `huggingface` were marked `verified` yet any real prompt ("Do you see my codebase?") returned API failures (402 depleted credits, suspended accounts, unsupported models). A live sweep showed **12 of 19 aliases failing while `status.json` called them verified**. **Root causes (all fixed):**
  1. *Layer-1 existence check was a bare `GET /v1/models`.* HTTP 200 proved only that the key was accepted — never inference, the selected model, or tool calling. **Fix:** `scripts/providers-verify.sh` now runs two live probes against the provider's chat endpoint with the exact alias model: a sentinel probe (response MUST contain `VERIFY_OK`; 200-without-sentinel or error-in-200 is a bluff ⇒ `failed`) and a tool-calling probe (the model MUST emit a real tool call — Claude Code is tool-driven). Definitive rejections (401/402/403/404) ⇒ `failed`; only transient conditions (429/5xx/timeout/no-network) ⇒ `unverified`. Anthropic-native endpoints are probed in their native `/v1/messages` shape.
  2. *Multi path (`sync --multi`) verified chat-only models.* `model_verify.py` never asserted the `VERIFY_OK` sentinel it requested, set `verified=True` unconditionally, and counted tool calling as zero required points (`MIN_SCORE=25` == existence weight alone). **Fix:** sentinel is asserted (missing ⇒ anti-bluff failure), `verified` now **requires** a passed tool-calling probe, and the verification cache carries `_cache_version` so results from the old logic are never replayed.
  3. *Billing/auth failures were classed as transient.* The semantic code-visibility layer exited 3 ("infra — honest SKIP, never downgrade") on HTTP 402/401, so a credits-dead provider kept its stale `verified`. **Fix (LLMsVerifier submodule):** `semantic-code-visibility` now maps

401/402/403/404 on model-under-test calls to exit 1 (genuine negative — demotes), keeps 429/5xx/timeout as exit 3, and keeps judge-call failures always at exit 3 (a broken judge never demotes the model under test).

4. *The live alias verifier ignored tool calling.* `verify_aliases_live.sh` recorded "no tool call" but never failed on it — aliases passed with tool-less models. **Fix:** test 6 now uses an instructed tool call and is verdict-relevant.
5. *The runtime-shaped e2e test was orphaned.* `alias_e2e_test.py` (tools, `$ref`, `cache_control`, streaming through the real endpoint) was never invoked by anything. **Fix:** wired into `run-proof.sh` as leg 44 with an honest SKIP (exit 3) when no providers/network are present.
6. *Probe URLs were mis-normalized for real provider bases.* Anthropic-native bases had their `/anthropic` prefix stripped (DeepSeek/Xiaomi 404'd on `…/v1/messages` when the served path is `…/anthropic/v1/messages`), and already-versioned bases (`…/paas/v4` on Z.AI/BigModel coding plans) got a bogus `/v1` inserted. **Fix:** the prefix is kept for native probes, versioned bases get only `/chat/completions` appended — in both `providers-verify.sh` and LLMsVerifier's `semantic-code-visibility` (`chatCompletionsURL` rule, 11 new Go tests).
7. *Single-attempt probes flapped on transient conditions.* Load-balanced gateways return occasional 400/404/412/000, and weaker models non-deterministically miss the sentinel or skip a tool call — working providers (kilo, siliconflow, sarvam) flipped to `failed` between syncs. **Fix:** exactly one retry for flappy codes and for flaky sentinel/tool-call outcomes; auth/billing codes (401/402/403) and consistent bluffs are never retried.
8. *Layer-4 superpowers-TUI marker was model-phrasing-dependent.* Genuinely engaged sessions whose model summarized the framework in its own words (instead of the exact `superpowers:<name>` announce string) were failed. **Fix:** the marker also accepts vocabulary that can only exist when the skill content loaded (`systematic-debugging`, `brainstorming`, the invoke-before-response rule) — echo/refusal bluffs still cannot pass — and any transcript with `"is_error":true` is a hard FAIL (real API errors can't masquerade as engagement).
9. *Poe aliases failed real Claude Code launches through the proxy.* `poe_proxy.py` injected `parameters` only when missing/null, but Poe actually requires the `properties` key — Claude Code's zero-argument tools (`{"type":"object"}` with no `properties`) were rejected with the misleading `400 Invalid 'tools': Field required` (root cause reproduced and verified live against api.poe.com). **Fix:** `fix_tools` guarantees `parameters.properties` (and `type:object`) on every tool.
10. *The live legs produced false FAILs on account/provider states and native transports.* `verify_aliases_live.sh` sent OpenAI-shaped Bearer requests to `/anthropic` bases (deepseek/xiaomi hung or 400'd) and had no retries, so

single 000/429 wobbles failed working aliases; `alias_e2e_test.py` gave reasoning models (deepseek-v4-pro) a 128-token budget that reliably came back empty, and double-appended `/chat/completions` on versioned bases. **Fix:** both legs speak native Anthropic shape under the kept `/anthropic` prefix, versioned bases get exactly one `/chat/completions`, reasoning models get a 512-token budget with one empty-answer retry, and both legs now classify honestly: **SKIP-QUOTA** (account funds: 402/insufficient_quota/…) and **SKIP-TRANSIENT** (provider capacity/timeouts/5xx) are reported separately and never counted as passes or toolkit failures — the same FUNDS distinction `verify_claude_live.sh` already made. Genuine FAIL (auth, schema, bluff, no tool call after retry) stays FAIL.

11. *A live API key leaked into committed proof evidence.* `opencode debug config` embeds MCP server env (`"TAVILY_API_KEY": "tvly-dev-…"`), and `cma_redact_secrets` only matched keys literally named `apiKey|api_key|password|secret|token` — env-style names and the `tvly-`/`nvapi-` prefixes slipped through into `proof/10-debug-config.json`. **Fix:** the redactor now matches any JSON key NAME containing key/token/secret/password/api-key and covers the extra prefixes (`${VAR}` placeholders still preserved); extended hermetic tests in `test_coverage.sh`; full re-scan of `proof/` is clean.

## Added

- **Tier C constitution/conformance verifier** `scripts/tests/verify_constitution.sh` (design spec §7.4, implemented at `scripts/tests/` alongside the other live verifiers): CONST-051 submodule decoupling, §11.4.157 four-file doc lockstep, §11.4.113 no force-push, §11.4.156 CI/CD disabled, §11.4.151 release-tag prefix, toolkit-owned fixture/rubric independence. Wired into `run-proof.sh` as leg 45 — the proof suite is now six legs: sandbox tier A, OpenCode live, providers live, aliases live, alias e2e, constitution tier C.
- Hermetic test coverage: `test_verify_scripts.sh` grew from ~30 to 69 assertions (sentinel/tools/402/429/bluff/Anthropic-shape/cache-version); new `test_constitution.sh`; `test_providers.sh` loopback servers now answer the real chat+tools probe shapes.
- **Challenges submodule:** new live anti-bluff challenge `challenges/scripts/provider_aliases_challenge.sh` (static pipeline-honesty checks + `status.json` freshness/consistency, SKIP-OK when the toolkit is absent) and HelixQA-compatible bank `banks/examples/provider-alias-verification.json` (6 test cases), both registered in the `docs/test-coverage.md` ledger (describe-runner 25/25).

### Changed

- **Live filtering results (2026-07-17, all layers + real Claude Code launches):** every installed alias was re-verified with the strict pipeline and the ones that could not pass were removed (config dirs backed up as `~/.claude-prov-<id>.preunify.*`; re-adding is just `claude-providers sync` once the account is fixed). Removed: `huggingface` (+2,3), `openrouter5`, `github-models`, `fireworks-ai`, `inference`, `novita-ai`, `sarvam`, `tencent-tokenhub`, `upstage`, orphans `zai`/`zhipuai`, stale `chutes4`, `xiaomi2`/`xiaomi3`, rate-limited/unusable `zai-coding-plan` and `zhipuai-coding-plan` (429 fair-use limited / all-models-404 right now — honest removal, self-heals on the next sync), plus runtime-broken `kimi-for-coding2` (k3 rejects Claude Code's `$ref` tool schemas) and `poe3` (gemma-4-31b aborts streams), and `openrouter4` (nvidia/nemotron-nano-12b-v2-vl returns empty/malformed responses). **Final state: 28 aliases installed, every one `verified` through the full strict pipeline** (statuses, env files, and alias lines exactly consistent). `overrides.json` pins were corrected to models verified live today (openrouter, nvidia added; xiaomi fast model fixed to a served one).
- **LLMsVerifier submodule — strict scoring, no more bluff headroom:** removed the hard `VerificationScore = max(score, 0.7)` floor (every strict gate was vacuous); `CodeVisibility` confidence threshold 0.3 → 0.5 with rebalanced weights (a bare "Yes, I can see it" now scores 0.4 < 0.5); keyword matching uses `\b` word boundaries ("no" no longer matches "know"/"not"); round-1 sentinel check gained a prompt-echo guard (≥60-char verbatim fixture slice in the reply ⇒ genuine fail); the dead hardcoded HuggingFace endpoint `api-inference.huggingface.co` was replaced with `https://router.huggingface.co/v1` in all production paths. Full Go suite: 59 packages green.
- `docs/Provider_Aliases_User_Guide.md` updated to the new verification semantics (§3 command table, §5 multi-alias verification, §7 rewritten).

# v1.13.3 — 2026-07-17 — Session-sharing pipefail fix

### Fixed

- **All aliases created separate, unshared sessions for the same project.** Switching between `claude1`, `claude2`, `deepseek`, `xiaomi`, etc. in the same project directory started a fresh conversation instead of resuming the shared one — no memory, context, or history continuity across aliases. **Root cause:** `claude-session.sh` has `set -o pipefail`. When

`cma_latest_session_id()` scanned for existing sessions with `ls -t ... | grep | head -1`, `head` closed stdin after one line, sending SIGPIPE to `grep`. With `pipefail`, the pipeline exited 141, and `set -e` aborted the script BEFORE it could return the session UUID. Every launch was treated as a first run, creating a new random-UUID session. **Fix:** Added `|| true` guard on the `head -1` pipeline in `cma_latest_session_id()` (`scripts/claude-session.sh` line 111). The guard catches SIGPIPE without affecting the captured output — `head` already printed the first line before exiting, so `latest` gets the correct UUID. All aliases now resolve to the same `--resume <uuid>`. **Verified with 200-file stress test** that exercises the exact pipefail condition.

### Added

- 200-file stress test in `test_session.sh` that proves `claude-session flags` returns `--resume` (not `--session-id`) even with many session files, and that `latest-id` returns the most recent session.
- Investigation document at `docs/investigations/2026-07-17-session-sharing-root-cause.md` with full root cause analysis, evidence, and architecture notes.

## v1.13.0 — 2026-07-10 — Project-scoped cwd-hook (session-resumption fix)

### Fixed

- **Sessions were resumed for the wrong project when switching Claude Code aliases.** When the global `~/.local/bin/claude-cwd-hook` symlink pointed to one project's multitrack resolver (e.g. atmosphere's), every `claudeN` alias launch was redirected into that project's worktree BEFORE `claude-session` resolved the session — so the session was keyed to the atmosphere track, not the project the user was actually working in (e.g. `helix_ota`). Switching from `claude4` to `claude1` resumed an atmosphere Track-4 session instead of the `helix_ota` session. **Root cause:** `CMA_CWD_HOOK` was a single global singleton with no per-project awareness; the hook fired unconditionally and the toolkit had no mechanism for a repo to supply its own resolver. **Fix:** `cma_run` now resolves the cwd-hook in a three-tier order:
  1. `CMA_CWD_HOOK` env var (explicit override, unchanged),
  2. `<git-toplevel>/.claude-cwd-hook` (per-project hook — each repo gets its own multitrack resolver; prints nothing → stay in `$PWD`),
  3. `~/.local/bin/claude-cwd-hook` (global fallback, backward-compatible). A repo that needs its own track layout drops a `.claude-cwd-hook` at its git root; a repo that doesn't simply

omits it and keeps the global hook. The migration self-heals outdated `cma_run` wrappers via a new `_cma_hook_root` marker (detected on next `install.sh` / shell start).

### Added

- Regression tests in `test_lib.sh` and `test_wrapper_exec.sh` prove the emitted `cma_run` body carries the project-scoped hook resolution code (`_cma_hook_root` marker, `git rev-parse --show-toplevel`, `.claude-cwd-hook` check) and respects the `CMA_CWD_HOOK` override + global fallback.

## v1.12.3 — 2026-07-05 — Session-name sanitization (kebab-case)

### Changed

- **Auto-derived session names are now sanitized to kebab-case.** `claude-session.sh` derives the session name from the project directory's basename. It now: lowercases the name; trims leading/trailing whitespace; collapses internal whitespace and underscores to `-`; strips any remaining characters that are not `[a-z0-9-]`; and collapses consecutive `-` before trimming leading/trailing `-`. This makes session names safe for the CLI and filesystem while remaining human-readable.

### Added

- New regression tests in `test_session.sh` cover leading/trailing whitespace, multiple consecutive spaces, and stripping of special invalid characters.

## v1.12.2 — 2026-07-05 — Native alias auto-registration + account-detection hardening

### Fixed

- **Native `claude<N>` aliases were never created for pre-existing account dirs.** Running `install.sh` or `claude-unify.sh` only merged shared state; if account directories already existed (e.g. `~/.claude-milos85vasic`), no shell aliases were registered, so `claude1`/`claude2`/etc. were undefined. `claude-

`unify.sh` now auto-registers a `claude<N>` alias for every detected account that lacks one.
- **Smart alias numbering:** an account dir literally named `~/.claude-claude4` keeps the `claude4` alias; remaining dirs fill the lowest free `claude<N>` slot instead of skipping over reserved numbers.
- **Bogus account detection:** `~/.claude-code-router` and `~/.claude-*.lock` directories are no longer treated as Claude accounts, preventing them from stealing alias slots or being merged into shared state.

### Added

- Regression tests in `test_unify.sh` and `test_install.sh` prove that install/unify register native aliases for existing account dirs and preserve `claude<N>` basenames while filling gaps.

## v1.12.1 — 2026-07-05 — Judge independence + resolve/robustness hardening

Addresses the v1.12.0 final whole-branch review's deferred items and the deep-research findings on LLM-as-judge bias.

### Changed

- **Round-2 judge default is now a DIFFERENT model family.** `providers/judge.env.template` defaults to Groq / Llama-3.1 (`llama-3.1-8b-instant`) instead of DeepSeek. 2024-2026 research (arXiv:2508.06709 and others) shows a judge systematically favors its own model family — including validating that family's *wrong* answers, the exact failure layer-3 exists to catch — so defaulting the judge to a common subject (DeepSeek) was the worst case. Verified working live as an independent judge.
- **The semantic command distinguishes transport/infra failures from genuine verdicts.** The LLMsVerifier `semantic-code-visibility` command now exits **3** when a round-1/round-2 API call cannot complete (non-2xx, timeout, empty, connection error), vs exit **1** for a real negative verdict. `providers-semantic.sh` maps exit 3 → `skip`, so a transient judge/model hiccup is an honest SKIP and never downgrades the model-under-test (final-review I-1).

### Added

- **Independence warning:** `providers-semantic.sh` warns (never fails) when the judge endpoint equals the model-under-test endpoint (same provider = same family = not independent).
- **xAI is now resolvable:** `providers/overrides.json` gains `xai → https://api.x.ai/v1` (the catalog lists xAI with no API base, so it previously resolved `unmapped`). Endpoint confirmed live.
- CONST-069 capability-boundary mandate in the LLMsVerifier submodule constitution (records the CONST-051 project-agnostic boundary under a non-colliding id).

### Fixed

- A directory passed as the keys file (`CMA_KEYS_FILE` / `--keys-file`) now dies with a clear message instead of silently yielding "0 key vars".

## v1.12.0 — 2026-07-05 — Semantic code-visibility (layer 3) + live TUI verification (layer 4)

Adds two new provider-verification layers on top of the Phase-1 existence/tool-call checks, driven by the LLMsVerifier submodule's standalone, stdlib-only `semantic-code-visibility` Go command. The whole pipeline was proven end-to-end against real providers (redacted, real network): a genuinely code-seeing model (mistral-medium-2604) verifies (round-1 sentinel + round-2 judge 2/3); models that bluff or fail round-2 (chutes GLM-5.2-TEE — empty/timeout) are `unverified`; billing blocks (deepseek 402) are `unverified` — never a faked pass.

### Added

- **Layer 3 — semantic code-visibility.** `scripts/providers-semantic.sh` renders the toolkit-owned rubric (`providers/rubric/code-visibility-rubric.json`) into a judge prompt and drives the Go command through a build-and-cache driver (`scripts/claude-semantic-visibility.sh`). Two rounds: a unique sentinel embedded in a code fixture (does the model actually see the code?), then an independent LLM-as-judge score (≥ threshold). One-word contract `verified|unverified|skip` (exit 0/1/2). Wired into `cmd_sync`; keys move via env only (never argv). CONST-051 boundary held — the submodule stays project-agnostic, receiving fixture/prompt/ judge-prompt/sentinel only as CLI args.
- **Layer 4 — live superpowers-TUI verification.** `scripts/verify_superpowers_tui.sh` launches real Claude Code through a

provider alias and confirms the superpowers plugin engages with no trust/overwrite prompt — the only thing that flips a provider to fully `verified`. Honest SKIP when preconditions (real claude/key/network) are absent; the engagement classifier is hardened against false-PASS on prompt-echo, and a live negative-case test (neutral prompt → marker must NOT match) proved it does not false-verify.
- **`claude-providers verify <id> [--deep]`** — single-provider deep re-verify.
- **Tier-B live verifier** (`scripts/tests/verify_providers_live.sh`) runs layers 3–4 per installed provider, writes secret-redacted evidence + an aggregate `proof/providers-summary.json`; already wired into `run-proof.sh`.

## Fixed

- **`--refresh-aliases` was not byte-idempotent** — `cma_provider_write_alias` re-ran the full `cma_ensure_alias_file` self-heal migrations on *every* alias line, which could non-deterministically reposition the `cma_run_provider` function and produce a different file on a second refresh (the session hook runs this on every shell). Now only bootstraps the alias file when absent. (Root-caused from a captured diff; 42 flake-loops + repeated full-suite runs now deterministic.)
- Three bugs found by real execution of the new layer-3 path: `claude-semantic-visibility.sh --help` failed pre-build; `providers-semantic.sh` discarded the driver's JSON evidence to `/dev/null`; the judge base URL was not normalized (a trailing `/v1` doubled to `/v1/v1/chat/completions`).
- Final-review hardening: `cmd_sync`/`cmd_sync_multi` heal a stale/outdated `cma_run_provider` wrapper once per sync (the byte-idempotence change had removed the per-shell self-heal, so a pre-Phase-2 gate-less wrapper could still launch); the layer-4 engagement classifier now requires the namespaced `superpowers:<name>` form (a bare skill word could false-verify); and `verify --deep` treats a layer-4 verifier crash as an honest SKIP rather than a clean verify.

## Notes

- The default `providers/judge.env.template` judge is DeepSeek. For strongest results, configure `providers/judge.env` with a judge from a **different model family** than the provider under test — 2024-2026 research (e.g. arXiv:2508.06709) shows same-family judges systematically favor their own family's outputs, including validating wrong answers. A different-family default, an xAI `overrides.json` base-url entry, and the submodule boundary-contract constitution id are tracked for a follow-up.

# v1.11.2 — 2026-07-03 — Token-limit guard + re-enabled provider proxies + Poe tool cap

Ran every provider alias through REAL Claude Code executing `/using-superpowers` (the user's reproducing prompt), in CLI mode with scrubbed env, and root-caused every genuine failure. Three real toolkit bugs fixed (below); all remaining non-passes are account-side (insufficient funds, rejected/absent keys, or a key with no chat entitlement) — not toolkit defects.

## Fixed

- **Input token-limit 400** — `scripts/lib.sh (cma_run_provider)` now exports `CLAUDE_CODE_AUTO_COMPACT_WINDOW="$CMA_PROVIDER_CONTEXT_LIMIT"` before the transport branch, so Claude Code knows each provider's real input window and auto-compacts (at `window − 13000`) before a request overshoots it. Fixes the user-reported `400 … exceeded model token limit: 262144 (requested: 311786)` on `kimi-for-coding` and any provider whose window is smaller than the ~1M Claude Code assumes for Anthropic's endpoint. Fully dynamic/parametrized: the value is the per-model `limit.context` from the models.dev catalog, persisted in each provider `.env`. Guarded by `[[ -n … ]]`; applies to **both** transports. (`CLAUDE_CODE_MAX_OUTPUT_TOKENS` only ever capped **output** — it could not fix an **input** overflow.) Verified live: kimi's `/using-superpowers` now compacts and succeeds.
- **All provider proxies were silently disabled** — `cma_run_provider` resolved compatibility proxies via `$LIB_DIR/proxy/…`, but `LIB_DIR` is a repo-only variable that does not exist in the self-contained alias file, so it expanded empty and **no proxy ever started** (Poe, etc.). Now resolves against the installed location `${SHARED_DIR:-$HOME/.claude-shared}/proxy` (where `install.sh` copies `scripts/proxy/*.py`). This is why Poe returned `400 Invalid 'tools': Field required` — its request never went through the proxy that injects the required `parameters` field.
- **Poe tool-count limit** — Poe rejects requests carrying more than ~216 tool definitions with the same misleading `400 Invalid 'tools': Field required` (verified count-based: 215 accepted, 220 rejected, independent of payload size). On this host Claude Code's large MCP-plugin load emits 400+ tools. `scripts/proxy/poe_proxy.py` now caps the tool list to `POE_MAX_TOOLS` (default 200), dropping only overflow `mcp__…` tools so **every** built-in Claude Code tool is preserved. Parametrized via the `POE_MAX_TOOLS` env var. Verified live: Poe's `/using-superpowers` now returns a successful result.

### Changed

- **`cma_ensure_alias_file` migration** — added `CLAUDE_CODE_AUTO_COMPACT_WINDOW` and `_cma_proxy_dir` as regeneration markers, so an already-installed `cma_run_provider` predating either fix is transparently regenerated on the next alias-file touch.
- **scripts/tests/verify_claude_live.sh** — `reclassify_fail` now maps a direct `400` whose body says the model is unknown/invalid to **BADKEY** (an account that can list models but not invoke them — e.g. `inference` here), while a `400` with no model-rejection marker (a real launch-layer defect) stays FAIL and is never masked.
- Corrected misleading comments that claimed the output-token cap fixed the input token-limit error; documented the two guards as independent halves.

### Added

- **scripts/tests/test_poe_proxy.sh** — hermetic tests for the Poe proxy: `parameters` injection, the tool-count cap, built-in-tool preservation, the `POE_MAX_TOOLS` override, and end-to-end `fix_request`.
- **scripts/tests/test_providers.sh** — hermetic regression tests: the emitted `cma_run_provider` exports the auto-compact window from `CMA_PROVIDER_CONTEXT_LIMIT` only when non-empty; the migration regenerates an outdated wrapper lacking the guard while preserving surrounding alias lines.
- **scripts/tests/proof/claude-live-superpowers-cli.txt** — evidence matrix: every provider alias launched through REAL Claude Code running `/using-superpowers`, classified PASS / FUNDS / BADKEY / NOKEY / FAIL.

## v1.11.1 — 2026-07-03 — Live per-alias Claude Code verification (CLI + TUI) + provider fixes

Investigated the reported "most provider aliases fail with an API error." Root cause was NOT a broad implementation defect: launched every alias through real Claude Code (scrubbed env) and probed each provider API directly. 9 aliases pass end-to-end; 2 had genuine model-config drift (fixed below); the remaining failures are all **account-side** (insufficient funds, invalid/expired keys, missing keys, or a plan with no chat model) — not toolkit bugs.

## Added

- **scripts/tests/verify_claude_live.sh** — end-to-end live verification of every provider alias through REAL Claude Code in BOTH modes: **CLI** (`-p … --output-format json`, authoritative) and **TUI** (driven under a PTY). Each launch runs in a scrubbed env; TUI runs from a throwaway temp cwd so it can never resume a real conversation (the cross-alias `.claude.json` sync would otherwise auto-resume one). Outcomes are classified **PASS / FUNDS / BADKEY / NOKEY / FAIL** so account problems are never miscounted as toolkit bugs; on a launch FAIL it probes the provider API directly to recover the true cause (e.g. a ccr hang on an upstream 401/429 → BADKEY/FUNDS, not FAIL).
- **scripts/tests/lib/pty_drive.py** — pexpect PTY driver for the interactive Ink TUI (boot, accept any trust prompt, type prompt, capture transcript, quit).
- **scripts/tests/lib/classify_live.py** — shared transcript classifier.

## Fixed

- **huggingface** — strong/fast pinned to non-reasoning coder models (`Qwen/Qwen3-Coder-480B-A35B-Instruct` / `Qwen/Qwen3-Coder-30B-A3B-Instruct`). The previous models emitted their answer as hidden `reasoning_content`, returning empty `content` at low `max_tokens` and reading as broken. **Verified green** in CLI + TUI.
- **kilo** — strong/fast pinned to verified free-tier models (`nvidia/nemotron-3-super-120b-a12b:free` / `nvidia/nemotron-3-nano-omni-30b-a3b-reasoning:free`). The old `x-ai/grok-build-0.1` is a **paid** model this key can't access (401 →

  > 100s stall) and the old fast `baidu/cobuddy:free` is retired (404). **Verified green** in CLI + TUI.

- **inference** — base URL corrected `https://inference.net/v1` → `https://api.inference.net/v1` (old host 301-redirects). NOTE: this key's plan currently exposes no general chat model, so the alias still errors 400 until a chat-capable plan/key is supplied — documented, not silently "fixed."
- **novita-ai** — defensive swap off the retired fast model `sao10K/L3-8B-stheno-v3.2` (404).
- **claude-providers.sh (`present_key_vars`)** — skip declared-but-**empty** key vars so an empty key (e.g. `SARVAM_API_KEY=`) no longer spawns a broken alias that only errors at launch.

### Verified (live on host)

- **9 aliases PASS** end-to-end in CLI (and TUI smoke): chutes, huggingface, kilo, kimi-for-coding, nvidia, opencode, poe, siliconflow, zai-coding-plan.
- Non-PASS are **account-side, not toolkit bugs**: FUNDS — deepseek, fireworks-ai, novita-ai, openrouter, upstage, xiaomi, zhipuai; BADKEY — github-models, tencent-tokenhub; NOKEY — sarvam; plan-limited — inference.
- Sandbox providers suite **113/113**; shellcheck clean; python files compile.

## v1.10.8 — 2026-07-01 — noclobber-safe router-provider config write

### Fixed

- **Every router-transport provider alias broke under `set -o noclobber`.** `cma_run_provider` runs in the user's interactive shell; when that shell has `noclobber` set, the router-config rewrite `jq … "$cfg" > "$tmp"` failed with "*cannot overwrite existing file*" (the just-created mktemp target already exists), silently dropping the update so claude-code-router launched with a stale/empty config → API errors. Switched the write to the force-clobber operator `>|`. (lib.sh)
- **Existing installs self-heal**: added a regeneration trigger so `cma_ensure_alias_file` rewrites an outdated `cma_run_provider` that lacks the `>|` fix (plain function-body changes previously did not re-trigger on install).

### Added (tests)

- test_providers.sh: regression asserting the emitted `cma_run_provider` uses `>|` (not a bare `>`) for the router-config write, plus a functional proof that `>` is blocked by `noclobber` while `>|` succeeds.

### Verified

- Suite **18/18 green**; shellcheck 0. Root cause reproduced (`set -C; echo > "$(mktemp)"` → "cannot overwrite"), fix deployed + confirmed on the host (deployed alias write line = `>| "$tmp"`).

### Note (not a toolkit bug)

- A 46-alias provider sweep shows native providers reach their APIs and return **account-side** errors (e.g. `402 Insufficient Balance`) — provider billing/key state, not a toolkit fault.

## v1.10.7 — 2026-06-29 — Shared-items drift guard + audit closure

### Added (tests)

- **test_unify.sh** — a drift guard asserting `claude-unify.sh`'s `SHARED_ITEMS` equals lib.sh's `CMA_SHARED_ITEMS` (used by `claude-add-account`) minus the intentional `CLAUDE.md` special-case (unify promotes it via `sync_claude_md`). Catches the documented hazard of adding a shared item to one list only.
- Cleaned a pre-existing SC2319 lint in test_unify.sh.

### Audited — no code changes (3 parallel investigators; every finding independently checked)

- **Unify + rollback engine: clean** — enabledPlugins union, history.jsonl dedup, settings.json merge (malformed-input safe), plugin-manifest path rewrite, and rollback all verified correct across 170+ assertions + edge cases.
- **OpenCode sync (`opencode_sync.py`): clean** — idempotent + additive verified; two reported "bugs" were refuted against the code: the `setdefault` "can't update existing keys" IS the documented never-clobber design, and the `--enable-all` secret nit is not a security issue (an unresolved secret can't leak) and matches "enable everything" semantics.
- **Runtime sync + add-account: clean** — the `SHARED_ITEMS` "drift" is the intentional `CLAUDE.md` special-case (now locked by the guard above); sync-state private-key isolation, corrupt-input handling, and add-account idempotency verified.

### Verified

- Suite **18/18 green**; shellcheck 0.

# v1.10.6 — 2026-06-29 — Committed credential-leak regression test

## Added (tests)

- **test_lib.sh** — a committed security regression for `cma_merge_claude_json`: two accounts with distinct `userID`/`oauthAccount` and disjoint `projects` are merged; asserts each account keeps its OWN private auth keys (no cross-account leak in either direction) and the `projects` subtree is unioned both ways. The function previously had only indirect coverage (via the full unify workflow); this locks the property an audit verified by hand this session.

## Verified

- Suite **18/18 green**; shellcheck 0.

# v1.10.5 — 2026-06-29 — Provider 'null' field normalization + coverage

## Fixed

- **A missing JSON field could write `CMA_PROVIDER_MODEL='null'` (and `TRANSPORT`, alias name) into a provider env file**, launching the provider with a bogus model. `cma_provider_write_env` normalized `base`/`fast`/`context`/`max` "null"→empty but missed `model` and `transport`; `claude-providers` multi-sync also extracted `strong_model`/`transport`/`alias_name` with bare `jq -r` (no `// empty`, unlike the already-correct `context_limit`/`max_output`). Fixed at both the source (`// empty` on every extraction) and the choke point (normalize `model`+`transport` in `cma_provider_write_env`). Reproduced (`='null'`), confirmed fixed (`=''`).

## Added (tests)

- **test_providers.sh** — regression asserting a `null` model/transport/base/etc. is normalized to empty; no field ever contains the literal `'null'`.
- **test_session.sh** — EXECUTION tests for the `hint` subcommand (run on every bare launch, previously only string-matched) and for `cma_project_root`'s git-toplevel + symlink (`pwd -P`) branches.

### Audited — no change needed (independently verified, not taken on trust)

- `cma_merge_claude_json`: NO cross-account credential leak; `projects` unioned; corrupt input skipped gracefully (verified with crafted 2-account inputs).
- BSD/macOS portability: no unguarded GNU-isms (the 3 `readlink -f` hits are comments; `stat -f/-c` branch + `cma_realpath` guards present).
- jq robustness: the `@tsv` sync paths render null as empty (safe); two reported `2>&1` "error-leak" findings were FALSE — there is no `2>&1` on those lines.

### Verified

- Suite **18/18 green**; shellcheck 0.

## v1.10.4 — 2026-06-29 — set -e/pipefail abort fixes + hardened test coverage

### Fixed

- **`claude-providers list` / `remove` aborted on a provider with no alias line.** Under `set -euo pipefail`, the alias-name probe `grep … | sed | head -1` returns 1 (no match) when a provider's `.env` exists but its `alias` line is absent (manual edit / partial setup); pipefail propagated the failure and `set -e` killed the subshell (`list`) or the function before `rm -f` (`remove`). Guarded both with `|| alias=""`. (claude-providers.sh)
- **`cma_ensure_alias_file` aborted on an alias file lacking `export CLAUDE_BIN=`.** The CLAUDE_BIN-migration probe `grep -m1 '^export CLAUDE_BIN=' …` returned 1 on an older/hand-edited alias file and aborted the function mid-run under `set -e`. Guarded with `|| _cur_cb=""`. (lib.sh)

### Changed (tests)

- **test_providers.sh** — replaced the AT-RISK fixed-window `grep -A40 '^cma_run()'` assertions (the `push` marker had drifted to within 9 lines of the window edge, the same brittleness that already broke `-A30` once) with full-body awk extraction; added EXECUTION regressions that run the real `claude-providers list`/`remove` against an alias-less provider and assert no abort.
- **test_coverage.sh** — added a regression that EXECUTES `cma_ensure_alias_file` against an alias file with no `export CLAUDE_BIN=` line and asserts it completes.
- **test_session.sh** — added EXECUTION tests for the `hint` subcommand (run on every bare launch, previously only string-

matched): exits 0, writes only to stderr, names the snake_case project, handles an empty label.

### Verified

- Suite **18/18 green**; shellcheck 0. All three aborts reproduced (RED) and confirmed fixed (GREEN); the providers fix proven RED on a guard-stripped copy. Found via 3 parallel investigator subagents, each finding independently reproduced before fixing.

## v1.10.3 — 2026-06-29 — Execution-level wrapper test coverage

### Added

- **`test_wrapper_exec.sh`** — the first hermetic test that actually *executes* the generated `cma_run` wrapper (every other suite only string-matches its emitted body, so a runtime bug — a `set -e` abort, a dropped `unset`, wrong call order — could ship past a green suite). It drives `cma_run` with a stub `CLAUDE_BIN` env-recorder plus stub `claude-session`/`claude-sync-state`, then asserts RUNTIME guarantees: provider-env isolation (a leaked `ANTHROPIC_BASE_URL`/`AUTH_TOKEN`/`MODEL` is genuinely cleared *before* claude runs), session flags reach claude on a bare launch, `sync-state pull` fires before launch and `push` after, explicit args pass through verbatim with no session-flag injection, plus a non-vacuity guard proving the stub claude really executed. Proven **RED** on a dropped `unset`, **GREEN** on the real wrapper.

### Verified

- Suite **18/18 green**; shellcheck 0.

## v1.10.2 — 2026-06-29 — Self-healing rc source lines + strict rc tests

### Fixed

- **Dangling `source "…/aliases.sh"` lines in rc files.** A transient or moved alias-file path could leave a `source` line in `~/.bashrc`/`~/.zshrc` pointing at a deleted file, so every new login shell printed `-bash: …/aliases.sh: No such file or directory`. `cma_ensure_alias_file` now **prunes** any rc `source`/`.` line whose `aliases.sh` target no longer exists (self-heal on the next install), and recognizes an existing source line across `./source` and

`$HOME`/~/absolute forms, so re-installs never accumulate duplicate source lines.

### Added

- **`test_rc_sourcing.sh`** (10 strict assertions) — reproduces the bug class the hermetic suite missed (it sandboxes `$HOME` and never inspected or *sourced* the rc files): prune drops dangling / keeps valid + comments + unrelated lines, ensure self-heals, **a fresh shell sources the rc with NO error** (the reported symptom), idempotent (exactly one source line after 3 calls), and cross-form dedup. Proven RED on the old behavior, GREEN on the fix.

### Verified

- Suite **17/17 green**; shellcheck 0.

## v1.10.1 — 2026-06-29 — Robust cma_run wrapper assertions

### Fixed

- **`test_claude.sh` used a fixed `grep -A30` window** to scan the `cma_run` body and silently missed the sync-state `push` marker once the body grew with the v1.10.0 apply-color calls (push slipped past line 30) — failing the suite against a v1.10.0-installed alias file even though the wrapper itself was correct. It now extracts the full function body (awk header → closing brace), robust to future growth.

### Verified

- Suite **16/16 green** against the v1.10.0 wrapper; shellcheck 0.

## v1.10.0 — 2026-06-29 — Auto-applied per-alias session color + coverage/wiring

The per-alias session color is now **auto-applied** (it was only a hint in v1.9.x), plus self-healing for a stale `CLAUDE_BIN` and several closed test-coverage gaps.

### Added

- **Auto-applied per-alias session color.** Each bare alias launch now writes the alias's color into the session as an `agent-color` record — the exact record Claude Code's `/color` writes — via the new `claude-session apply-color`, called by `cma_run`/`cma_run_provider` (before launch to colour a resumed session; after exit to colour a freshly-created one). Deterministic `md5(label) mod 8` over red/blue/green/yellow/purple/orange/pink/cyan: each alias gets a stable, distinct colour, and switching the same session between aliases re-colours it. Verified **LIVE** on claude 2.1.195 — written, idempotent, persists across `--resume`. (Prompt-bar rendering must be confirmed visually: `/color` is TUI-only and `claude -p '/color x'` is a no-op, so record injection is the only non-interactive mechanism. See docs/SESSION_COLOR.md.)
- Test coverage: `test_install.sh` (executes `install.sh` in a sandbox — symlinks, alias file, idempotency), `test_verify_scripts.sh` (`model_verify.py`
  - `providers-verify.sh`), `test_session` apply-color tests (incl. the `set -e` regression), `test_coverage` B7 (`CLAUDE_BIN` resolver), B8 (`CLAUDE_BIN` migration), B9 (apply-color wired into both wrappers). `run-proof.sh` now also runs the previously-orphaned `verify_aliases_live.sh`.

### Fixed

- **Stale `CLAUDE_BIN` self-heals.** Existing installs whose alias file pointed `CLAUDE_BIN` at a non-existent path (e.g. `~/.local/bin/claude` where npm put claude in `~/.npm-global/bin` — the amber.local case) now rewrite it to a resolved, executable claude on the next install/ensure.
- A `set -e`/`pipefail` bug where `apply-color` aborted before writing on a session that had no existing `agent-color` record.

### Verified

- Suite **16/16 ALL GREEN**; **shellcheck 0**. Color injection proven **LIVE** on real claude 2.1.195 (write / idempotent / persist-across-`--resume` / recolour-on-alias-switch).

## v1.9.2 — 2026-06-29 — Hermetic CLAUDE_BIN resolver test

### Fixed

- **`test_coverage.sh` B7 "fallback when nowhere" was not hermetic** and failed on hosts that have a real `/usr/local/bin/`

`claude` (caught live on thinker.local): the resolver *correctly* returns the system claude there, but the test wrongly assumed "claude nowhere" was achievable under a sandboxed HOME/PATH (it can't mask absolute system paths). Dropped that one assertion; the load-bearing discovery cases (explicit `CLAUDE_BIN`, `~/.npm-global/bin` discovery) stay covered. Runtime behavior unchanged.

### Verified

- Suite **14/14 green on all five hosts** (this host, mistborn, thinker, amber, nezha); shellcheck 0.

## v1.9.1 — 2026-06-29 — CLAUDE_BIN resolves across per-host install locations

A patch found during the live multi-host rollout of v1.9.0.

### Fixed

- **Alias launches failed where claude was installed outside `~/.local/bin`.** `npm i -g @anthropic-ai/claude-code` lands in different prefixes per host (`~/.npm-global/bin`, Homebrew, `~/.local/bin`); the toolkit's fixed `CLAUDE_BIN` default mis-pointed on those hosts, so every `claudeN`/provider launch failed "No such file or directory" (amber.local needed a manual symlink to work). `cma_resolve_claude_bin` now prefers an explicit `CLAUDE_BIN`, then `$PATH`, then the known locations (`~/.local/bin`, `~/.npm-global/bin`, `/opt/homebrew/bin`, `/usr/local/bin`), with a `~/.local/bin` fallback.

### Verified

- Suite **14/14 green**; **shellcheck 0**. `test_coverage.sh` B7 covers explicit / npm-global / fallback resolution. v1.9.0's auto-session naming confirmed installed + **live-validated** (create-named + legacy-rename) on all five hosts: this host, mistborn, thinker, amber, nezha.

# v1.9.0 — 2026-06-29 — Per-project auto-sessions that actually work live + zero-coverage tests

A minor release that makes the v1.8.0 "auto session-per-project" feature do what it promised. As shipped, opening any alias gave an **unnamed** session; three root causes — all reproduced and fixed against the real `claude 2.1.195` binary, then proven **LIVE end-to-end** — are corrected here, plus test coverage for two zero-coverage utilities and a documentation refresh.

## Fixed

- **Per-project auto-session naming never actually named the session — now proven LIVE.** Three independent root causes:
  - **Legacy/unnamed sessions were never renamed.** The launcher only `--resume`'d an existing session and never passed `--name`, so a session created by an older wrapper or by plain `claude` stayed unnamed forever. Fix: always pass `--name` on resume too. Proven live — `claude --resume <id> --name <x>` renames a previously-unnamed session (custom-title `<NONE>` → `<x>`), contradicting the docs but confirmed empirically.
  - **The session-existence check used a run-collapsing slug.** It collapsed runs of non-alnum to one `-` (`s/[^A-Za-z0-9]+/-/g`), but claude slugs **per char**, so paths with consecutive non-alnum segments (hidden dirs, `/tmp/.private`, `__pycache__`) false-negatived the lookup and **re-created** instead of resuming. Fix: per-char slug (`s/[^A-Za-z0-9]/-/g`), matching the real on-disk dir names.
  - **`cma_run` self-heal regenerated only on a missing `unset ANTHROPIC_` marker.** Wrappers predating auto-session carried that marker but not the `claude-session` one, so they never regained the integration. Fix: regenerate when **either** marker is missing.

## Added

- **`docs/SESSION_COLOR.md`** — resolves the previously dangling reference, documenting per-project auto-session naming and the honest `/color` limitation: in `claude 2.1.195`, `/color` is **TUI-only** (no CLI flag, no `settings.json` key, no env var — verified against the binary and the docs), so the toolkit can only print a deterministic per-alias hint, never auto-apply it.
- **`test_toon.sh` (9 assertions) and `test_bootstrap.sh` (39 assertions)** — both utilities previously had **zero** coverage. toon (hermetic, SKIP-if-no-node): `toon.mjs` encode/decode round-trip, the `toon_encode.py` python→mjs chain, and non-zero

exit on invalid JSON. bootstrap (hermetic): `claude-bootstrap --count 2 --yes` in a sandbox `$HOME` asserting account dirs, shared symlinks, private-file isolation, alias lines, and the documented refuse-to-clobber re-run behavior.
- **`test_coverage.sh` B6** asserts the emitted `cma_run` / `cma_run_provider` bodies actually carry the auto-session integration (bare-launch guard, `claude-session flags`, `eval set --` apply, color hint) — the session script's own unit tests can't see the wrapper — plus a **self-heal regression**: a stale `cma_run` missing the `claude-session` marker is regenerated (exactly one `cma_run()`, provider-env isolation retained, aliases preserved). `test_session.sh` updated for `--name`-on-resume and a per-char-slug regression (a `/.cfg/` path must resume, not re-create).
- **Docs refresh.** README rewritten as a project landing page; `scripts/README` refreshed with the full current script inventory; the long-form guide gained "Per-project auto-session & per-alias color" and "TOON utility" sections; the `/color` notes pinned to the verified `claude 2.1.195` (superseding older `2.1.178` references).

### Verified

- Full suite **14/14 ALL GREEN**; **shellcheck 0**. The session fix proven **LIVE end-to-end** on the real `claude 2.1.195` binary — fresh create, resume, and legacy-unnamed rename all confirmed. New tests are non-vacuous (concrete expected values / negative controls). Installed live on this host and validated.

## v1.8.1 — 2026-06-29 — Merge-engine correctness + portability hardening

A patch release: an adversarial correctness audit of `claude-unify`'s merge engine plus a BSD/GNU portability pass over the test + proof tooling. All fixes/hardening — **no new features**. Housekeeping: a divergent mirror lineage that re-created `v1.7.11` (`1e975e5`) was merged back into `main` resolved to **OURS** (local already carries v1.7.11 → v1.8.0 and later fixes that supersede it), leaving a tree byte-identical to HEAD so all four mirrors converge on one lineage; the `containers` submodule was fast-forwarded to latest `main` (`71d3256` → `67ed35a`).

### Fixed

- **`history.jsonl` merge fused records across a source missing its trailing newline.** `merge_history_jsonl` `cat`'d sources into a temp first, gluing one file's last line onto the next file's first line → two entries collapsed into one invalid-JSON line. Fix:

feed files straight to `awk` (fresh record per file). Regression **R1** (RED before, GREEN after).

- **`enabledPlugins` union dropped "any true".** The `jq` used `+`/`*` (rightmost-wins), so a plugin enabled in an earlier account but `false` in the lexically-last account ended up disabled for everyone — contradicting the documented "any true survives" guarantee. Fix: OR-of-true reduce over every account. Regression **R2**.
- **A single malformed `settings.json` aborted the whole unify — and naive guarding then risked silent config loss.** The multi-file `jq -s` ran unguarded under `set -e` (settings is item 15 of 16), halting mid-run. Merely skipping the merge was worse: `link_to_shared` still replaced each valid account's real `settings.json` with a symlink to a never-written target (a dangling link → silent loss, exit 0). Final fix (hardened after adversarial review): validate each file with `jq empty` and merge only the valid ones (a malformed sibling is excluded, not fatal), and `link_to_shared` refuses to create a link when the shared target is absent. Regression **R3** (asserts the valid account's settings stay readable, not just that unify exits 0).
- **Directory-merge conflicts were resolved by lexical account name, not recency.** `merge_dir_into_shared`'s second pass overlaid only `ACCOUNTS[-1]` (alphabetically-last) while claiming to bias toward the "most recently active" account, so a stale account sorting last could clobber fresher `memory/*.md`. Fix: overlay every account with `rsync -au` so the newest-mtime file wins each conflict, independent of name/order. Regression **R4**.
- **Rollback left dangling symlinks.** Unify symlinks every shared item into each account; for an item an account never had there is no `.preunify` backup, so rollback's restore loop never visited it and the symlink dangled once `SHARED_DIR` moved aside. Fix: after restoring backups, remove any leftover symlink whose target points into `SHARED_DIR` (skipping the shared store itself). Regression **R5**.
- **Rollback restored a non-deterministic backup.** `find -print0` was unsorted, so when a path had several `.preunify.*` backups an arbitrary one was restored. Fix: `sort -z` (timestamps are `YYYYMMDDHHMMSS` = lexical-chronological) so the oldest — the true pre-unify original — wins. Regression **R6**.
- **`test_unify.sh` B2 was a vacuous PASS.** It called `cma_realpath` (a `lib.sh` function) without sourcing `lib.sh`, so the call errored to empty and the assertion compared `"" == ""` — the symlink target was never verified. Fix: source `lib.sh` + `set +e` (matching every sibling test that uses lib functions directly). Now prints the real resolved `SHARED_DIR/plugins/cache` path.
- **Portability: 3 GNU-only constructs broke the test/proof tooling on macOS** (the shipped runtime toolkit was already clean). `readlink -f` (no `-f` on BSD) in `assert_symlink_to/test_unify.sh` returned empty → spurious symlink pass/fail, fixed with a self-contained `_assert_realpath` in `assert.sh` + `cma_realpath` in `test_unify.sh`; `sed -E 's/\x1b…//'` (`\xNN` is GNU-

sed-only) in `run-proof.sh`/`verify_opencode_live.sh` left ANSI in, skewing `grep -c` counts, fixed by building the ESC byte via `printf '\033'`; unguarded `timeout` (GNU coreutils) in `verify_opencode_live.sh`, fixed by resolving `timeout`/`gtimeout` once and degrading if absent.
- **De-vendored `node_modules`.** `node_modules/@toon-format/toon` was committed by an accidental "Auto-commit" yet load-bearing (`scripts/toon.mjs` imports the bare specifier; `toon_encode.py` shells out to it) with nothing ever running `npm install`. Removed from the tree (`git rm --cached` + gitignore `/node_modules/`). Proven by fresh-clone simulation: `ERR_MODULE_NOT_FOUND` before, encodes correctly after.
- **`mktemp` portability.** Standardized every bare `mktemp [-d]` to the templated `mktemp [-d] "${TMPDIR:-/tmp}/cma.XXXXXX"` form CLAUDE.md prescribes (BSD `mktemp` requires a template; only GNU tolerates a bare call) across `lib.sh`, `claude-unify`/`providers/opencode-sync/session/install`, and the test harness (the sandbox keeps its `cma-test.` prefix for the cleanup safety check).

## Added

- **B5 token-limit guard coverage** (`test_coverage.sh`). The v1.8.0 `context_limit`/`max_output` path (`cma_provider_write_env` → `CMA_PROVIDER_CONTEXT_LIMIT`/`CMA_PROVIDER_MAX_OUTPUT` → `cma_run_provider` exporting `CLAUDE_CODE_MAX_OUTPUT_TOKENS`) shipped with **zero** tests — the only v1.8.0 fix lacking one. 4 cases / 6 concrete-value assertions: round-trip (`262144`/`32768`), `null`→empty normalization, 7-arg back-compat, and the emitted wrapper carrying the export.
- **`npm install` step in `install.sh`** (soft — warns, never hard-fails, when `npm` is absent; core unify/add-account needs no Node), so a fresh clone gets `@toon-format/toon` without a vendored tree. `curl-install.sh` inherits it via delegation.
- **+16 regression assertions** — 6 in `test_coverage.sh` (B5) and 10 in `test_unify.sh` (R1–R6 above), each written RED-before / GREEN-after.
- **Documented two deliberate merge/sync trade-offs** in-code so they are explicit rather than silent: `cma_merge_claude_json` replaces (not element-unions) array values; `claude-sync-state` pull/push is last-writer-wins (no portable mutex is worth its stale-lock failure modes for a per-launch hook).

## Verified

- Full suite **12/12 ALL GREEN**; **shellcheck 0**; all `.py` compile under `python3 -W error`. Each bugfix proven **RED-before / GREEN-after**; the de-vendor proven via fresh-clone simulation (`node scripts/toon.mjs` + `toon_encode.py`); ESC-strip verified functionally; the post-merge tree confirmed byte-identical to

HEAD. Installed live on this host and validated against all existing aliases (3 native + 44 provider) + `claude-list-accounts`.

# v1.8.0 — 2026-06-29 — Alias isolation + token-limit guard + per-project auto-sessions

A systematic-debugging pass fixing three reported issues plus a new session-per-project feature. Every root cause was reproduced and the fix proven with physical evidence before shipping.

## Fixed

- **CRITICAL — aliases cross-contaminated API endpoints across sessions.** `cma_run_provider` exports `ANTHROPIC_BASE_URL`/`AUTH_TOKEN`/`MODEL`/`SMALL_FAST_MODEL` into the interactive shell, and native `cma_run` did **not** clear them — so running a provider alias (e.g. `xiaomi`) and then a native alias (`claude1`) in the same shell made the native one inherit the provider's endpoint (`api.xiaomimimo.com`). `cma_run` now `unset`s those four vars before launch. Proven live: after a leaked xiaomi env, native launch shows `ANTHROPIC_BASE_URL=<unset>`. Existing installs auto-regenerate the wrapper (migration keyed on the new `unset ANTHROPIC_` marker).
- **Token-limit 400 ("exceeded model token limit: 262144").** The models.dev catalog's per-model `limit.context` / `limit.output` were read for ranking but never emitted, so Claude Code overshot a provider's real context window. `providers_resolve.py` now emits `context_limit` + `max_output`; `providers_generate.py` and `cma_provider_write_env` write `CMA_PROVIDER_CONTEXT_LIMIT` / `CMA_PROVIDER_MAX_OUTPUT` into each `.env`; and `cma_run_provider` exports `CLAUDE_CODE_MAX_OUTPUT_TOKENS` from it. Proven: kimi-for-coding now resolves `context_limit=262144 max_output=32768` from the live catalog.
- **"workspace has not been trusted" warning on launch.** Confirmed NOT a merge bug (trust propagates across accounts correctly); the warned project was simply never trusted under any account. The launch wrapper now writes `projects[<root>].hasTrustDialogAccepted=true` for the launching project via the new `claude-session` helper.

## Added

- **Auto session-per-project (`claude-session.sh`).** Every bare alias launch (native or provider) now resumes — or, the first time, creates — one long-lived Claude session per project root:

stable `--session-id` (UUID derived from the git-root path), `--name` set to the root dir basename in lowercase snake_case (`Android 15` → `android_15`). Explicit args/flags are always respected verbatim. Verified against the real `claude` CLI: `--session-id` creates, `--resume` resumes.
- **Per-alias color hint.** A deterministic alias→color mapping over Claude Code's real palette (`red blue green yellow purple orange pink cyan`); printed as a `/color <x>` tip on launch. (Investigated thoroughly: `/color` is a TUI-only command with no CLI flag / settings key / writable persistence, so it cannot be auto-applied — the toolkit suggests it rather than faking it.)
- **`test_session.sh`** — 27 hermetic assertions for name/id/color/flags/trust/ git-root behavior. **`run-all.sh` is now 12 files / 60 assertions, ALL GREEN.**

### Verified

- Full suite **12/12 ALL GREEN**; **shellcheck 0**; all `.py` compile under `python3 -W error`. All four items proven end-to-end against the live catalog and the emitted alias file.

## v1.7.12 — 2026-06-28 — One-line curl installer

### Added

- **`curl-install.sh`** — one-line bootstrap installer:

```
curl -fsSL https://raw.githubusercontent.com/vasic-digital/
        claude-toolkit/main/scripts/curl-install.sh | bash
```

  Detects platform (Linux/macOS) and shell, auto-installs missing hard dependencies (jq, rsync, awk) via the system package manager (apt/dnf/apk/pacman/brew), clones (or pulls if already present) the repo with all submodules recursively to `~/claude-toolkit`, runs `install.sh`, and prints next-steps. Idempotent; re-runnable. Install dir overridable via `CLAUDE_TOOLKIT_DIR` env var.
- **README.md** — curl one-liner added at the top of the Install section.
- **`test_curl_install.sh`** — 22 hermetic tests covering syntax, permissions, URL correctness, submodule cloning, idempotency, platform detection, dependency checks, error handling, and next-steps output.

### Verified

- `bash -n` + `shellcheck 0` on `curl-install.sh` and `test_curl_install.sh`.
- `run-all.sh` **11/11 ALL GREEN** (was 10; +`test_curl_install.sh`).

## v1.7.11 — 2026-06-28 — Round-4: coverage-gap regression tests, toon recursion guard, arg validation

Fourth audit round: found the codebase is converging (export-docs, test harness, add/list/remove/rollback all verified clean); shipped 10 targeted coverage tests closing the same shallow-coverage class that let the v1.7.10 enable-plugins bug ship; plus 2 LOW code findings.

### Fixed

- **`toon_encode.py` `fallback_encode`**: unbounded recursion on deeply nested JSON. A crafted input with >1000 levels would blow Python's default stack and exit with an unhandled traceback instead of encoding. Added a `_depth` guard; beyond 64 levels emits compact JSON as a safe fallback.
- **`toon.mjs` encode-file / decode-file**: running `toon.mjs encode-file` with no argument produced a confusing `TypeError` from Node's `fs.readFileSync`. Now prints `Error: encode-file requires a filename argument` + exit 1.

### Added — coverage-gap regression tests (the same class that let enable-plugins

ship)

- **B9 (HIGH) — `cma_ensure_alias_file` migration path** (`test_coverage.sh`): builds a realistic old `cma_run_provider()` body lacking `claude-sync-state`, calls `cma_ensure_alias_file`, asserts the body is migrated, the following `alias claude1=` survives, and `cma_run_provider()` appears exactly once.
- **B3 (HIGH) — `_cma_q` bash quoting in `cma_provider_write_env`** (`test_coverage.sh`): sources a `.env` with a model name containing a literal single quote and asserts it round-trips intact; also asserts an injection payload does NOT execute on source (mirrors the already-tested Python `q()`).
- **B1 (HIGH) — `absorb_default_plugins`** (`test_unify.sh`): creates a real plugin file under `$HOME/.claude/plugins/cache/` before unify; asserts it lands in `$SHARED_DIR/plugins/cache/`.

- **B2 (HIGH) — `link_default_plugin_subdirs`** (`test_unify.sh`): asserts `$DEFAULT_DIR/plugins/cache` becomes a symlink into `$SHARED_DIR/plugins/cache` after unify, and that re-running unify doesn't create a second backup.
- **B4 (MEDIUM-HIGH) — `sync_claude_md` seed branches** (`test_unify.sh`): branch (b) seeds `$DEFAULT_DIR/CLAUDE.md` and asserts it wins; branch (c) removes it and gives an account a `CLAUDE.md`, asserts that one wins.

### Verified

- `run-all.sh` **10/10 ALL GREEN** (coverage now 39+10=49 assertions; unify now 43+7=50); **shellcheck 0**; all `.py` compile under `python3 -W error`; `node --check toon.mjs` clean; toon_encode 500-level-nest no longer crashes; toon.mjs missing-arg gives clean error + exit 1.

## v1.7.10 — 2026-06-28 — Round-3 audit: enable-plugins bug fix, path-traversal guards, proxy robustness

Third audit round (deep dive on the less-covered surface: opencode_sync, claude-unify merge, the poe proxy, bootstrap). Fixes verified centrally.

### Fixed

- **`cma_enable_plugins` silently enabled NO plugins when given 3 or more** (`lib.sh`). The jq `--arg` index was derived as `${#args[@]}/2`, but each iteration appends **three** elements — so for the default 4 always-on plugins it produced arg names `p0,p1,p3,p4` while the jq program referenced `$p0..$p3`; `$p2` was undefined, jq failed, `2>/dev/null` swallowed the error, and `enabledPlugins` was left empty. Replaced the derived index with a dedicated counter. Proven live: `cma_enable_plugins a b c d` now yields all four `true` (was empty); a ≥3-plugin regression test was added.

### Security (defense-in-depth)

- **Path traversal via unvalidated provider id** in `claude-providers.sh` `cmd_show` / `cmd_remove`: `$id` was interpolated into `<dir>/$id.env` and then `cat`/`rm -f`'d without validation. Now rejected unless it matches `[A-Za-z0-9._-]` (blocks `../`), matching `cma_provider_write_alias`.
- **`opencode_sync.py` `${CLAUDE_PLUGIN_ROOT}` path traversal**: a malicious installed plugin could set an arg like `$

`{CLAUDE_PLUGIN_ROOT}/../../../tmp/evil.js` and `--enable-all-local` would have OpenCode exec the traversed path. Expansion now lexically contains the result to the plugin dir; an escaping value is left unexpanded (fails safe).
- **`cma_write_alias` now rejects whitespace in the config dir** (`lib.sh`): an unquoted space silently word-split the alias into a bogus command — now a clear error instead of a broken alias.

### Robustness

- **`poe_proxy.py`**: `resolve_refs` gained a recursion-depth guard (a circular `$ref` previously crashed the request handler with `RecursionError`); the success-path `gzip.decompress` is now guarded like the error path, so a corrupt gzip body no longer propagates.

### Verified

- `run-all.sh` **10/10 ALL GREEN** (coverage now 32 assertions incl. the enable-plugins + injection regressions); **shellcheck 0**; all `.py` compile under `python3 -W error`.
- `cma_enable_plugins` fix proven live (4 plugins → all `true`); opencode containment + id validation proven with PoCs. The model-verification / alias- write path is unchanged from v1.7.9's live-proven 137 models / 32 aliases.

### Audit (round 3) — verified clean

`cma_merge_claude_json` private-key isolation, eval-token provenance, `cma_validate_alias`, proxy bind (localhost only) + no key logging, `_cma_q` escaping, `merge_settings_json` atomic write, history dedup, rollback NUL-safe traversal, bootstrap `--dir-of` injection filter. (`opencode_sync --enable-all` intentionally bypasses the needs-secret guard — operator opt-in, documented.)

## v1.7.9 — 2026-06-28 — Hardening round 2: injection-safe alias writes, broadened secret redaction, docs accuracy, shellcheck 0

A second multi-agent audit + hardening pass on top of v1.7.8 (adversarial security audit + doc-accuracy audit + lint sweep, fixes verified centrally).

## Security

- **Provider id / config dir can no longer inject shell via the alias file** (`lib.sh` `cma_provider_write_alias` / `cma_write_alias`). Both interpolate values into `alias name="…"` lines that the shell **re-parses on invocation**, and jq `@tsv` does not escape `"`. They now reject shell metacharacters (provider id restricted to `[A-Za-z0-9._-]`; config dir rejects `" $ \ \ ; & | < > ( )` and newline). Proven: `foo"; touch …` ` payload is rejected, no command runs, the hostile alias is never written.
- **Keys-file read no longer breakable by a quote in the path** (`claude-providers.sh` `cmd_sync_multi`). The old `bash -c "set -a; source '$keysf'; …"` let a single quote in the keys-file path break out of the string. Replaced with an isolated subshell `( set +e; set -a +u; . "$keysf"; set +a; eval … )` — the same safe pattern `cmd_sync` already used. Proven with a `do n't/` path.
- **`.env` value quoting** (`providers_generate.py`): `q()` now POSIX single-quote-escapes embedded quotes (mirrors `lib.sh` `_cma_q`), so a catalog value containing a quote can't inject when `cma_run_provider` sources the `.env`. Proven: an injection payload is neutralized to a literal.
- **xtrace secret leak** (`lib.sh` `cma_run_provider`): the indirect key read is now wrapped in `set +x`/restore so an active `set -x` in the user's shell can't echo the key to the terminal or a redirected log.
- **Broadened secret redaction + guard**: `cma_redact_secrets()` (`verify_opencode_live.sh`) and the committed-proof scan guard (`test_lib.sh`) now also catch `sk-ant-`, `hf_`, `AIza`, `xoxb-`/`xoxp-`/`xoxs-`, `pc-`, `re_`, `secret_`, and JWTs — regardless of JSON field name — closing the gap where arbitrary MCP env-var names (e.g. `NOTION_API_KEY`) slipped through the original six-name allowlist.

## Fixed

- **`install.sh` used `readlink -f`** (absent on BSD/macOS) for its symlink up-to-date check — missed by the v1.7.7 sweep. Now uses `cma_realpath`; the `test_lib.sh` guard scans `install.sh` too.
- **`verify_aliases_live.sh` hardcoded one developer's account dirs**, producing false FAILs on every other host. Now discovers accounts dynamically and skips dirs that don't exist.
- Dead code / cruft: `providers_generate.py` (unused import, dead vars, lambda→def, a no-op `provider_id + ('' if … else '')`); `model_verify.py` (unused `import hashlib`); `model_verify.py` docstring `--key` → `CMA_PROBE_KEY`.

## Docs

- Long-form doc + `README`s + `CLAUDE.md` corrected against the code: macOS rc-file caveat (`~/.zshrc` only), the test table now

lists all 10 suites, the full installed-command list (+`claude-providers`/`claude-sync-state`/ `claude-bootstrap`), repo-relative paths (was `~/Documents/scripts/`), a new `claude-bootstrap` section, the `CMA_PROBE_KEY` security model in §11, and a refreshed date stamp.

### Quality

- **shellcheck: 93 → 0** across all scripts. Added `.shellcheckrc` (`external-sources=true`) which resolves the sourced-file warnings properly; fixed the `$?`-after-condition (SC2319) test idioms, SC2015/SC1090/SC1003; the one remaining reserved no-op flag carries a justified inline disable.

### Verified

- `scripts/tests/run-all.sh` **10/10 ALL GREEN**; **shellcheck 0**; every `.py` compiles under `python3 -W error`.
- Injection PoCs (provider id, config dir, `.env` value, keys-file path) all proven neutralized; broadened redaction proven against `AIza`/`hf_`/JWT/etc.
- Live `sync --multi`: **137 models verified, 32 aliases** across 8 providers (opencode 4, poe 33, chutes 7, huggingface 6, nvidia 30, openrouter 14, siliconflow 38, xiaomi 5), zero `CMA_PROBE_KEY`/ unbound errors — identical to the v1.7.8 baseline, so the new key-read path is non-regressive end-to-end.
- 4-host byte-parity + 10/10 suite re-verified after deploy.

## v1.7.8 — 2026-06-28 — Secret hygiene (argv + committed-proof leaks), dead-code fix, coverage tests

Security + robustness follow-up found by a parallel multi-agent audit of v1.7.7. Four independent subagents fixed disjoint file sets; integration + the full suite + live multi-model verification were run centrally.

### Security

- **API key no longer passed on `argv`** (`model_verify.py` + `claude-providers.sh`). `cmd_sync_multi` invoked `model_verify.py --key "$token"`, placing the secret verbatim in `/proc/<pid>/cmdline` and `ps aux` output — readable by any user on a multi-user host. The key now flows via the `CMA_PROBE_KEY` environment variable (set per-command, not exported); `model_verify.py` reads it from the environment and errors clearly if unset. The `--key` flag is removed entirely.

- **API key no longer passed to `curl` on `argv`** (`verify_aliases_live.sh`). Six live-probe calls used `-H "Authorization: Bearer $key"`. The header is now written to a `mktemp`'d, `chmod 600` config file consumed via `curl --config` (portable on GNU + BSD curl) and removed via an `EXIT/INT/TERM` trap.
- **Leaked secrets purged from committed proof artifacts** (committed in 24bc379, rolled into this release): the OpenCode live verifier wrote resolved `opencode debug config` / `mcp list` output — which contained a real provider key and a DB connection-string password — verbatim into the committed proof dir. The three artifacts are redacted; the generator (`verify_opencode_live.sh`) now redacts via `cma_redact_secrets()` before writing (raw dump → `.raw` temp → redacted file → `.raw` removed). **Operator follow-up still required:** rotate the leaked key and decide on a git-history scrub — the values remain in history on all four remotes.

## Fixed

- **Unreachable code** in `verify_aliases_live.sh`: `exit $failed` sat *before* the Claude-alias test function and its caller, making them dead (shellcheck SC2317). `exit $failed` moved to the final statement.
- **Fragile `$?` capture** in `test_list.sh`: `grep …; [[ $? -ne 0 ]]` then `assert_eq 0 $?` read `$?` from the wrong command. Now captures `rc=$?` immediately.
- **Unquoted glob** in `claude-sync-state.sh:67`: `"$HOME"/${ACCOUNT_PREFIX}prov-*/` → `"$HOME/${ACCOUNT_PREFIX}"prov-*/` so only the intended `*` globs.
- **`SyntaxWarning: invalid escape sequence '\ '`** in `providers_resolve.py`: the usage docstring's `\` line-continuations are now a raw string (`r"""`).

## Added

- **`test_coverage.sh`** — 11 new hermetic tests (19 assertions) covering previously-untested `lib.sh` behavior: `cma_ensure_alias_file` (fresh / idempotent / old-format migration preserving unrelated lines), `cma_can_prompt` (`CMA_NONINTERACTIVE=1` and no-tty both non-interactive), `cma_enable_plugins` (JSON shape + additive + the jq `//` falsy-vs-null upgrade), `cma_link_shared_items` (every `CMA_SHARED_ITEMS` entry becomes a symlink into `$SHARED_DIR`, idempotent), and `stats-cache.json` newest-by-mtime selection.
- **Proof regression guard** (`test_lib.sh`): scans `scripts/tests/proof` for provider-key prefixes and URL `user:password@` creds, counting suspect lines so a failure never re-echoes a secret.

### Verified

- `scripts/tests/run-all.sh` — **10/10 ALL GREEN** locally (was 9; +`test_coverage.sh`).
- Live multi-model verification (`claude-providers.sh sync --multi`, real HTTP probes with the host's real keys): **137 models verified, 32 aliases generated** across 8 providers (opencode 4, poe 33, chutes 7, huggingface 6, nvidia 30, openrouter 14, siliconflow 38, xiaomi 5). Zero `CMA_PROBE_KEY`-unset and zero `unbound variable` errors — the env-var key path works end-to-end. (Providers with 0 verified are external: dead/paid keys, HTTP 401/402/403, WAF blocks — not toolkit regressions.)
- The new proof secret-scan guard immediately earned its keep: on first cross-host run it flagged a **stale, pre-redaction proof dir on all three remote hosts** (3 files with literal secrets), which were then re-synced with the redacted artifacts.
- `model_verify.py` / `providers_resolve.py` compile clean under `python3 -W error`.

## v1.7.7 — 2026-06-28 — Portable realpath (BSD portability hardening), set -u edge fix, regression tests

Follow-up hardening release found by a parallel multi-agent audit of v1.7.6.

### Fixed

- **`readlink -f` → portable `cma_realpath`** at three sites: `claude-unify.sh` (`already_linked_to_shared` and `merge_settings_json`) and `claude-list-accounts.sh` (the link check). `readlink -f` is absent on older macOS and on other BSDs (FreeBSD/NetBSD); there the checks silently fail — making `claude-unify` re-link every shared item on each re-run (accumulating stale `.preunify.*` backups) and `claude-list-accounts` report linked accounts as "not linked". **Honest scope:** modern macOS (Sequoia) and GNU coreutils DO support `readlink -f`, so on the current fleet this was a *latent* bug with no active symptom — but it broke the toolkit's stated BSD portability. Replaced with a new pure-bash `cma_realpath` (single-arg `readlink` symlink-walk + `pwd -P`), verified to produce output identical to `readlink -f` on macOS.
- **`set -u` empty-array edge in `cma_enable_plugins`** — `jq "${args[@]}"` with an empty `args` is an "unbound variable" error on bash 3.2 (reachable via `CMA_ALWAYS_ON_PLUGINS=""` from the non-re-exec'd `claude-providers.sh`). Guarded with `${args[@]+"${args[@]}"}`.

### Added

- **`cma_realpath`** portable canonicalizer in `lib.sh`.
- **Regression tests** (`test_lib.sh`): `cma_realpath` resolves a symlink chain and is identity on a real path; plus a guard asserting NO runtime script *invokes* `readlink -f`.

### Verified

- `scripts/tests/run-all.sh` — **9/9 ALL GREEN on all four hosts**: nezha, thinker, amber (Linux), mistborn (macOS, re-exec to bash 5.3, BSD userland).
- `cma_realpath` output confirmed byte-identical to `readlink -f` on macOS.

### Audit findings (v1.7.6 — no code change required)

- Disabled providers are EXTERNAL, not toolkit bugs (toolkit correctly disabled them on failed verify): `github-models` → HTTP 401 (dead GitHub PAT), `upstage` → HTTP 403 from AWS WAF (egress-IP block).
- `api_keys.sh` across all 4 hosts: **0 dangling refs, 0 duplicates, 0 malformed**; key parity confirmed (mistborn's 2 host-local Kimi-Platform keys preserved).
- Cross-host integrity: all 11 toolkit scripts byte-identical to the released tag on every host.
- Known/deferred: published tags `v1.2.0` (gitlab) and `v1.5.0` (gitlab/gitverse/gitflic) point to older commits than local — reconciling needs a force tag push; left for a maintainer decision.

## v1.7.6 — 2026-06-28 — Always-non-interactive execution, alias-file integrity, macOS/bash-3.2 portability, 4-host rollout

### Fixed

- **Alias-file corruption from a mis-firing migration** — `cma_ensure_alias_file`'s "outdated `cma_run_provider`" migration grepped for `claude-sync-state pull`, but the emitted on-disk text is `…/claude-sync-state" pull` (a quote precedes the space), so the guard **never matched** and the migration fired on *every* alias write. Its `awk` then chopped everything from `cma_run_provider()` to EOF — destroying previously-written provider aliases and any `claudeN` aliases that follow the function block. This silently corrupted the alias file on multi-provider /

multi-account hosts. Detection is now scoped to the function body and matches the bare command name (quote/space agnostic), and the migration removes **only** the function block, preserving alias lines. This was the single root cause of the failures across `test_providers.sh`, `test_claude.sh`, and `test_add_remove.sh`.
- **`set -u` abort while sourcing the keys file** — provider sync sourced `~/api_keys.sh` inside a `set -euo pipefail` subshell. A dangling reference in the user's keys file (e.g. `export SARVAM_API_KEY=$ApiKey_Sarvam_AI_India`) aborted the source **mid-file** under `nounset`, leaving every key defined *after* it unexported — so those providers silently failed verification ("unverified") and stderr was spammed with "unbound variable". Keys are now sourced with `nounset` disabled (subshell-local in sync; save/restore around the alias-file `cma_run_provider`). Installed alias files are auto-migrated to the `nounset`-safe wrapper on next sync.
- **macOS / bash-3.2 portability of the test harness** — `tests/run-all.sh` used `mapfile` (bash 4+), so the **entire suite failed to run on stock macOS**. Replaced with a portable `read` loop and guarded empty-array expansion under `set -u`. Same fix applied to `test_lib.sh` and `tests/lib/sandbox.sh` (empty `${arr[@]}` expansions are unbound on bash 3.2). The suite now runs green on macOS bash 3.2.

## Added

- **`CMA_NONINTERACTIVE` + automatic TTY detection** — a new `cma_can_prompt` helper makes every prompt (`claude-add-account`, `claude-remove-account`, `claude-bootstrap`) fall back to its non-interactive default whenever no terminal is available (CI, SSH without a PTY, the test sandbox) or when `CMA_NONINTERACTIVE=1` is exported. Toolkit execution is now **always non-interactive off a terminal**. Destructive account removal still refuses (rather than guessing) without `--yes` when it cannot confirm.
- **Regression tests** for non-interactive `claude-add-account` and for alias-line survival across repeated account adds.
- **`test_export.sh` graceful SKIP** when its prerequisites (pandoc + a PDF engine) are absent — matching the existing SKIP convention for optional-dependency features.

## Multi-host rollout (nezha · mistborn.local · thinker.local · amber.local)

- Distributed `~/api_keys.sh` to every host via a **no-loss merge** (host-local keys preserved — e.g. mistborn kept its 2 Kimi-Platform keys; amber created fresh) and wired **both** `.bashrc` and `.zshrc` to source it on every host.

- Installed/updated the toolkit on all four hosts and configured `claude1/claude2/claude3` on each; installed Claude Code on amber.
- Ran live provider/model detection on every host — **17–20 active providers each**, models verified via HTTP probes, **0 unbound errors**.

### Verified

- `scripts/tests/run-all.sh` — **9/9 files, ALL GREEN on all four hosts**: nezha (Linux), thinker (Linux), amber (Linux), mistborn (macOS / bash 3.2).
- Cross-host: both rc files source `api_keys.sh`; `claude1/2/3` + `poe`/`deepseek`/`xiaomi` aliases present on every host.

## v1.7.5 — 2026-06-28 — Cross-provider / resume session visibility fix

### Fixed

- **Cross-provider `/resume` session loss** — when switching between provider aliases (e.g., `deepseek` → `opencode` → `kimi-for-coding`), `/resume` would sometimes show empty session history. Root cause: the `cma_run_provider` function in the alias file was **missing sync-state pull/push calls** that were present in `lib.sh`. The alias file is what actually runs when a user invokes an alias, so the sync never happened.
- **Migration for outdated alias files** — added automatic detection and regeneration of outdated `cma_run_provider` functions in `lib.sh`. If the function exists but lacks `claude-sync-state pull`, it's removed and rewritten with the correct implementation.
- **Router transport transformer config** — added `transformer: {use:["cleancache","streamoptions"]}` to the alias file's router transport section (was only in `lib.sh`), ensuring `cache_control` stripping works for all router-transport providers.

### Root Cause Analysis

The `cma_run_provider` function in `lib.sh` (lines 225-333) correctly includes sync-state pull/push calls, but the alias file's copy of the function was outdated and explicitly stated "cross-account claude-sync-state is intentionally NOT run." This meant:

1. Sessions created under provider A had their `lastSessionId` written only to A's `.claude.json`
2. When switching to provider B, B's `.claude.json` still had its own (different) `lastSessionId`

3. `/resume` read B's `lastSessionId` and couldn't find A's session

After fix: all providers/accounts share the same merged `lastSessionId` via sync-state.

### Verified

- **Local host**: all providers show identical `lastSessionId` after sync (confirmed)
- **mistborn.local**: 76 projects merged across all accounts/providers (confirmed)
- **Migration**: `install.sh` correctly detects and fixes outdated alias files on both hosts

### Tests

- Cross-alias session visibility (Section 5): **ALL PASS**
- Existing test suite: session-related tests pass

## v1.7.4 — 2026-06-26 — Kimi provider fix + AWS IaC MCP disabled by default

### Fixed

- **Kimi Code provider base URL** in `scripts/providers/overrides.json` — changed from `/coding/v1` to `/coding/` so native transport works correctly.
- **AWS IaC MCP timeout** — removed `aws-dev-toolkit/awsiac` from the default OpenCode MCP allowlist in `scripts/claude-opencode-sync.sh`. The server consistently timed out on connection and is now configured but disabled by default.

### Changed

- Regenerated `Claude_Multi_Account_Fine_Tuning.{html,pdf,docx}` from current markdown.
- Refreshed proof artifacts in `scripts/tests/proof/`.

### Tests

- Local: **9/9 ALL GREEN**
- Live OpenCode verification: **9 passed, 0 failed**, 27/27 enabled MCPs connected
- Provider alias verification: **5 passed, 0 failed**

# v1.6.6 — 2026-06-21 — TOON integration for token-efficient prompts

## Added

- **TOON (Token-Oriented Object Notation)** integration — saves ~40% tokens vs JSON for structured data in LLM prompts by declaring fields once in arrays.
- **`scripts/toon.mjs`** — Node.js TOON utility (encode/decode/demo)
- **`scripts/toon_encode.py`** — Python wrapper for TOON encoding
- **`docs/TOON_Integration.md`** — comprehensive guide on using TOON with Claude Code
- **`package.json`** — `@toon-format/toon` v2.3.0 dependency

## Token Savings

- File listings: ~39% fewer tokens
- Tool definitions: ~40% fewer tokens
- User records: ~42% fewer tokens
- Accuracy: 76.4% (vs JSON's 75.0%)

## Note

TOON formats message CONTENT for token savings. API transport remains JSON (providers require it). HTTP/3 and compression require provider-side support.

## Tests

- 8/8 ALL GREEN

# v1.6.5 — 2026-06-21 — Poe proxy fix (alias file + install)

## Fixed

- **Poe proxy not starting from alias** — proxy logic was only in `lib.sh`, not in the alias file's `cma_run_provider` function. The alias file is what actually runs when a user invokes an alias. Added proxy detection + auto-start to the alias file.
- **install.sh: SHARE_DIR → SHARED_DIR** — wrong variable name caused `unbound variable` error on nezha (Linux, `set -u`).
- **install.sh: auto-copy proxy scripts** to `~/.local/share/.../proxy/` during install.

### Verified

- All 3 Poe aliases work: `poe`, `poe2`, `poe3`
- Deployed to both local host and nezha.local

### Tests

- 8/8 ALL GREEN

## v1.6.4 — 2026-06-21 — Poe proxy fix for tool compatibility

### Fixed

- **Poe tool format error** — Poe requires `parameters` in every tool function definition. Claude Code sometimes omits it (valid in Anthropic format, invalid for Poe). Added `poe_proxy.py` that auto-fixes tools before forwarding to Poe API.
- **Proxy auto-start** — `cma_run_provider` now auto-starts compatibility proxies for providers that need them (detected by `scripts/proxy/<provider>_proxy.py`).

### Verified

- All 3 Poe aliases work through proxy: `poe`, `poe2`, `poe3`

### Tests

- 8/8 ALL GREEN

## v1.6.3 — 2026-06-21 — Poe provider (382 models, 3 aliases)

### Added

- **Poe provider** — universal AI platform with 382 models from all major providers. OpenAI-compatible API at `https://api.poe.com/v1`. Chat, code, image gen, video gen, TTS, STT, and more.
- **3 aliases**: `poe` (claude-sonnet-4.6 + gpt-5.4-mini), `poe2` (gpt-5.5 + deepseek-v4-pro-e), `poe3` (grok-4 + gemini-3.1-pro)
- **key-aliases**: `POE_API_KEY` + `ApiKey_Poe` → `poe`
- **Tool calling verified** on claude-sonnet-4.6, gpt-5.4-mini, deepseek-v4-pro-e, grok-4

- **382 models categorized**: 130 chat/reasoning, 16 code, 40 image gen, 17 video gen, 12 TTS, 1 STT, 166 other
- **Documentation**: full Poe section in Provider_Aliases_User_Guide.md

### Verified

- API endpoint responds correctly
- Authentication works
- Tool calling confirmed
- All 3 aliases tested through ccr with "Do you see our codebase?" — all YES

## v1.6.2 — 2026-06-21 — Chutes provider documentation + model update

### Changed

- **Chutes provider models updated** — catalog was stale. Chutes now offers 13 TEE (Trusted Execution Environment) models. Updated strong=`zai-org/GLM-5.2-TEE`, fast=`Qwen/Qwen3.6-27B-TEE`.
- **Chutes documentation** added to Provider_Aliases_User_Guide.md with full model table, TEE explanation, pay-per-use note, and setup instructions.

### Verified

- Chutes API endpoint responds correctly
- All 13 TEE models accessible (require funded account for actual inference)
- OpenAI-compatible format confirmed at `https://llm.chutes.ai/v1`

## v1.6.1 — 2026-06-21 — cache_control fix + E2E tests

### Fixed

- **`cache_control` parameter error** — Claude Code sends `cache_control` (Anthropic-specific) in its API requests. ccr forwarded this to OpenAI-compatible endpoints which reject it with HTTP 422. Fixed by adding ccr's built-in `cleancache` transformer to every provider config, which strips `cache_control` before forwarding to the provider.

### Added

- **`alias_e2e_test.py`** — end-to-end alias verification script that tests each alias by sending requests through ccr and verifying responses work without errors.

### Verified working (all aliases tested with "Do you see our codebase?")

- `opencode` (north-mini-code-free): YES
- `opencode2` (big-pickle): YES
- `opencode3` (nemotron-3-ultra-free): YES
- `deepseek` (native transport): YES
- `deepseek2` (router transport): YES
- `xiaomi` (native transport): YES
- `zai-coding-plan` (router transport): YES

## v1.6.0 — 2026-06-21 — Multi-alias provider system

### Added

- **Multi-alias provider system** — every provider can now have multiple aliases (`provider`, `provider2`, `provider3`...) exposing ALL working models, not just the top 2. Verified via live HTTP probes with anti-bluff detection.
- **`model_verify.py`** — comprehensive model verification & scoring engine. Tests every model for a provider via HTTP probes, scores on 7 dimensions (existence 25pts, tool_call 20pts, reasoning 15pts, context_window 15pts, streaming 10pts, latency 10pts, free_tier 5pts). Anti-bluff detection prevents false positives (HTTP 200 with error body, empty responses, boilerplate errors). 24h verification cache to avoid re-testing.
- **`providers_generate.py`** — multi-alias generation from verified models. Pairs models into alias groups of 2 (strong + fast), handles odd count (last model reused for both positions), single model (used for both positions). Generates env files, shell aliases, and overrides.json entries.
- **`claude-providers.sh --multi`** — new flag for `sync` that triggers the full verification + multi-alias generation pipeline. Additional flags: `--max-aliases` (default 5), `--min-score` (default 25), `--verify-concurrency` (default 5).
- **Endpoint normalization** — `/anthropic` endpoints auto-converted to `/v1` for OpenAI-compatible probing during verification.

- **Submodules updated** to helix_translate-2.3.1: LLMsVerifier (ModelVerifier, Seed, xiaomi provider), challenges (anti-bluff §11.4, chaos/stress tests), containers (deploy-stack).

### Changed

- Probe `max_tokens` increased from 32 to 128 — reasoning models need more tokens for chain-of-thought + response (was causing false anti-bluff rejections).
- `User-Agent` header added to HTTP probes (some APIs require it).

### Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures.

### Usage

```
# Standard sync (2 models per provider, as before)
claude-providers sync

# Multi-alias sync (verify ALL models, create multiple aliases)
claude-providers sync --multi

# With options
claude-providers sync --multi --max-aliases 10 --min-score 20
```

## v1.5.1 — 2026-06-20 — Linux stat fix + nezha deployment

### Fixed

- **`stat -f %m` on Linux** — the mtime cache check in `claude-providers.sh` used `stat -f %m || stat -c %Y` as an `||` chain. On Linux, `stat -f` succeeds (returning filesystem info, not mtime), so both outputs merged into garbage (`"File: ...1781634386"`), causing `File: unbound variable` under `set -u`. Fixed with `case "$(uname -s)"` to pick the correct flag per platform.

### Deployment

- **nezha.local** (Linux x86_64) deployed and verified: 19 providers activated, 100/100 provider tests pass, 5/5 live verifier pass, cross-alias sync confirmed. Evidence in `scripts/tests/proof/90-nezha-deployment.txt`.

### Full test suite

- macOS: 8/8 ALL GREEN
- Linux (nezha): 7/8 pass (export fails: pandoc not installed — pre-existing)

## v1.5.0 — 2026-06-20 — Cross-alias session visibility

### Added

- **Cross-alias session visibility** — sessions created under ANY alias (`claudeN`, `deepseek`, `opencode`, `xiaomi`, etc.) are now visible from every other alias via `/resume`. Memory, project settings, and session data are fully shared across all accounts and providers.
- **`claude-sync-state.sh` extended** — now discovers provider dirs (`~/.claude-prov-*`) alongside account dirs for its `.claude.json` merge. Provider sessions participate in the same lightweight jq merge that keeps account sessions in sync.
- **`cma_run_provider` sync-state hooks** — the provider wrapper now calls `claude-sync-state pull` before launch and `claude-sync-state push` after exit, matching the `cma_run` pattern. Previously provider sessions were intentionally excluded from sync; now they participate fully.
- **Sandbox test coverage**: 10 new assertions proving cross-alias merge (sessions from account→provider, provider→account, account→account all visible after sync). Providers test 90 → 100 assertions.
- **Live verification**: `lastSessionId` for a real project confirmed identical across all dirs (3 accounts + 1 provider). 61 projects merged in every `.claude.json`. Evidence in `scripts/tests/proof/80-cross-alias-sessions.txt`.

### Changed

- `scripts/claude-sync-state.sh` — provider dirs included in merge targets
- `scripts/lib.sh` — `cma_run_provider` wrapper updated with sync-state pull/push
- Alias file `aliases.sh` — updated `cma_run_provider` function (re-installed)

### Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures. Providers test includes 10 new assertions for cross-alias session visibility.

### How it works

1. `claude-sync-state pull` merges every account's + provider's `.claude.json` into the launching dir before Claude Code starts (including `lastSessionId`, `allowedTools`, MCP config, etc.).
2. Claude Code launches with the merged state — `/resume` sees all sessions.
3. `claude-sync-state push` merges the post-session `.claude.json` back out after exit, so the next alias to launch picks up the new session.
4. The `sessions/` directory was already shared via symlink — this release ensures `.claude.json` project settings are also merged.

### Performance

- Adds ~1-2 seconds overhead per provider launch (jq merge of `.claude.json` across all dirs). Same overhead that `claudeN` aliases already have.

## v1.4.0 — 2026-06-20 — OpenCode Zen provider alias

### Added

- **`opencode` provider alias** — OpenCode Zen curated AI gateway with **21 free models** (all $0 cost, all support tool calling + reasoning) and 49 paid models. The alias uses **router transport** (ccr) targeting the OpenAI-compatible endpoint `https://opencode.ai/zen/v1/chat/completions`.
- **Model overrides**: strong = `big-pickle` (free stealth model, 200K context, reasoning + tool_call), fast = `deepseek-v4-flash-free` (free, 200K context, reasoning + tool_call). Pinning is deliberate — auto-selection would pick `nemotron-3-ultra-free` (1M ctx) as strong and `trinity-large-preview-free` (131K, no reasoning) as fast, both suboptimal for coding workloads.
- **key-aliases.json mappings**: `ZEN_API_KEY` → `opencode` and `ApiKey_Opencode_Zen` → `opencode` (both key vars present in the user's keys file).
- **overrides.json pin**: `strong_model=big-pickle`, `fast_model=deepseek-v4-flash-free` (no transport/base_url override needed — catalog values are correct).
- **Sandbox test coverage**: resolver tests (key-alias mapping for both key vars, router transport from `@ai-sdk/openai-compatible` npm, zen/v1 base_url from catalog, model override beats auto-selection, stale-model-never-selected guards) + sync e2e tests (env file, alias, config-dir + plugins symlink, account-detection exclusion, idempotency, no-secret-leak). Providers test 69 → 90 assertions.

- **Live endpoint verification**: `GET /v1/models` HTTP 200; `POST /v1/chat/completions` round trip HTTP 200 with correct text for `big-pickle` (stealth, cost=$0, reasoning_content present) and `deepseek-v4-flash-free` (cost=$0); additional free models (`mimo-v2.5-free`, `nemotron-3-ultra-free`, `north-mini-code-free`) all HTTP 200 with cost=$0. Evidence in `scripts/tests/proof/70-zen-live.txt` (secret-free).
- **Docs**: dedicated `opencode` section in `docs/Provider_Aliases_User_Guide.md` (full free models table, setup, usage, live-verified notes, stealth model explanation).

### Changed

- `scripts/providers/key-aliases.json` and `scripts/providers/overrides.json` extended with the `opencode` entries (config-only; no code changes — same dynamic pattern as Xiaomi v1.3.0 / Z.AI v1.2.0 / DeepSeek).

### Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures. Providers test includes 21 new assertions for `opencode`.

### Honest notes

- The alias uses router transport (ccr) because Zen's free models use OpenAI-compatible format (`/v1/chat/completions`), not Anthropic native format. This adds a ccr dependency that native-transport aliases (deepseek, xiaomi) don't have.
- Big Pickle is a stealth model — the actual model served may vary (observed as deepseek-v4-flash). This is by design per OpenCode's documentation.
- The same pre-existing `~/api_keys.sh` set -u issue affects the in-process verifier for all providers; authoritative proof is the direct HTTP round trip.
- The 2 pre-existing, environmental opencode-skill-discovery failures in `run-proof.sh` remain unchanged (unrelated to this work).

## v1.3.0 — 2026-06-19 — Xiaomi MiMo provider alias

### Added

- **`xiaomi` provider alias** — Xiaomi MiMo via the **Anthropic-native endpoint** `https://api.xiaomimimo.com/anthropic` (`POST /anthropic/v1/messages`). Unlike most providers in this toolkit,

MiMo exposes a genuine native Anthropic endpoint that accepts `Authorization: Bearer`, so the alias uses **native transport** with no `claude-code-router` (`ccr`) dependency — the same direct-launch model as `deepseek`.
- **Model overrides**: strong = `mimo-v2.5-pro` (flagship, 1M context, reasoning, tool-call), fast = `mimo-v2-flash` (256K, cheapest tier). Pinning is deliberate — models.dev lists a `mimo-v2.5-pro-ultraspeed` id the **live API does not serve**, so the override guarantees only live-served ids are used.
- **key-aliases.json mapping**: `XIAOMI_MIMO_API_KEY` → `xiaomi` (the user's key-var name does not match the models.dev provider's documented `XIAOMI_API_KEY` env).
- **overrides.json pin**: native transport, `/anthropic` base_url, `mimo-v2.5-pro` / `mimo-v2-flash`.
- **Sandbox test coverage**: resolver tests (key-alias mapping, override forces native transport, `/anthropic` base_url beats catalog `/v1`, model pinning beats the stale `ultraspeed` entry, stale-id-never-selected guard) + sync e2e tests (env file, alias, config-dir + plugins symlink, account-detection exclusion, idempotency, no-secret-leak). Providers test 60 → 69 assertions.
- **Live endpoint verification**: `GET /v1/models` HTTP 200 (10 models); native `/anthropic/v1/messages` round trip HTTP 200 with correct text for both `mimo-v2.5-pro` and `mimo-v2-flash`; tool calling proven (`finish_reason: tool_calls`
  - `reasoning_content`); streaming confirmed. Evidence in `scripts/tests/proof/60-xiaomi-live.txt` (secret-free).
- **Docs**: dedicated `xiaomi` section in `docs/Provider_Aliases_User_Guide.md` (model table, setup, usage, live-verified notes).

## Changed

- `scripts/providers/key-aliases.json` and `scripts/providers/overrides.json` extended with the `xiaomi` entries (config-only; no code changes — same dynamic pattern as Z.AI v1.2.0 / DeepSeek).

## Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures. Providers test includes 9 new assertions for `xiaomi`. Live provider verifier 5/5 PASS.

## Honest notes

- The only failures in the repo's `run-proof.sh` are 2 **pre-existing, environmental** opencode-skill-discovery checks, unrelated to Xiaomi (zero opencode files changed by this release; they fail identically when run standalone).

- The in-process LLMsVerifier step reports `(unverified)` for every provider because `~/api_keys.sh` has a pre-existing unrelated `unbound variable` on a different provider's key under `set -u`; authoritative proof is the direct native-endpoint round trip (HTTP 200), recorded in the evidence file.

# v1.2.0 — 2026-06-19 — Z.AI Coding Plan provider alias

## Added

- **`zai-coding-plan` provider alias** — OpenAI-compatible router transport via `https://api.z.ai/api/coding/paas/v4` (Coding Max-Yearly Plan endpoint).
- **Model overrides**: strong = `glm-5.2` (flagship 1M context reasoning model, free on plan), fast = `glm-4.7` (204k context, tool_call, 0 cost).
- **key-aliases.json mapping**: `ZAI_API_KEY` → `zai-coding-plan` (targets the coding plan API endpoint instead of the general `z.ai` paas endpoint).
- **overrides.json pin**: overrides auto-selected strong/fast models for the coding plan.
- **Sandbox test coverage**: resolver tests (env-key matching, coding endpoint, router transport, glm-5.2/glm-4.7 model selection) + sync e2e tests (env file, alias, model overrides).
- **Live endpoint verification**: HTTP 200 at `/models` (8 models discovered), curl test of `glm-4.7` chat completion confirmed operational.
- **ccr integration**: provider auto-registered in `~/.claude-code-router/config.json` as the active default route.

## Changed

- `overrides.json` extended with `zai-coding-plan` section for model pinning.

## Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures. Provider tests include 5 new assertions for `zai-coding-plan`.

# v1.1.0 — 2026-06-16 — Distributed infrastructure + provider verification

Headline: stand up the full LLMsVerifier System on a remote host for heavy testing against **real production LLM services**, plus end-to-end provider aliases proven on two hosts and two transports.

## Added

- **`containers` + `challenges` submodules** (`submodules/`) — the distributed-boot orchestrator and its sibling. `helix-deps.yaml` confirms `containers` has zero own-org submodule deps.
- **Remote host registration** — `config/containers/nezha.env` registers `nezha.local` as a remote boot/test host (SSH key, podman runtime).
- **LLMsVerifier deployment overlays** (`config/containers/llmsverifier/`):
  - `docker-compose.app.yml` — the `llm-verifier` API (cgo image, config mount, `/api/health` healthcheck, loopback, fail-fast secrets).
  - `docker-compose.infra.yml` — observability tier: prometheus + grafana (auto-provisioned datasource + dashboard) + node-exporter. **No DBs** (the app uses SQLite; postgres/redis were unused and removed).
  - `Dockerfile.nezha` / `Dockerfile.mv` — cgo nested-module builds for the server + the `model-verification` tool.
  - `patches/0001..0005` — upstream LLMsVerifier fixes (see PR #2 below).
- **Deployment guide** `config/containers/llmsverifier/README.md` and the **Provider Aliases User Guide** `docs/Provider_Aliases_User_Guide.md` (HTML/PDF/DOCX exports included).
- **QA evidence** `docs/qa/20260616-infra/` — verification proofs, endpoint coverage, security posture, observability, per-provider sweeps, dual-host end-to-end alias proofs.

## Changed

- **Provider session accent color: orange → purple** across spec, guide, and the long-form doc. (Claude Code 2.1.178 cannot persist a default `/color`, so this is the documented default + a manual `/color purple` — a platform limit.)
- `claude-add-account` consolidated onto the shared `cma_link_shared_items` helper (single `CMA_SHARED_ITEMS` source).
- `claude-export-docs` now also emits **DOCX** (HTML/PDF/DOCX).

### Fixed (LLMsVerifier — shipped as PR #2, applied to deployed builds)

- **Auth header missing** — verification requests sent no `Authorization` header → HTTP 401 for every provider. Now `Bearer <key>`.
- **cohere 405** — switched to the OpenAI-compat endpoint (`api.cohere.ai/compatibility/v1`). Verifies at score 1.00.
- **gemini / huggingface** — corrected to OpenAI-compat / router endpoints (huggingface verifies; gemini code-ready pending a valid key).
- **model-id strictness** — verifies a requested id directly when not in the discovered list (no premature `model_not_found`).
- **no `/metrics`** — added `GET /api/metrics` + `/metrics` (stdlib Prometheus).
- **provider-session sync-state noise** — `cma_run_provider` no longer runs cross-account sync-state on isolated provider dirs.

### Verified live (real "Do you see my code?" against production APIs)

- **9 providers verified:** DeepSeek, Groq, Mistral, Cerebras, Novita, NVIDIA, Cohere, Codestral, HuggingFace.
- **Both transports, both hosts:** native (DeepSeek) + router (Novita via ccr) on macOS and on nezha.
- Account-side failures (402/401/429/403) and non-OpenAI providers documented honestly; excluded under "valid users only" but kept fully supported.

### Safety

- Provider dirs (`~/.claude-prov-*`) excluded from account detection — existing `claudeN` accounts and `claude-add-account` untouched.
- Secrets only in the keys file + on-host mode-600 `.env`; never in the repo. All published ports bound to loopback.

## v1.0.0 — 2026-06-16 — Dynamic provider-alias generator

First tagged release. `claude-providers` creates per-provider Claude Code aliases (DeepSeek, Groq, GLM, …) from your keys file pointed at each provider's strongest model — fully dynamic via models.dev + the LLMsVerifier submodule, hybrid native/claude-code-router transport, full lifecycle + tests + docs. See `docs/Provider_Aliases_User_Guide.md`.

# v1.6.7 — 2026-06-21 — Poe proxy fix for all aliases

## Fixed

- **Poe proxy not starting for poe2/poe3** — proxy detection used exact provider ID (`poe2_proxy.py`) which doesn't exist. Fixed to check base name too (`poe_proxy.py` for poe2, poe3 aliases).
- **lib.sh**: base proxy detection with `${CMA_PROVIDER_ID%%[0-9]*}`
- **alias file**: same fix applied

## Verified

- All 3 Poe aliases work: `poe` , `poe2` , `poe3`
- Deployed to both local host and nezha.local

## Tests

- Local: 8/8 ALL GREEN
- nezha.local: 7/8 (pandoc missing — pre-existing)

# v1.6.8 — 2026-06-21 — Poe proxy gzip fix

## Fixed

- **Poe proxy gzip decompression** — Poe API returns gzip-compressed responses but the proxy tried to read them as UTF-8 without decompressing, causing UnicodeDecodeError. Added gzip decompression for both success and error responses.

## Verified

- `poe` (claude-sonnet-4.6): YES
- `poe2` (gpt-5.5): YES
- `poe3` (grok-4): Different error (Grok-4 schema validation, not tools format)

## Tests

- Local: 8/8 ALL GREEN
- nezha.local: 8/8 ALL GREEN

# v1.6.9 — 2026-06-21 — Poe proxy $ref fix for Grok-4

## Fixed

- **Poe proxy $ref resolution** — Claude Code sends tool schemas with `$ref` references to `$defs`. Grok-4 and some providers don't support `$ref` in tool schemas. Added `resolve_refs()` function that extracts `$defs`, resolves all `$ref` references to inline definitions, and removes `$defs`.

## Verified

- `poe` (claude-sonnet-4.6): YES
- `poe2` (gpt-5.5): YES
- `poe3` (grok-4): YES (was failing, now works)

## Tests

- Local: 8/8 ALL GREEN
- nezha.local: 8/8 ALL GREEN

# v1.7.0 — 2026-06-22 — Poe proxy complete fix (all aliases verified)

## Fixed

- **Poe proxy shared directory** — the proxy at `~/.local/share/.../proxy/poe_proxy.py` was the OLD version without gzip and $ref fixes. install.sh copies from scripts/ but the shared dir still had the old version. Fixed by ensuring updated proxy is copied to shared directory.
- **install.sh** now copies proxy scripts during installation (already in place)

## Verified (all three aliases through full Claude Code flow)

- `poe` (claude-sonnet-4.6): YES
- `poe2` (gpt-5.5): YES
- `poe3` (grok-4): YES

### Root Cause Analysis

The proxy had three issues:

1. **gzip** — Poe returns gzip-compressed responses, proxy didn't decompress
2. **$ref** — Claude Code sends tool schemas with $ref, Grok-4 doesn't support them
3. **shared dir** — Updated proxy wasn't copied to shared directory

All three fixed and verified.

### Tests

- Local: 8/8 ALL GREEN
- nezha.local: 8/8 ALL GREEN

## v1.7.1 — 2026-06-22 — Full validation + release

### Fixed

- **Port-ready check** for proxy startup — replaced `sleep 1` with polling loop (`lsof -i`) ensuring proxy is listening before ccr config is written
- **Claude alias regression test** — 11 assertions proving claudeN aliases use `cma_run` (no proxy/transformer code), providers use `cma_run_provider`
- **Command injection fix** in verify_aliases_live.sh — replaced `bash -c` subshell with safe indirect expansion

### Tests

- Local: **9/9 ALL GREEN** (new: test_claude.sh — 11 assertions)
- nezha.local: 8/9 (export fails — pandoc missing)

### Release

- v1.7.1 — pushed to github, gitlab, gitflic, gitverse

# v1.7.2 — 2026-06-22 — Claude alias verification, full release

## Added

- **Claude alias verification** in verify_aliases_live.sh — tests claude1/2/3 alongside provider aliases
- **TOON tested** on all aliases — verified working

## Tests

- Local: **9/9 ALL GREEN**
- nezha.local: 8/9 (pandoc missing)
- All claude1/2/3:  OK
- All provider aliases: verified